DANIEL SITARU **MARIAN URSĂRESCU**

OLYMPIAD PROBLEMS

GEOMETRY

VOLUME I

ABOUT AUTHORS

Daniel Sitaru, born on 9 August 1963 in Craiova, Romania, is teacher at National Economic College "Theodor Costescu" in Drobeta Turnu - Severin. He published 36 mathematical books, last five of these "Math Phenomenon", Algebraic Phenomenon", Analytical Phenomenon", "The Olympic Mathematical Marathon" and "699 Olympic Mathematical Challenges" (the last one with Mihály Bencze), were very appreciated world wide. He is the founding editor of "Romanian Mathematical Magazine", an Interactive Mathematical Journal with 4.600.000 visitors in the last two years (www.ssmrmh.ro).

Many problems from his books were published in famous journals such as "American Mathematical Monthly", "Crux Mathematicorum", "Math Problems Journal", "The Pentagon Journal", "La Gaceta de la RSME", "SSMA Magazine". He also published an impressive number of original problems in all mathematical journals from Romania (GMB, Cardinal, Elipsa, Argument, Recreaţii Matematice). His articles from "Crux Mathematicorum" and "The Pentagon Journal" were also very appreciated.

Marian Ursărescu, was born on 1st of June 1965, in Focşani. He graduated from A.I. Cuza University, Faculty of Mathematics, in 1988. He is a teacher of mathematics from 1988 at "Roman Vodă" National College in Roman. Starting from 1990 until now, he had 47 pupils that participated on the Mathematical National Olympiad, which from 28 had obtained prizes and Olympic mentions.

He published over 100 problems and articles in Mathematical National Gazette. Also, he published several problems and articles in mathematical magazines such as "Mathematical Recreations", "Romanian Mathematical Magazine", "Let's understand math." A lot of his proposed problems had been selected in various mathematical contests, olympiads and mathematical books. He co-authored "Functional Equations" together with M. O. Drâmbe.

FROM AUTHORS

In July 2016 was founded "Romanian Mathematical Magazine" (RMM) (www.ssmrmh.ro) as an Interactive Mathematical Journal.

Same date was founded "Romanian Mathematical Magazine"-Online Mathematical Journal (ISSN-2501-0099) and "Romanian Mathematical Magazine"-Paper Variant (ISSN-1584-4897).

In three years the website of RMM was visited by over 5,000,000 people from all over the world. With over 7,000 proposed problems posted, over 11,000 solutions and many math articles and math notes RMM is a big chance for young mathematicians from whole world to be known as great proposers and solvers. This book is a small part of RMM-Interactive Journal.

Many thanks to RMM-Team for proposed problems and solutions.

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PROBLEMS

TRIANGLE

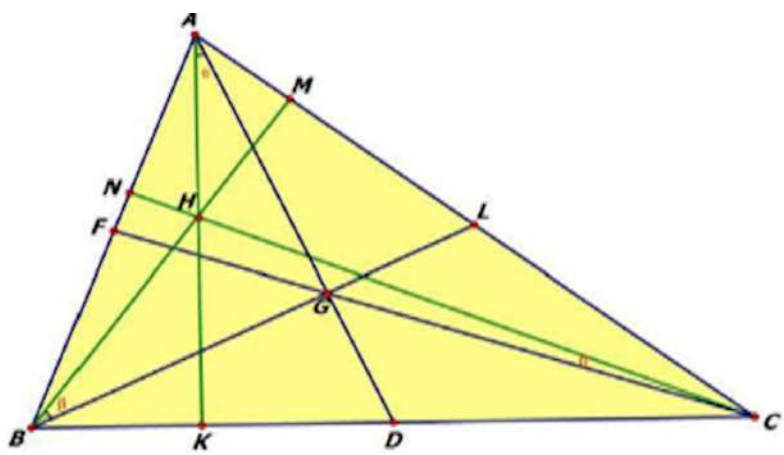
1.1 ***If in*** $\Delta\, ABC: A_1, A_2 \in (BC), B_1, B_2 \in (AC), C_1, C_2 \in (AB), \frac{BA_1}{CA_1} = \frac{CB_1}{AB_1} = \frac{AC_1}{BC_1} = \frac{CA_2}{BA_2} = \frac{AB_2}{CB_2} = \frac{BC_2}{AC_2}$ ***then:*** $AA_1^2 + BB_1^2 + CC_1^2 = AA_2^2 + BB_2^2 + CC_2^2$

Daniel Sitaru

1.2 ***In*** $\Delta\, ABC:\ max(A, B, C) = 135° \Leftrightarrow \frac{s+r}{R+r} = \sqrt{2}$

Mehmet Şahin

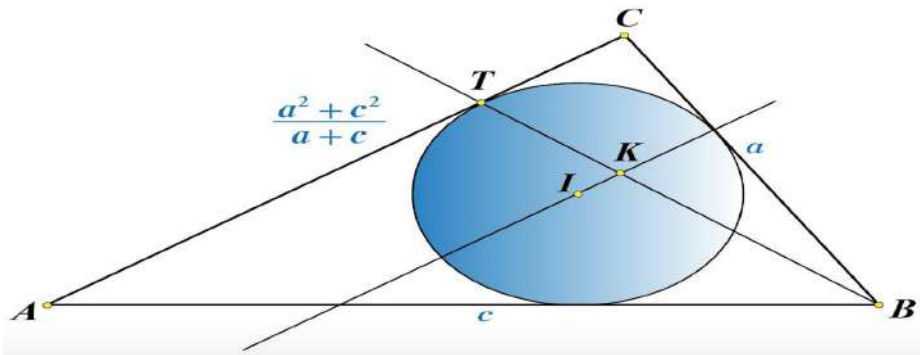
1.3 ***If in*** $\Delta\, ABC$ ***acute,*** $a > b > c$, H ***– orthocenter, G – centroid,*** $(\sphericalangle HAG) = \alpha, m(\sphericalangle HBG) = \beta, m(\sphericalangle HCG) = \theta$ ***then:*** $\tan\beta = \tan\alpha + \tan\theta$



Mehmet Şahin

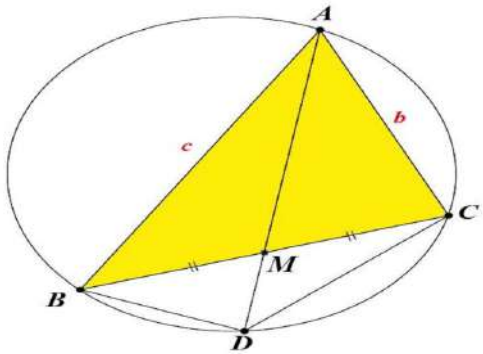
1.4 I***-Incenter of*** ABC, K***-symmedian point of*** ABC, T***-tangency point with incircle***

AC = ***Contraharmonic Mean*** $\{a, c\} = \frac{a^2+c^2}{a+c}$. ***Prove: 1.*** $IK \parallel AC$ ***2.*** B, K, T ***are collinear***



Abdilkadir Altintaş

1.5 ***ABC non-isosceles triangle, AM – median from A, D – point of intersection of AM with circumcircle, BC is contraharmonic mean of AB and AC. $AC = b, AB = c, BC = \frac{b^2+c^2}{b+c}$. Prove: $AD = BD + DC$***

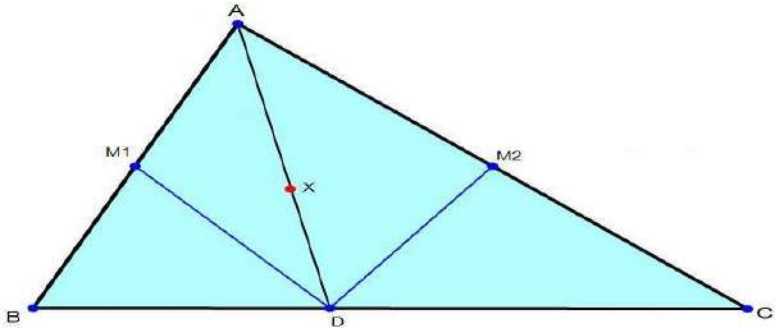


K.R.S. Sastry, Abdilkadir Altintaş

1.6 ***ABC non-isosceles triangle with sides a, b, c, AX intersects the BC at D, M_1, M_2 midpoints of sides AB and AC, $DM_1 = DM_2$. Prove:***

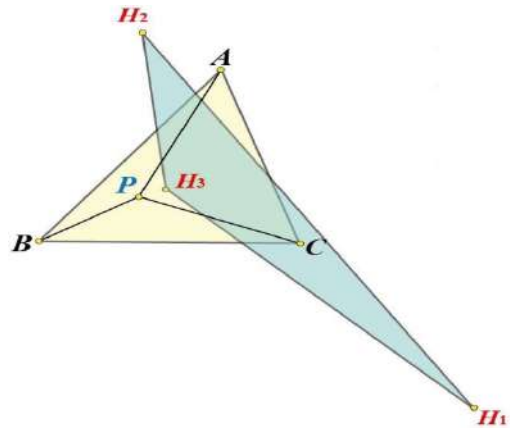
1. If X = Symmedian point $b^2 + c^2 = 2a^2$, 2. If X = Incenter $(b + c)^2 = 2a^2$

3. If X = Gergonne point $b + c = 2a$



Abdilkadir Altintaş

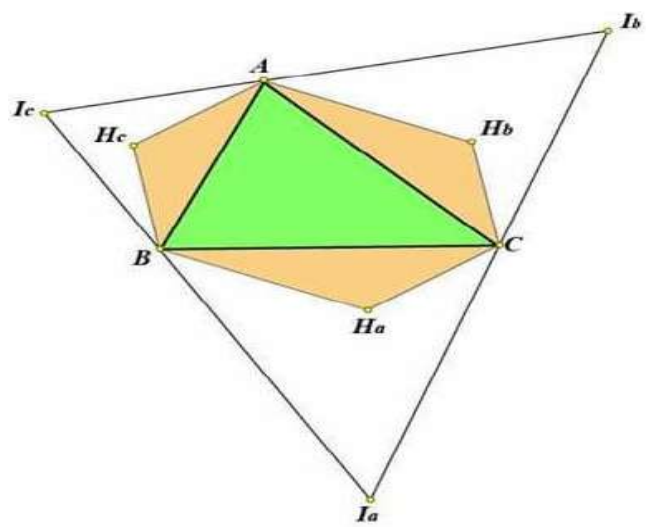
1.7 ***P, any point in the plane of ABC does not belong to side lines AB, BC, CA, H_1, H_2, H_3 orthocenters of PAB, PBC, PCA, Prove: $[ABC] = [H_1H_2H_3]$.***



Abdilkadir Altintaş

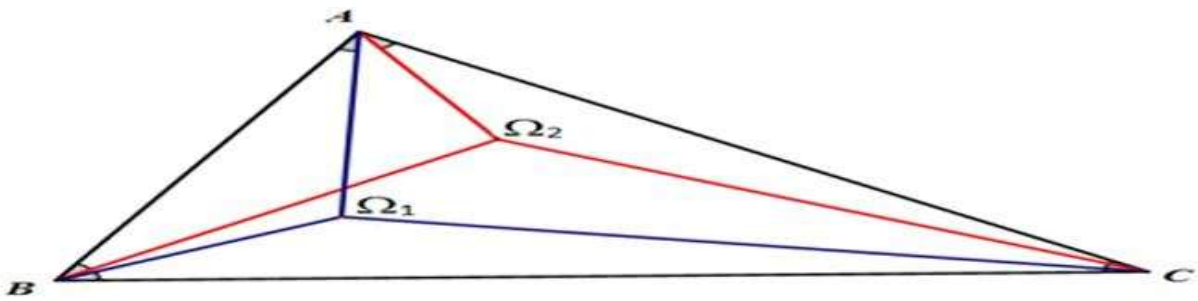
1.8 ***I_a, I_b, I_c – ex-centers of ABC, H_a, H_b, H_c – orthocenters of I_aBC, I_bCA, I_cAB, r, R – inradius and circumradius respectively. Prove:***

$$\frac{[AH_cBH_aCH_bA]}{[I_aI_bI_c]} = \frac{r}{R}$$



Abdilkadir Altintaş

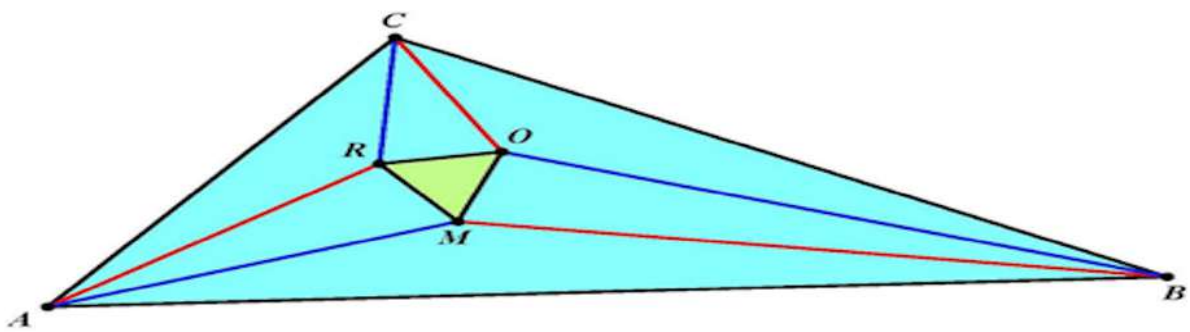
1.9 ***If Ω_1, Ω_2 - Brocard's point in $\Delta\ ABC$ then: $A\Omega_1 \cdot B\Omega_1 \cdot C\Omega_1 = A\Omega_2 \cdot B\Omega_2 \cdot C\Omega_2$***



Ali Can Gullu

1.10 ***If $\Delta\ MRO$ - is the MORLEY triangle of $\Delta\ ABC$ then:***

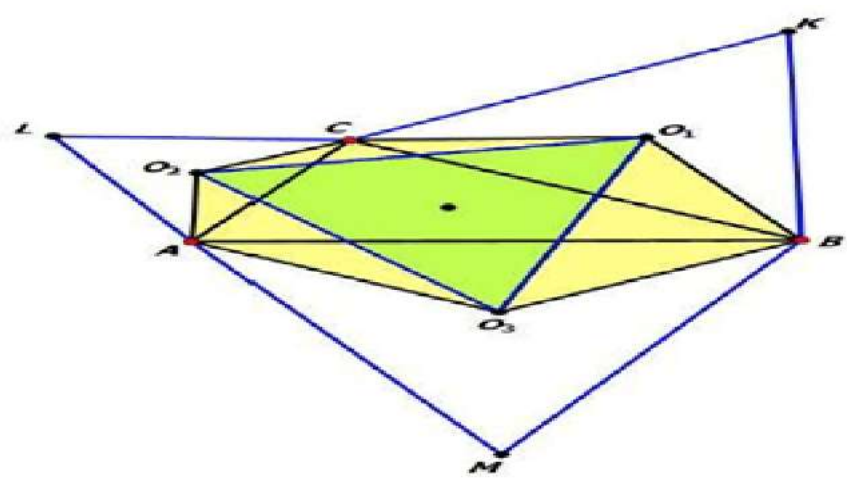
$$OC \cdot RA \cdot MB = OB \cdot RC \cdot MA$$



Ali Can Gullu

1.11 ***If O_1, O_2, O_3 - centroids in equilateral triangles BKC, CLA, AMB then:***

$$s[AO_3BO_1CO_2] = 2S[O_1O_2O_3]$$



Ali Can Gullu

1.12 **If in $\Delta\, ABC, m(\sphericalangle C) = 60^{\circ}$ then:**

$$\cos A \cdot \cos B = \left(\frac{r}{R}\right)^2 + \frac{r}{R} - \frac{1}{2}$$

Adil Abdullayev

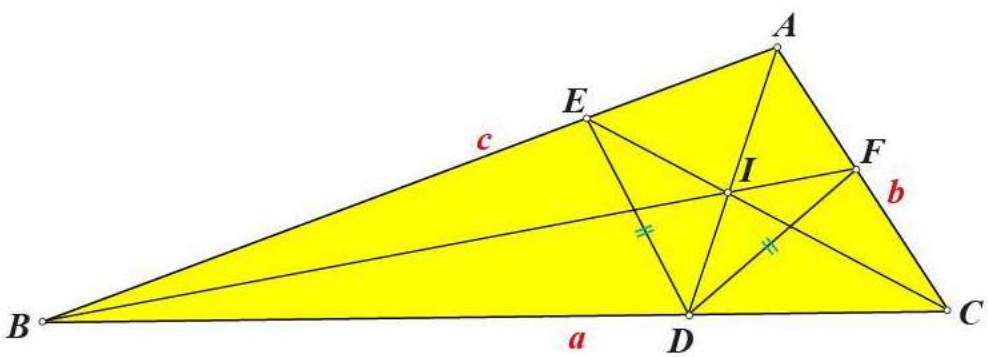
1.13 ***If in $\Delta ABC, AD, BE, CF - concurrent\ cevians, D \in (BC), E \in (AC), F \in (AB)$ then :***

$$\sum \frac{1}{\frac{1}{\frac{AF}{FB} + \frac{AE}{EC}} + 1} = 2$$

Aatman Supkar

1.14 ***ABC scalene triangle, AD, BE, CF angle bisectors, $DE = DF$***

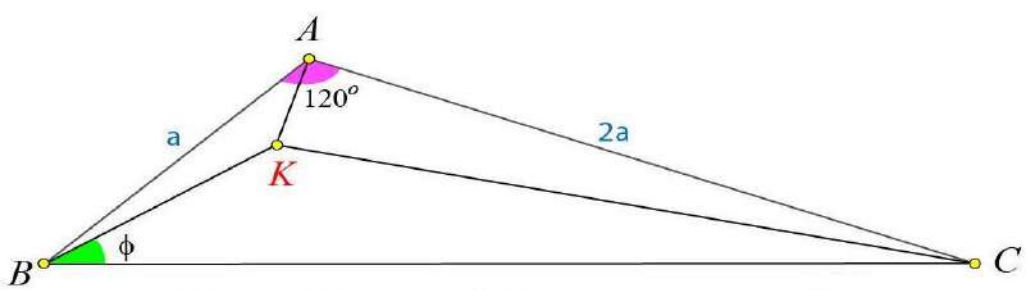
Prove: ***1.*** $\frac{a}{b+c} = \frac{b}{c+a} + \frac{c}{a+b}$, ***2.*** $\angle BAC > 90$



Abdilkadir Altintaş

1.15 ***K – symmedian point of ABC, $AC = 2AB$. $m(BAC) = 120°$. Prove:***

$$AK = \frac{a\sqrt{3}}{6}, \varphi = 30°, CA \perp AK$$



Abdilkadir Altintaş

1.16 ***In $\Delta\ ABC$ the following relationship holds:***

$$8R\sum\frac{\sin\frac{A}{2}}{w_a} = 2 + \sum\frac{a}{s-a}$$

Adil Abdullayev

1.17 **If in $\Delta\ ABC, m(\sphericalangle) = 90°, M \in (BC), MP \parallel AC, MN \parallel AB, MP = MN = k$ then:**

$$\left(\frac{b}{c}+1\right)^2 + \left(\frac{c}{b}+1\right)^2 = \left(\frac{a}{k}\right)^2$$

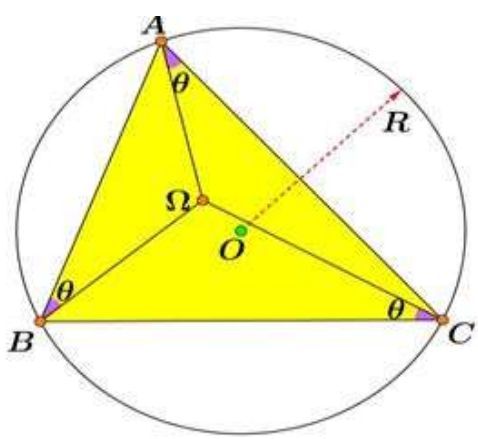
Aatman Supkar

1.18 **In $\Delta ABC, H - orthocentre,\ H - lies\ on\ the\ incircle \leftrightarrow cosAcosBcosC = \frac{r^2}{4R^2}$**

Adil Abdullayev

1.19 ***In ΔABC the following relationship holds:***

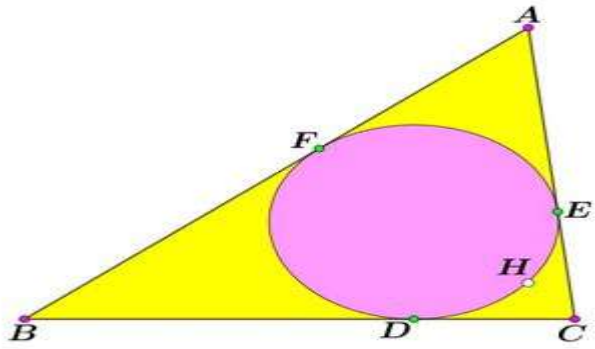
$$\frac{1}{R^2 - \Omega O^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$



Abdilkadir Altintaş, Miguel Ochoa Sanchez

1.20 ***If in ΔABC the orthocentre lies on the incircle then :***

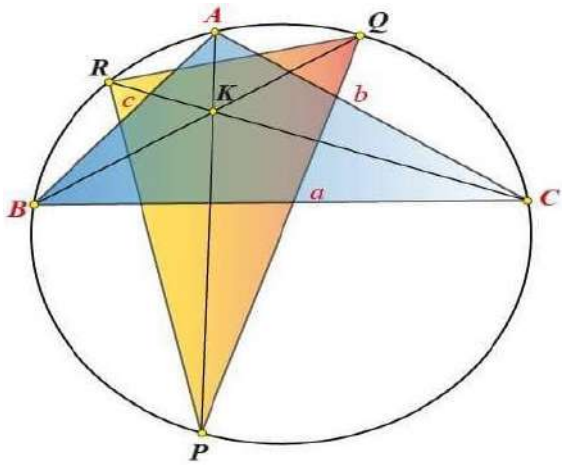
$$tanA + tanB + tanC = \frac{2s}{r}$$



Adil Abdullayev, Miguel Ochoa Sanchez

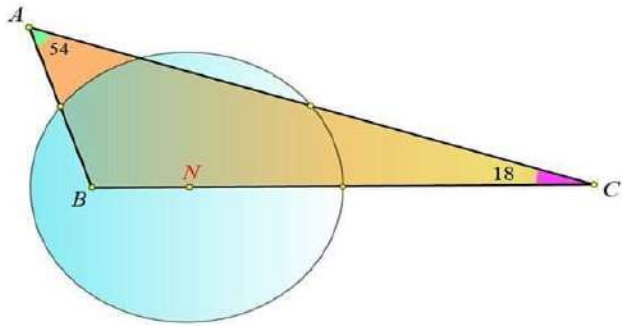
1.21 ***ABC is a triangle with sides a, b, c, K – symmedian point of ABC, P, Q, R points of intersection of AK, BK and CK with circumcircle of ABC. Prove:***

$$a^2 + b^2 = 5c^2 \Rightarrow PR^2 + QR^2 = PQ^2$$



Abdilkadir Altintaş

1.22 ***Prove: N – nine-point center of ABC lies on BC.***



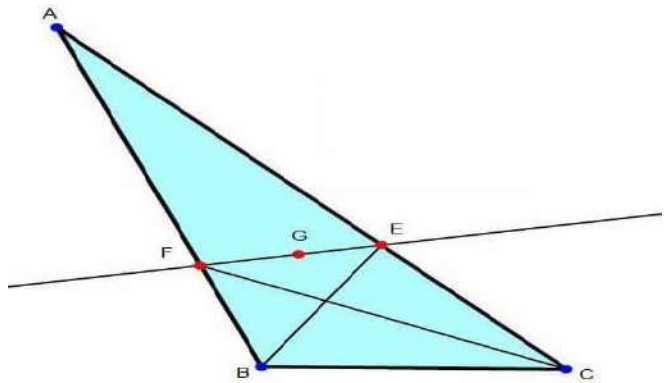
Abdilkadir Altintaş

1.23 ***In ΔABC the following relationship holds:***

$$\left(\sum_{cyc}\tan\frac{A}{4}\right)\left(\sum_{cyc}\cot\frac{A}{4}\right)=\left(\sum_{cyc}\frac{w_b+w_c}{h_a}\right)^2-\left(\sum_{cyc}\frac{r_b+r_c}{a}\right)^2$$

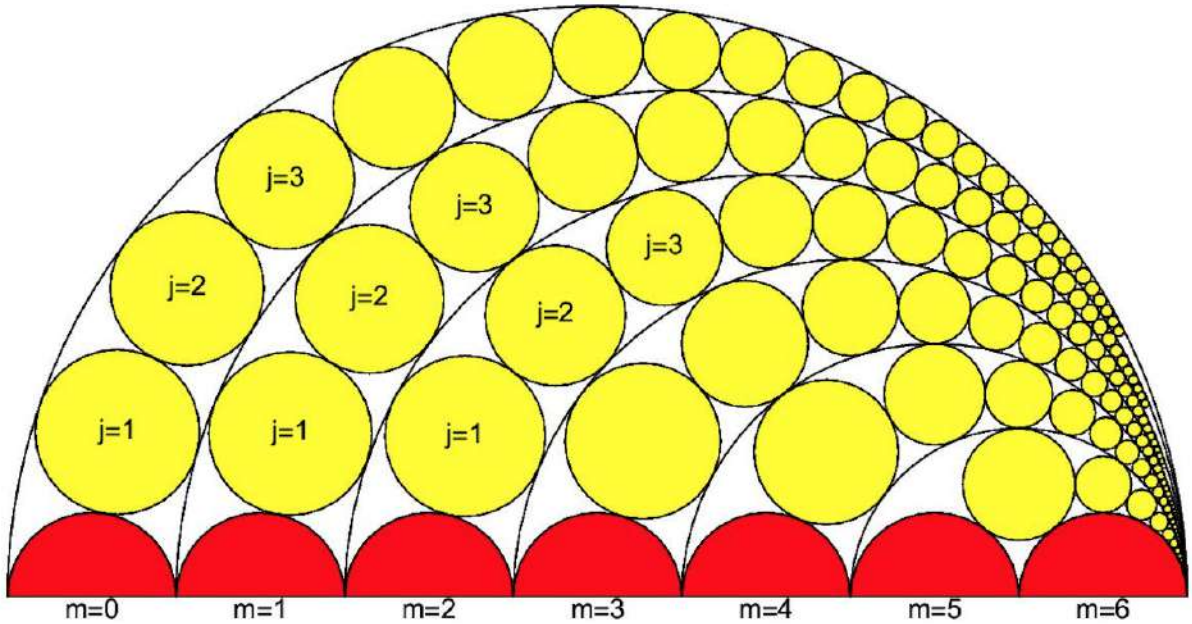
Bogdan Fuștei

1.24 ***ABC triangle with side lengths a, b, c. BE and CF internal angle bisectors G centroid of ABC, F, G, E collinear.Prove:*** $\frac{1}{a}=\frac{1}{b}+\frac{1}{c}$



Abdilkadir Altintaş

1.25



$n = 7$

$$r_{n,m,j} = \frac{(n-m)^2 - (n-m)}{(m-n)^2 - (n-m) + j^2} \cdot \frac{R}{n}$$

Francisco Javier Garcia Capitan

1.26 ***In $\Delta\ ABC$ the following relationship holds:***

$$r_a - r_b = 4R\sin\frac{A-B}{2}\cos\frac{C}{2}$$

Adil Abdullayev

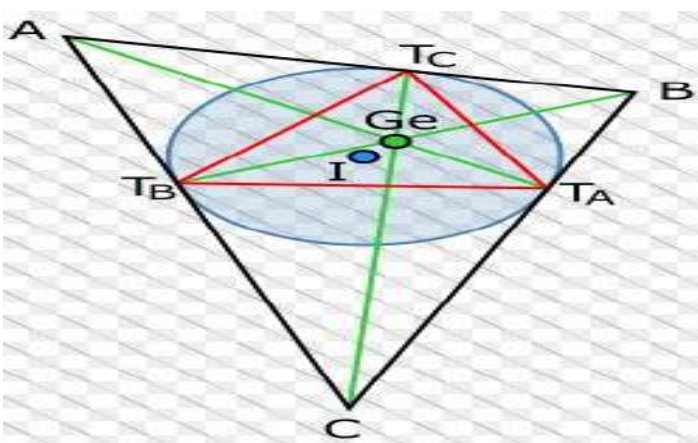
1.27 ***If $a, b, c - sides\ in\ \Delta ABC, \sqrt{a}, \sqrt{b}, \sqrt{c} - sides\ in\ \Delta A'B'C'$ then :***

$$\frac{\sqrt{a}}{\cos A'} + \frac{\sqrt{b}}{\cos B'} + \frac{\sqrt{c}}{\cos C'} = 2(r_a + r_b + r_c)\sqrt{\frac{R}{S}}$$

Mehmet Şahin

1.28 ***In $\Delta ABC, m_1, m_2, m_3$ - medians in Gergonne's triangle. Prove that:***

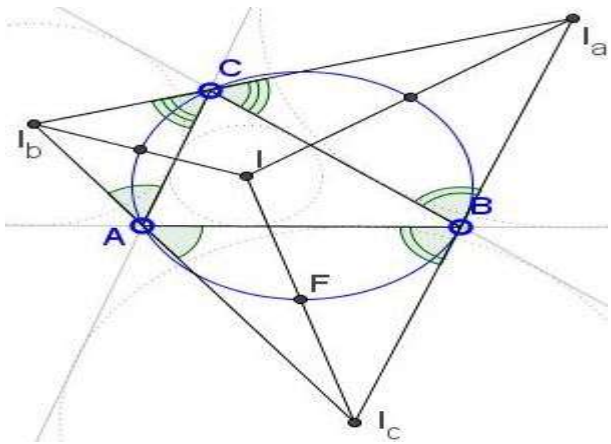
$$m_1^2 + m_2^2 + m_3^2 = \frac{3r^2}{2R}(r_a + r_b + r_c)$$



Mehmet Şahin

1.29 ***Let $\Delta PQR, \Delta I_aI_bI_c$ be the contact respectively the excentral triangle of ΔABC. If $S_1 = S[ARQ], S_2 = S[BPR], S_3 = S[CPQ], S_0 = S[PQR]$,***

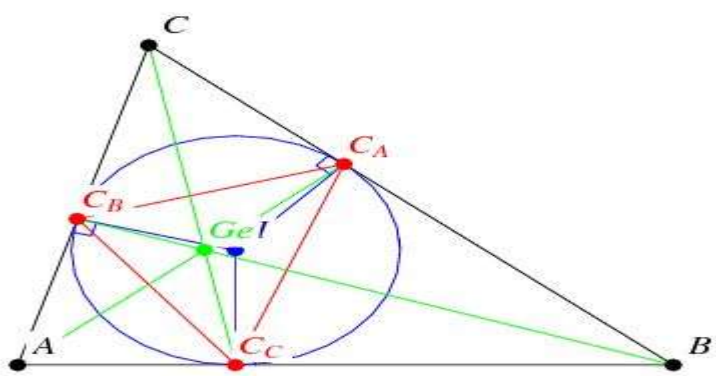
$A = S[I_aI_bI_c]$ then: $4A^2S_1S_2S_3 = S^5$, $S_0A = S^2$



Mehmet Şahin

1.30 ***If in*** $\Delta ABC, h_1, h_2, h_3 - are\ altitudes\ of\ \Delta C_A C_B C_C,$ ***where*** AC_A, BC_B, CC_C***-Gergonne's cevians then:***

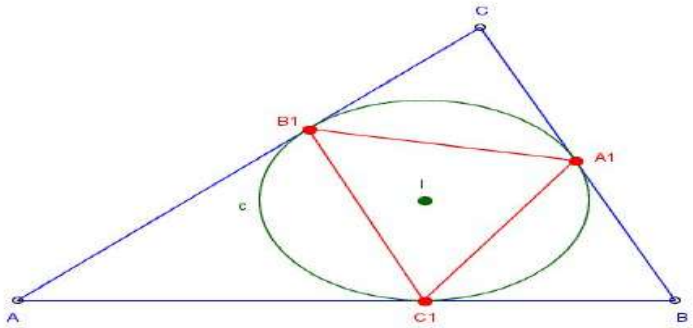
$$\left(\frac{h_1}{h_a}\right)^2 + \left(\frac{h_2}{h_b}\right)^2 + \left(\frac{h_3}{h_c}\right)^2 = 1 - \frac{r}{2R}$$



Mehmet Şahin

1.31 ***Let*** $h_1, h_2, h_3, m_1, m_2, m_3$ ***be the altitudes respectively the medians of intouch triangle in*** ΔABC***. Prove that:***

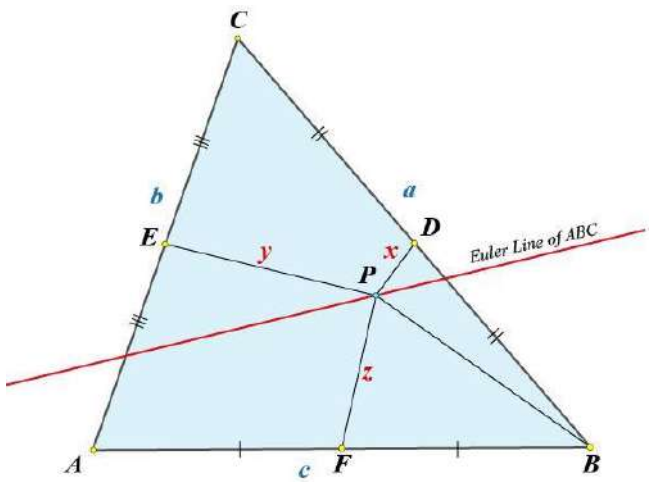
$$\frac{\frac{1}{h_1^2} + \frac{1}{h_2^2} + \frac{1}{h_3^2}}{\frac{1}{h_a^2} + \frac{1}{h_b^2} + \frac{1}{h_c^2}} = \frac{4R^2}{r^2} \cdot \frac{m_1^2 + m_2^2 + m_3^2}{m_a^2 + m_b^2 + m_c^2}$$



Mehmet Şahin

1.32 ***ABC be triangle with side lengths a, b, c, D, E, F midpoints of BC, AC, AB respectively P, any point on Euler line of ABC, $PD = x, PE = y, PF = z$***

Prove: $$mb^2 + nc^2 = (m+n)a^2 \Rightarrow my^2 + nz^2 = (m+n)x^2$$

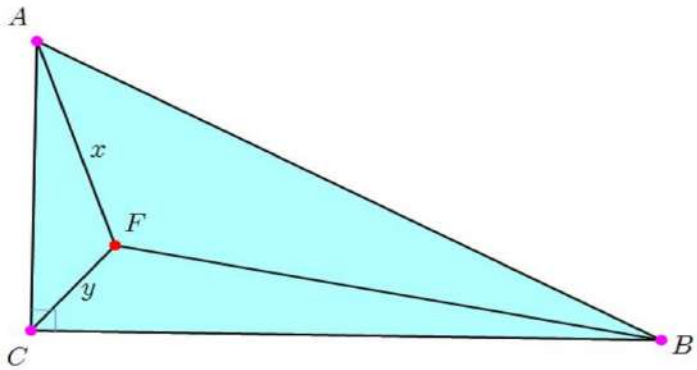


Abdilkadir Altintaş

1.33 ***ABC right angled triangle at C, F – Fermat point of ABC, i.e.***

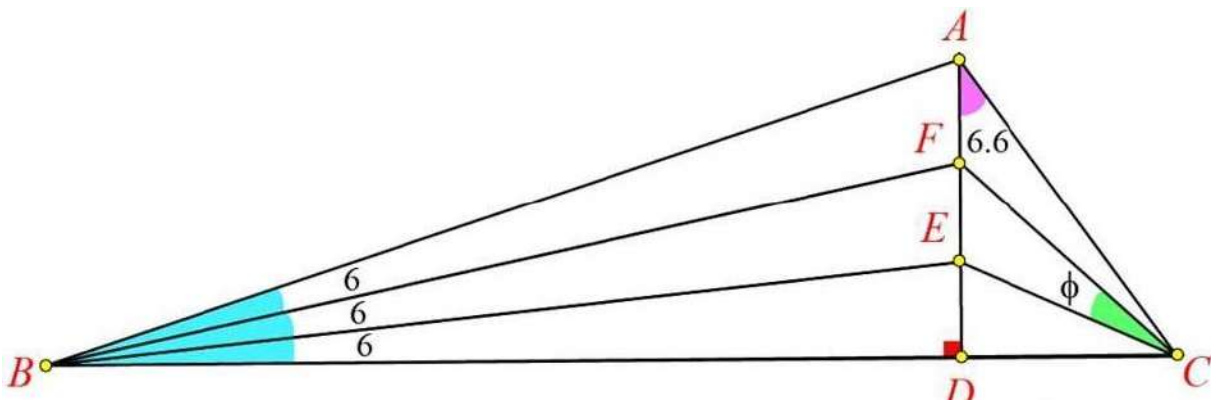
$$m(AFC) = m(CFB) = m(AFC) = 120°, AF = x, CF = y, x > y$$

Prove: $FB = \frac{y(x+2y)}{x-y}$



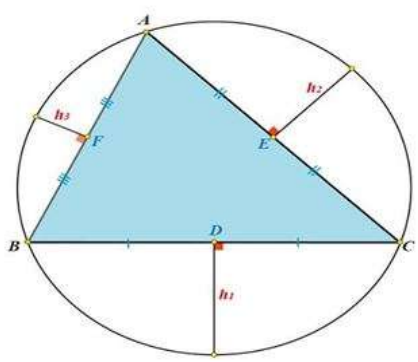
Abdilkadir Altintaş

1.34 ***Angles are in degrees,*** $AD \perp BC$. ***Prove:*** $\phi = 6 + 6 + 6$.



Abdilkadir Altintaş

1.35 ***Prove that:*** $2\left(h_1^2 + h_2^2 + h_3^2\right) = 8R^2 + r^2 - s^2$



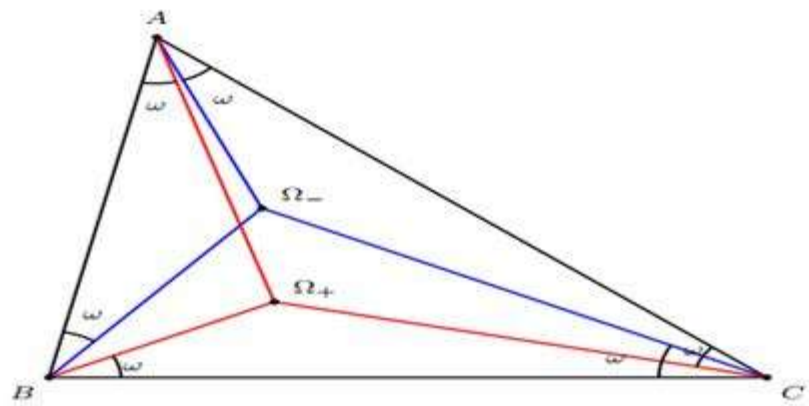
Abdilkadir Altintas

1.36 ***In*** $\Delta ABC, m(\sphericalangle B) = 60°, F$ ***–Fermat point. If*** $AF = x, BF = y, CF = z$ ***then:*** $x^2 = yz$.

Abdilkadir Altintaş

1.37 Ω_+, Ω_- ***are Brocard points. Prove that:***

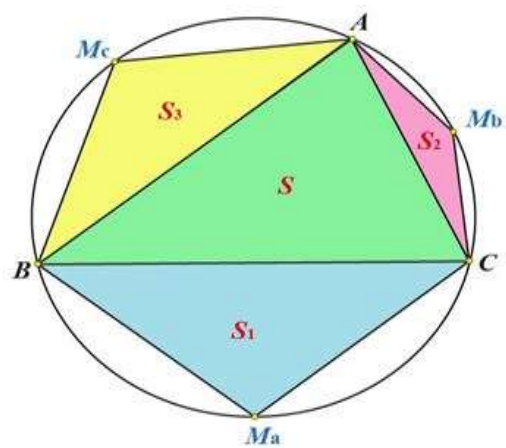
$$A + \omega = \frac{\pi}{2} \Leftrightarrow tan^2 A = tanB \cdot tanC$$



K.R.S.Sastry

1.38 M_a, M_b, M_c ***–midpoints of arcs. Prove that:***

$$\frac{S_1 + S_2 + S_3}{S} = \frac{R}{r} - 1$$



Abdilkadir Altintaş

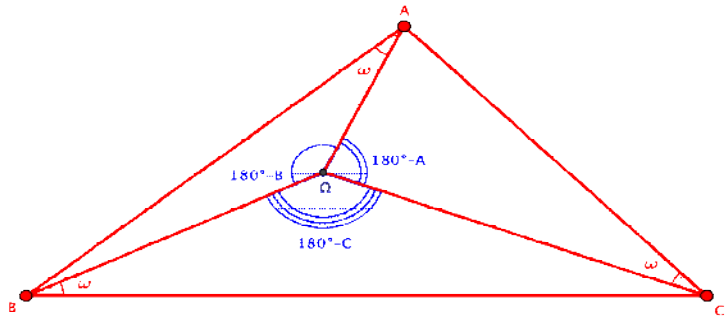
1.39 ***If in*** $\Delta ABC, I - incentre, ID \perp BC, IE \perp AC, IF \perp AB$ ***then:***

$$\frac{1}{S[IEF]} + \frac{1}{S[IDF]} + \frac{1}{S[IDE]} \geq \frac{4(5R - r)}{rS}$$

Mehmet Şahin

1.40 ***Prove that:***

$$\frac{1}{A\Omega^2}+\frac{1}{B\Omega^2}+\frac{1}{C\Omega^2}=\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)\left(\frac{a^2}{b^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}\right)$$



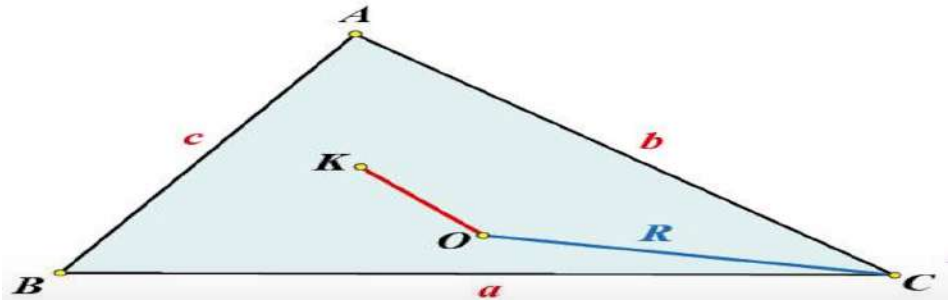
Mehmet Şahin

1.41 ***In ΔABC the following relationship holds:***

$$\left(\frac{b}{c}+\frac{c}{b}\right)cosA+\left(\frac{c}{a}+\frac{a}{c}\right)cosB+\left(\frac{a}{b}+\frac{b}{a}\right)cosC=3$$

Theodoros Sampas

1.42 ***O – circumcenter, R – circumradius, K – symmedian point. Prove:***



$$R^2-OK^2=\frac{3a^2b^2c^2}{(a^2+b^2+c^2)^2}$$

Abdilkadir Altintaş

1.43 ***In ΔABC the following relationship holds:***

$$\sqrt{\frac{h_bh_c}{rr_a}}+\sqrt{\frac{h_ch_a}{rr_b}}+\sqrt{\frac{h_ah_b}{rr_c}}=\frac{w_b+w_c}{a}+\frac{w_c+w_a}{b}+\frac{w_a+w_b}{c}$$

Bogdan Fuştei

1.44 ***In ΔABC, AD, BE, CF –altitudes, AL, BK, CM –medians, $\alpha = \sphericalangle(DAL)$,***

$\beta = \sphericalangle(EBK), \gamma = \sphericalangle(FCM)$. If $a > b > c$ then: $tan\beta = tan\alpha + tan\gamma$

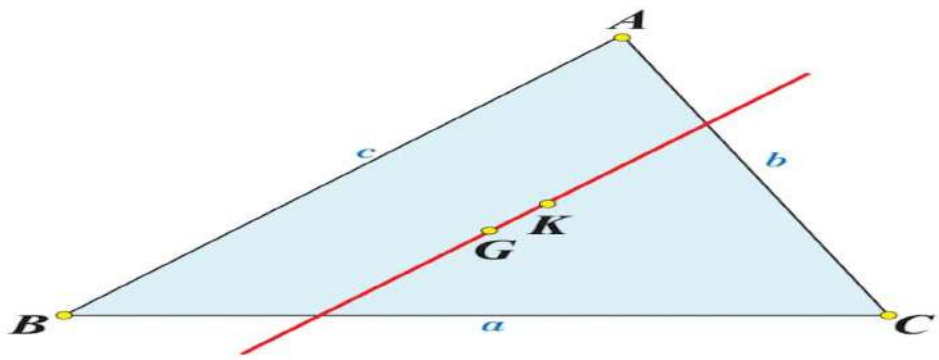
Mehmet Şahin

1.45 ***Let be ΔDEF –medial triangle of ΔABC with sides a, b, c, I –incenter and S_p –Spieker center (incenter of ΔDEF). If a, b or $c = \frac{s}{4}$ then I –lies on Spieker circle.***

Abdilkadir Altintaş

1.46 ***G – centroid of ABC, K – symmedian point of ABC, $a^2 + b^2 = 2c^2$***

Prove: $GK \parallel AB$

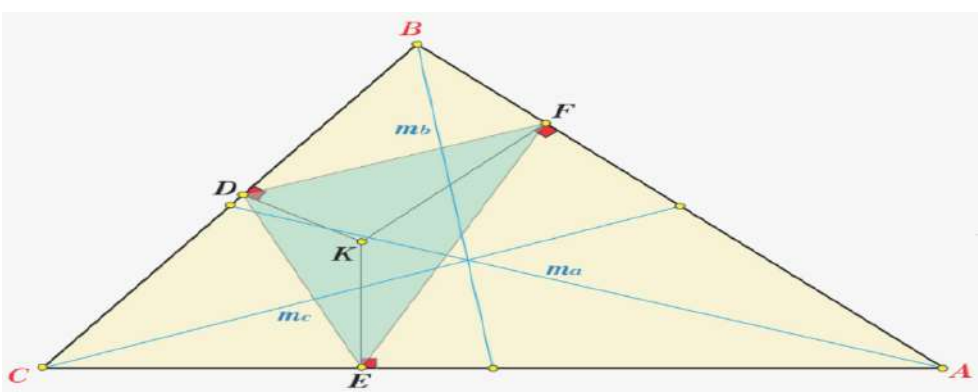


Abdilkadir Altintaş

1.47 ***K – symmedian point of ABC, DEF, pedal triangle of ABC from K***

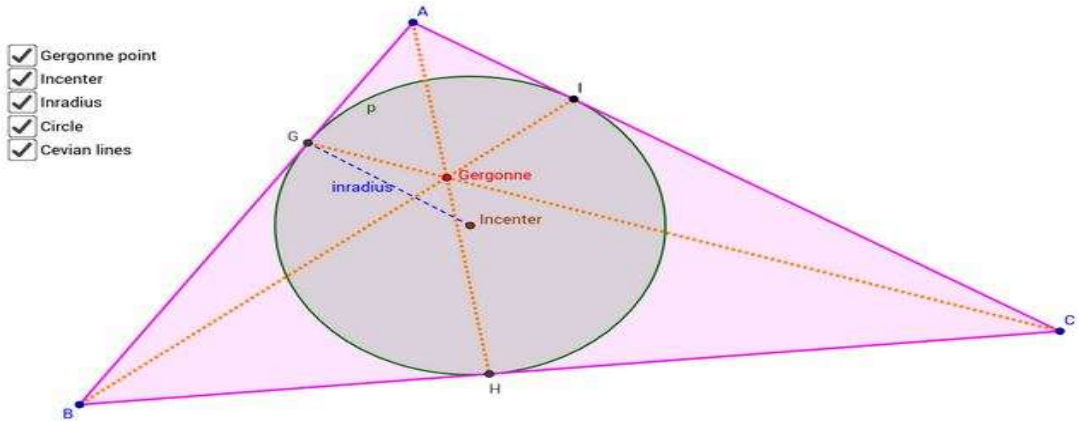
m_a, m_b, m_c median lengths of ABC.Prove that

$$\frac{m_a}{EF} = \frac{m_b}{DF} = \frac{m_c}{DE}$$



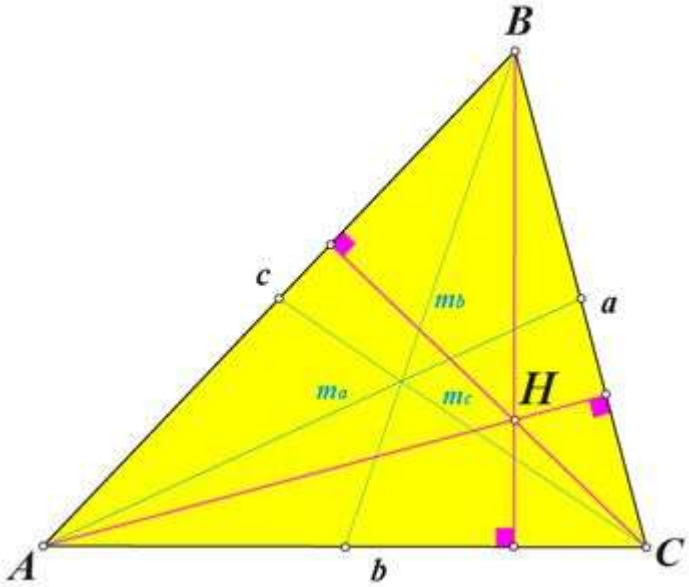
Honsberger, 1995

1.48 ***Let ΔGIH be the Gergonne's triangle of $\Delta\ ABC$ with I – incentre. Denote R_A, R_B, R_C – circumradii of $\Delta\ AGI, \Delta BHG, \Delta CHI$ and R_a, R_b, R_c – circumradii of $\Delta BIC, \Delta CIA, \Delta AIB$. Prove that: $R_aR_AR_bR_BR_cR_C = r^3R^3$***



Mehmet Şahin

1.49 ***AUTOMEDIAN TRIANGLE*** $(2b^2 = a^2 + c^2), H - orthocenter$



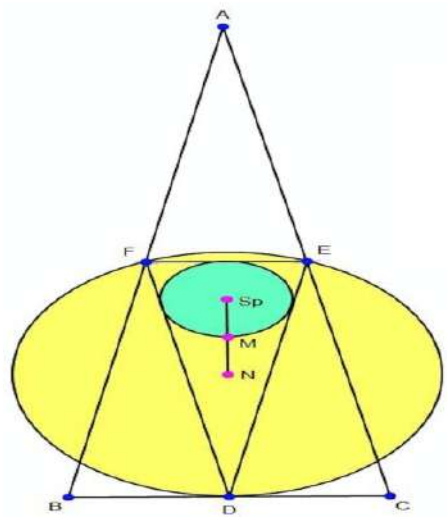
Prove that:

$$2cotB = cotA + cotC, \qquad 2cos2B = cos2A + cos2C$$

$$2m_b^2 = m_a^2 + m_c^2, \qquad 2BH^2 = AH^2 + CH^2$$

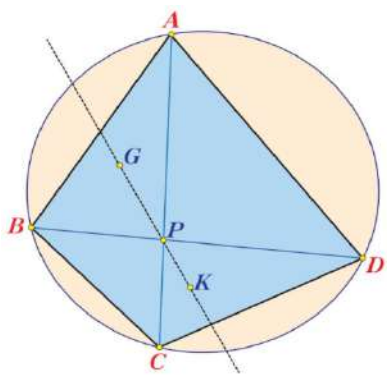
Abdilkadir Altintaş

1.50 $AC > BC, AB = AC$, M – *midpoint of Spieker center and Nine point Center,* M – *lies on Spieker circle. Prove:* $\sqrt{\frac{AC}{BC}} = \varphi$ *(Golden Ratio)*



Abdilkadir Altintaş

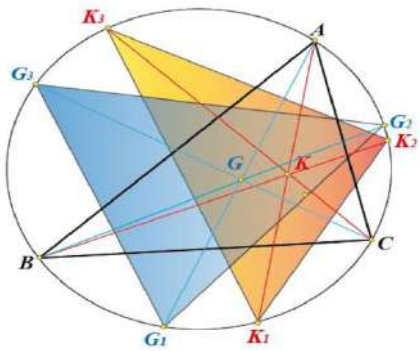
1.51 *Given:* $ABCD$ – *cyclic quadrilater,* P – *intersection of diagonals,* G – *centroid of* APB, K – *symmedian point of* PCD *. Prove:* G, P, K – *collinear*



Abdilkadir Altintaş

1.52 *Let* ABC *be triangle with centroid* G *and symmedian point* K. $G_1G_2G_3$ *is circumcevian triangle of* G *and* $K_1K_2K_3$ *is circumcevian triangle of* K. *Prove:*

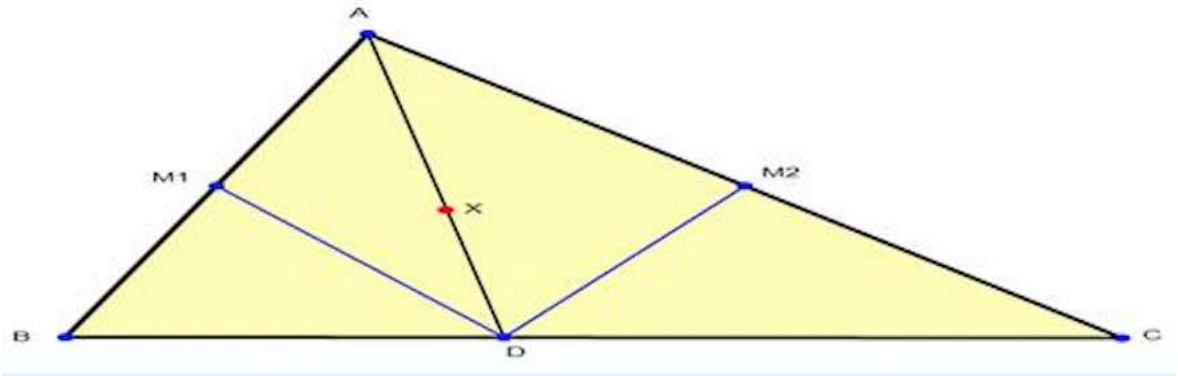
$$\frac{[G_1G_2G_3]}{[K_1K_2K_3]} = \frac{(a^2 + b^2 + c^2)^3}{27a^2b^2c^2}, \qquad [G_1G_2G_3] \geq [K_1K_2K_3]$$



Abdilkadir Altintaş

1.53 ***ABC non-isosceles triangle with sides a, b, c, AX intersects the BC at D, $M1, M2$ midpoints of sides AB and AC, $DM1 = DM2$. Prove:***

If X = Symmedian point then $b^2 + c^2 = 2a^2$



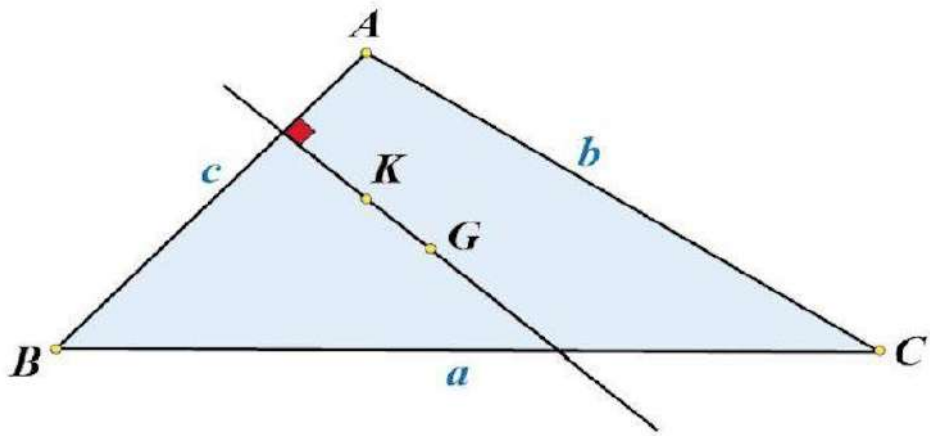
Abdilkadir Altintaş

1.54 ***In $\Delta ABC, K$ −Lemoine's point, BM −median, CN-internal bisector, $M \in (AC)$, $N \in (AB)$, $\frac{\angle A}{4} = \frac{\angle B}{2} = \frac{\angle C}{1} \Rightarrow N, K, M$ −collinears.***

Abdilkadir Altintas

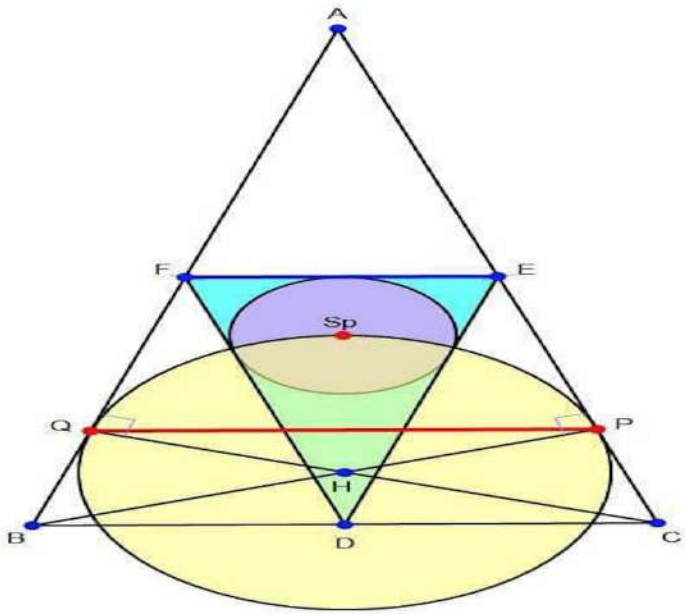
1.55 ***ABC triangle $a \neq b, K$ – symmedian of ABC, G – centroid of $ABC, KG \perp AB$***

Prove: $a^2 + b^2 = 5c^2$



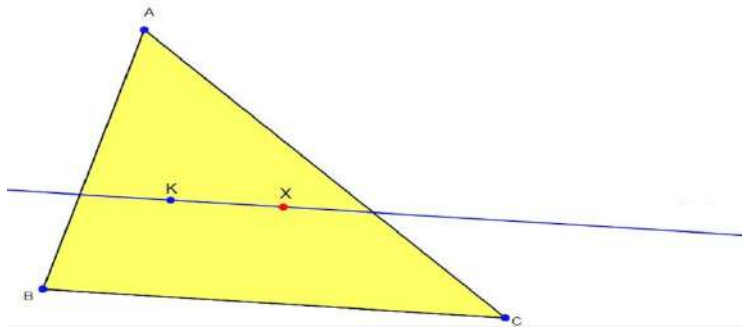
Abdilkadir Altintaş

1.56 ***ABC isosceles triangle $AB = AB$ and $AC > BC$, S_p – Spieker center (Incenter of Medial Triangle), H – orthocenter, BP, CQ – altitudes, P, Q, SP lie on the same circle centered at H. Prove: $\frac{PQ}{EF} = \varphi$ (Golden Ratio)***



Abdilkadir Altintaş

1.57 ***K – symmedian point of ABC. If $X = G$ where G is centroid of ABC $a^2 = \frac{b^2+c^2}{2}$. $X = I$ where I is incentre of ABC $a^2 = \left(\frac{b^2+c^2}{b+c}\right)^2$. $X = O$ where O is circumcenter of ABC $a^2 = \frac{b^4+c^4}{b^2+c^2}$***

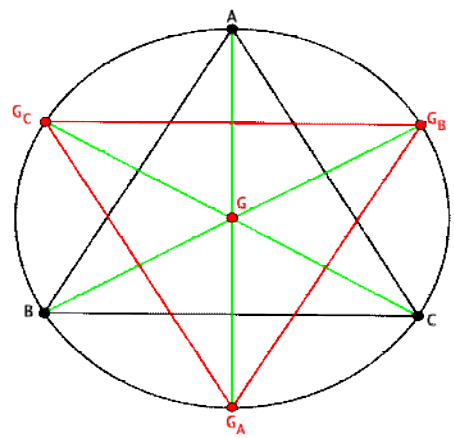


Abdilkadir Altintaş

1.58 ***Prove that in a triangle which sides are:*** $\pi^2 - e^2, e^2, \pi e$ ***one of angles is two times bigger than the other angle.***

Jovan Mikic

1.59 G ***– centroid of*** ΔABC***. Find:*** $\frac{[G_aG_bG_c]}{[ABC]} = f(a,b,c) = ?$



Thanasis Gakopoulos

1.60 ***FUSTEI'S IDENTITIES : In*** ΔABC ***the following relationship holds:***

$n_a^2 + g_a^2 + 2r_br_c = 4m_a^2, n_a$ ***– Nagel's cevian,*** $h_a = \frac{4m_a^2 - n_a^2 - g_a^2}{r_b + r_c}, g_a$ ***– Gergonne's cevian***

$$\frac{1}{r} = \sum_{cyc} \frac{r_b + r_c}{4m_a^2 - n_a^2 - g_a^2}$$

Bogdan Fuștei

1.61 ***In*** ΔABC ***the following relationship holds:***

$$\sum_{cyc} \frac{1}{r_a w_a^2 (a^2 + 4r_b r_c)} = \frac{1}{2abcS}$$

In scalene ΔABC the following relationship holds:

$$\sum_{cyc} \sqrt{\frac{m_a^2 - h_a^2}{w_a^2 - h_a^2}} = 2 + \frac{R}{r}$$

Bogdan Fuștei

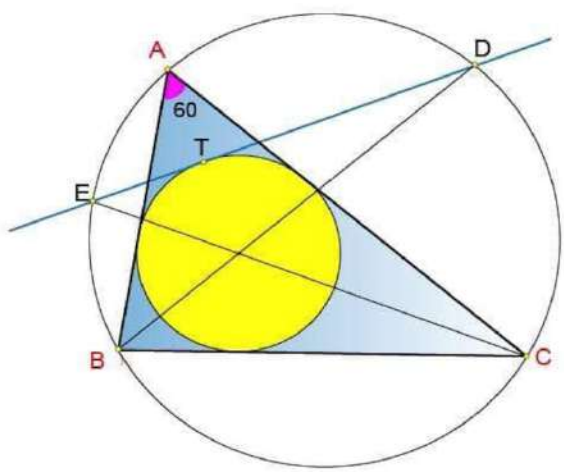
1.62 ***In ΔABC, CE, BF – internal bisectors, $E \in (AB)$, $F \in (AC)$, $BE = m$, $CF = n$.***

Prove that:

$$\frac{1}{r} + \frac{2\cot A}{a} = \frac{1}{\sin A}\left(\frac{1}{m} + \frac{1}{n}\right)$$

Adil Abdullayev

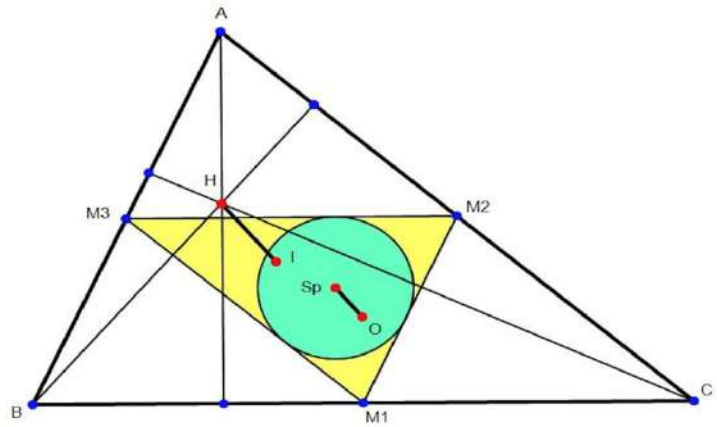
1.63 ***$\angle A = 60°$, D and E are points of intersections of angle bisectors from B and C with circumcircle. Prove: DE is tangent to incircle of ABC.***



Abdilkadir Altintaş

1.64 ***H – orthocenter of ABC, O – circumcenter, Sp – Spieker Point, I – incenter.***

Prove: $IH \parallel OSp$, $IH = 2 \cdot OSp$



Abdilkadir Altintaş

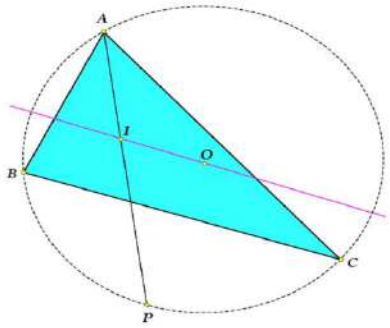
1.65 ***In $\Delta\ ABC, BF, CE$ – simedians, $F \in (AC), E \in (AB), O$ – circumcentre***

$$E, O, F \text{ – collinear} \Leftrightarrow \cot A = \cot B + \cot C$$

Marian Ursărescu

1.66 ***I – incenter of ABC,O – circumcenter,P – point of intersection of AI with circumcircle, R – circumradius, r – inradius, $IO \parallel BC$.Prove:***

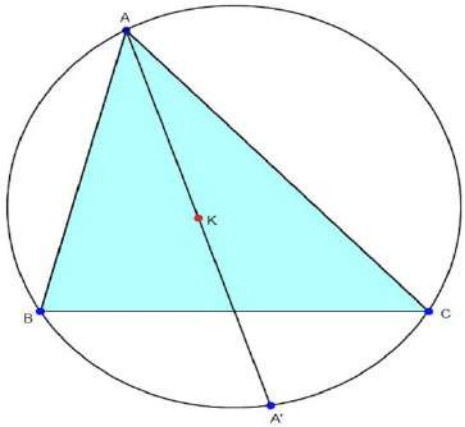
1. $\frac{AI}{AP} = \cos A, AI \cdot IP = 2Rr = \sqrt{R \cdot AI \cdot BI \cdot CI},$
2. $\sin^2\frac{B}{2} + \sin^2\frac{C}{2} = \frac{1}{2}$
3. $\cos^2\frac{B}{2} + \cos^2\frac{C}{2} = \frac{3}{2}$
4. $\tan^2\frac{A}{2} = \frac{R-r}{R+r}$



L. Bankoff

1.67 ***K – symmedian point of ABC, A' - point of intersection of AK with circumcircle***

$$2a^2 = b^2 + c^2 \Rightarrow AK = KA'$$



Abdilkadir Altintaş

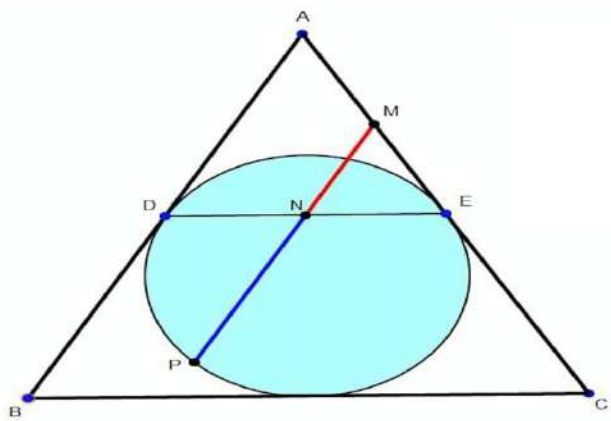
1.68 ***In $\Delta\ ABC$ the following relationship holds:***

$$\frac{R-r_a}{h_a}+\frac{R-r_b}{h_b}+\frac{R-r_c}{h_c}=1-\frac{R}{r}$$

Mehmet Şahin

1.69 ***ABC – equilateral triangle, D, E – touch points of incircle, M – midpoint of AE,***

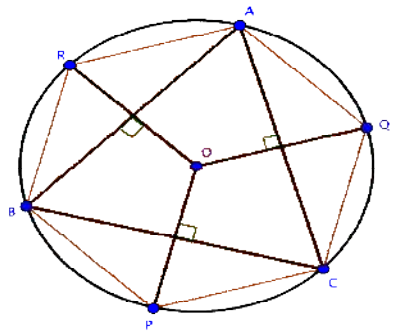
$$MP \parallel AB \rightarrow \frac{PN}{NM} = \textbf{Golden ratio.}$$



Abdilkadir Altintaş

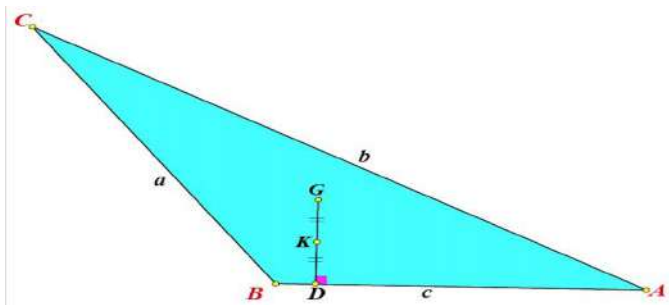
1.70 ***In acute $\Delta\ ABC, O$ – circumcenter, I_a, I_b, I_c – excenters. Prove that:***

$$A[ARBPCQ] = \frac{1}{2}A[I_aI_bI_c]$$



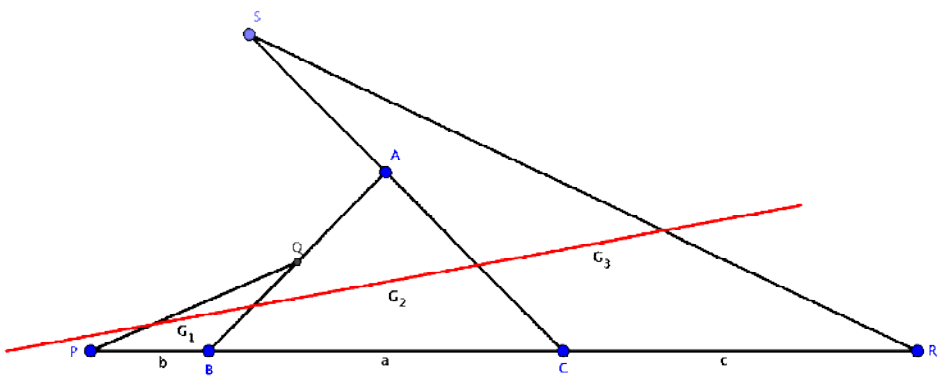
Mehmet Şahin

1.71 ***ABC be a triangle with sidelengths a, b, c, G – centroid of ABC, K – symmedian point of ABC, $GD \perp AB$. Prove: $a^2 + b^2 = 5c^2 \Leftrightarrow K$ – mid-point of GD***



Abdilkadir Altintas

1.72 ***$\Delta\ ABC$ – equilateral, $\Delta\ BPQ$ – isosceles, $\Delta\ CRS$ – isosceles, $BC = a, BP = b, CR = c$***



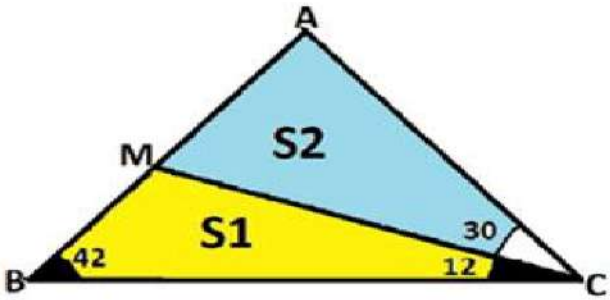
Prove: G_1, G_2, G_3 are collinear $\Leftrightarrow 3a^2 = ab + bc + ca$

Thanasis Gakopoulos

1.73 ***In $\Delta\ ABC, a \leq b, b = c, R = 6, r = 2$. Find a, b, c.***

Daniel Sitaru

1.74



Prove that: $\frac{S2}{S1} = \phi$

Muhammad Ozcelik

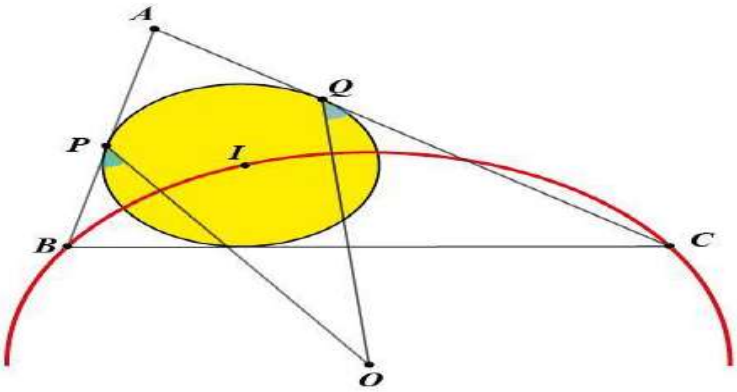
1.75 ***ΔDEF pedal triangle of I incentre in ΔABC, R_a, R_b, R_c – circumradii of $\Delta AEF, \Delta BFD, \Delta CDE$, $\varphi_a, \varphi_b, \varphi_c$ – circumradii in $\Delta\ BIC, \Delta CIA, \Delta AIB$. Prove that:***

$$\frac{R_a \cdot R_b \cdot R_c}{\varphi_a \cdot \varphi_b \cdot \varphi_c} = sin\frac{A}{2} \cdot sin\frac{B}{2} \cdot sin\frac{C}{2}$$

Mehmet Şahin

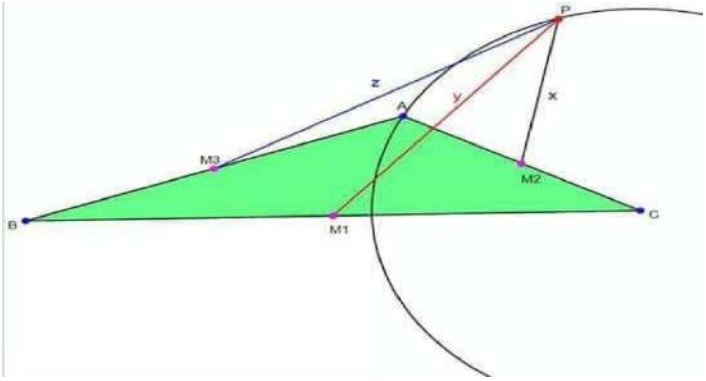
1.76 ***I – incenter of ABC,O – center of circle passing through B, I and C, P and Q tangency points of incircle. Prove:***

$$\angle BPO = \angle OQC$$



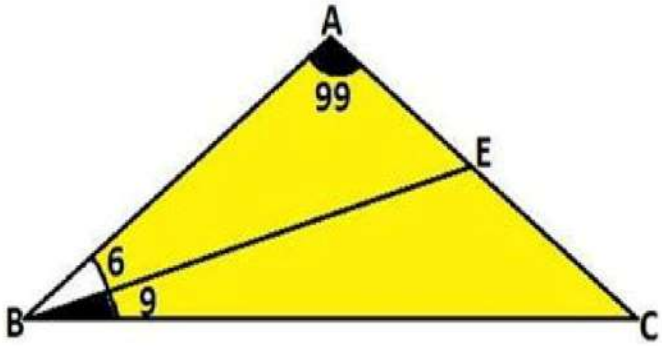
Abdilkadir Altintaş

1.77 ***M_1, M_2, M_3 - midpoints of sides, P – any point on circle centered at C and passes through A. Prove: $c^2 = a^2 - 3b^2 \Rightarrow z^2 = x^2 + y^2$***



Abdilkadir Altintaş

1.78



Prove that: $\frac{EC}{AE} = \phi$ ***(Golden ratio)***

Muhammed Ozcelik

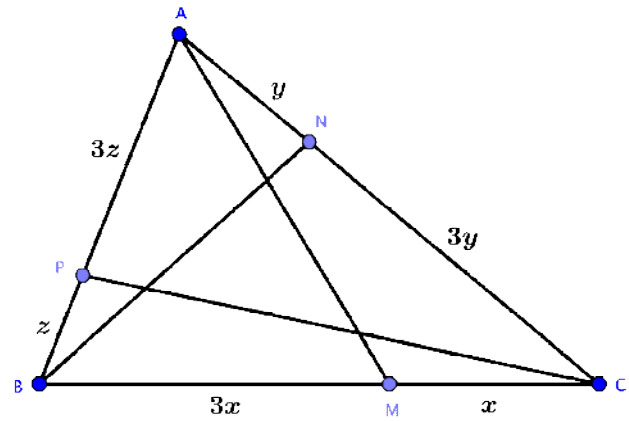
1.79 ***In*** ΔABC ***the following relationship holds:***

$$\sum \frac{r_b + r_c}{r_a} = s \sum \frac{a}{r_a^2}$$

Mehmet Şahin

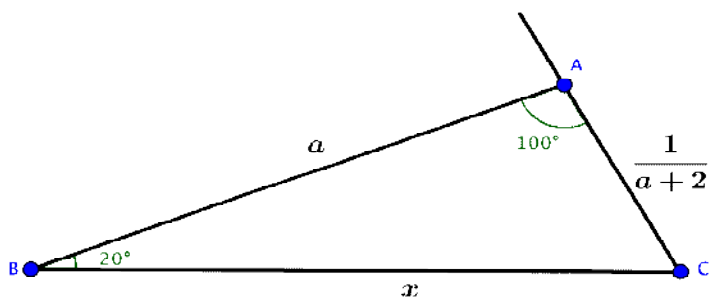
1.80 ***If in*** $\Delta\ ABC, M \in (BC), N \in (CA), P \in (AB), \frac{MB}{MC} = \frac{NC}{NA} = \frac{PA}{PB} = 3$ ***then find:***

$$\Omega = \frac{AM^2 + BN^2 + CP^2}{a^2 + b^2 + c^2}$$



Daniel Sitaru

1.81



$$BC = x = ?$$

Murat Oz

1.82 ***In ΔABC the following relationship holds:***

$$\frac{r_a}{a} + \frac{r_b}{b} + \frac{r_c}{c} = \frac{s^2 + (r_a + r_b + r_c)^2}{2[I_aI_bI_c]}, \qquad \Delta I_aI_bI_c - excentral\ triangle$$

Mehmet Şahin

1.83 ***In ΔABC the following relationship holds:***

$$\frac{a^2}{r_b + r_c} + \frac{b^2}{r_c + r_a} + \frac{c^2}{r_a + r_b} = 4R - 2r$$

Mehmet Şahin

1.84 ***In ΔABC the following relationship holds:***

$$\frac{a}{r_b+r_c}+\frac{b}{r_c+r_a}+\frac{c}{r_a+r_b}=\frac{2(r_a+r_b+r_c)}{a+b+c}$$

Mehmet Șahin

1.85 ***In ΔABC the following relationship holds:***

$$\frac{r_b+r_c}{r_a}+\frac{r_c+r_a}{r_b}+\frac{r_a+r_b}{r_c}=\frac{2[I_aI_bI_c]}{[ABC]}-2, \qquad \Delta I_aI_bI_c-excentral\ triangle$$

Mehmet Șahin

1.86 ***In ΔABC the following relationship holds:***

$$\frac{r+r_a}{h_a}+\frac{r+r_b}{h_b}+\frac{r+r_c}{h_c}=\frac{[I_aI_bI_c]}{[ABC]}, \qquad \Delta I_aI_bI_c-excentral\ triangle$$

Mehmet Șahin

1.87 ***In ΔABC the following relationship holds:***

$$r_a\left(\frac{1}{b}+\frac{1}{c}\right)+r_b\left(\frac{1}{c}+\frac{1}{a}\right)+r_c\left(\frac{1}{a}+\frac{1}{b}\right)=\frac{s}{r}$$

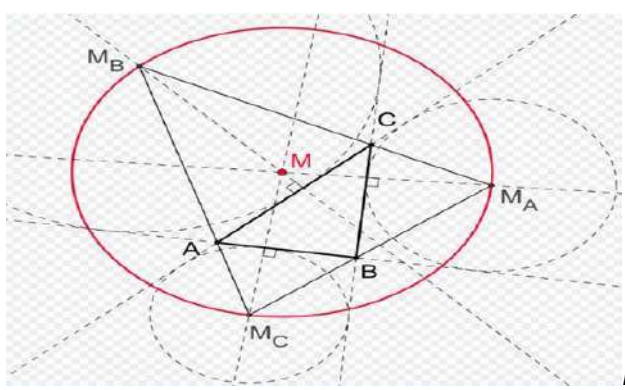
Mehmet Șahin

1.88 ***In ΔABC the following relationship holds:***

$$\frac{a^3}{r_b+r_c}+\frac{b^3}{r_c+r_a}+\frac{c^3}{r_a+r_b}=2[I_aI_bI_c]-4S, \qquad \Delta I_aI_bI_c-excentral\ triangle$$

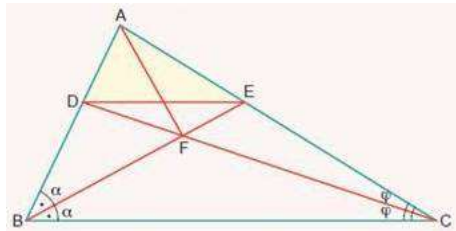
Mehmet Șahin

1.89 ***M_A, M_B, M_C – excenters, M – Bevan's point. Prove that AM, BM, CM are sides of a triangle.***



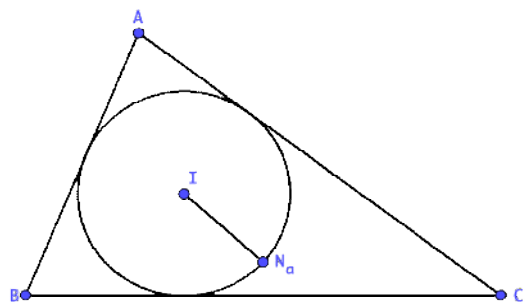
Mehmet Șahin

1.90 ***If*** $\left[\overset{\Delta}{ADE}\right] = |AF|^2$ ***then show that*** $[AB] \perp [AC]$ ***holds.***



Alpaslan Ceran

1.91 $AB = k, BC = k + 3, AC = k + 2, k > 0,$ I ***– incenter of*** ΔABC, N_a ***– Nagel's point of*** ΔABC, r ***– inradius of*** ABC, $N_a \in (I, r)$. ***Find:*** $S(abc)$ ***(area)***



Thanasis Gakopoulos

1.92 ***In*** ΔABC, O ***– circumcenter,*** r_1, r_2, r_3 ***– inradii of*** $\Delta BOC, \Delta COA, \Delta AOB$.***Prove that:***

$$2\sum_{cyc(A,B,C)}(\sec A + \tan A) = \frac{a}{r_1} + \frac{b}{r_2} + \frac{c}{r_3}$$

Mehmet Şahin

1.93 ***Find*** $k \in \mathbb{R}$ ***such that in any scalene*** ΔABC ***the following relationship holds:***

$$\sum_{cyc}\frac{n_a^2 - m_a^2}{(b-c)^2} = k + \frac{1}{2}\sum_{cyc}\frac{b+c}{a}, n_a, n_b, n_c$$ ***– Nagel's cevians***

Bogdan Fuştei

1.94 ***In*** ΔABC, I ***– incentre,*** R_a, R_b, R_c ***– circumradii of*** $\Delta BIC, \Delta CIA, \Delta AIB$.***Prove that:***

$$\left(\frac{ar_a}{R_a}\right)^2 + \left(\frac{br_b}{R_b}\right)^2 + \left(\frac{cr_c}{R_c}\right)^2 = 2s^2\left(2 - \frac{r}{R}\right)$$

Mehmet Şahin

1.95 *In ΔABC the following relationship holds:*

$$\frac{R+r_a}{bc}+\frac{R+r_b}{ca}+\frac{R+r_c}{ab}=\frac{3}{2r}-\frac{1}{2R}$$

Mehmet Şahin

1.96 *In ΔABC the following relationship holds:*

$$\frac{r_a^2}{r_a^2+s^2}+\frac{r_b^2}{r_b^2+s^2}+\frac{r_c^2}{r_c^2+s^2}=1-\frac{r}{2R}$$

Adil Abdullayev

1.97 *In ΔABC, $K-Lemoine's\ point$, $KD\perp BC, KE\perp CA, KF\perp AB$,*

$KD=x, KE=y, KF=z$. ***Prove that:*** $xh_am_a^2+yh_bm_b^2+zh_cm_c^2=3S^2$

Mehmet Şahin

1.98 *In ΔABC the following relationship holds:*

$$\frac{2R-r_a}{h_a}+\frac{2R-r_b}{h_b}+\frac{2R-r_c}{h_c}=1$$

Mehmet Şahin

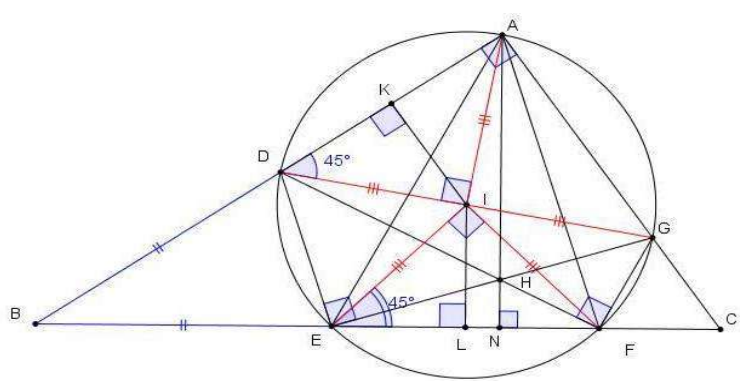
1.99 *In $\Delta\ ABC$ the following relationship holds:*

$$\sum\frac{1}{b\cos B+c\cos C-a\cos A}=\frac{R}{2S\cos A\cos B\cos C}+\frac{1}{a\cos A+b\cos B+c\cos C}$$

Daniel Sitaru

1.100 $m(\sphericalangle A)=90^\circ$, *the drawn circle has center in I – incenter of ΔABC and radii AI.*

Prove that H is orthocenter of ΔAEF.



Rahul Sethi

1.101 ***$\Delta\ DEF$ – pedal triangle of I – incenter of acute ΔABC, h_1, h_2, h_3 – altitudes in $\Delta\ DEF$.Prove that:***

$$\frac{\cos\frac{A}{2}}{h_1}+\frac{\cos\frac{B}{2}}{h_2}+\frac{\cos\frac{C}{2}}{h_3}=\frac{r_a+r_b+r_c}{S}$$

Mehmet Şahin

1.102 ***In acute $\Delta\ ABC, \Psi=\begin{vmatrix}\sin A & \sin B & \sin C\\ \sin 2A & \sin 2B & \sin 2C\\ \sin 3A & \sin 3B & \sin 3C\end{vmatrix}$, O – circumcentre, I – incentre, H – orthocentre. Prove that:***

$$S[OIH]=\frac{R^6}{2abcs}\cdot|\Psi|$$

Daniel Sitaru

1.103 ***In ΔABC the following relationship holds:***

$$\frac{r+r_a}{h_a-r}+\frac{r+r_b}{h_b-r}+\frac{r+r_c}{h_c-r}=2\left(\frac{r_a}{h_a}+\frac{r_b}{h_b}+\frac{r_c}{h_c}\right)$$

Mehmet Şahin

1.104 ***In ΔABC the following relationship holds:***

$$\frac{1}{2}\left(\frac{h_a}{AI}+\frac{h_b}{BI}+\frac{h_c}{CI}\right)=\cos\frac{A}{2}\cos\frac{B}{2}+\cos\frac{B}{2}\cos\frac{C}{2}+\cos\frac{C}{2}\cos\frac{A}{2}\ ,I-incenter$$

Bogdan Fuştei

1.105 ***In $\Delta ABC, K-Lemoine's\ point, KD\perp BC, KE\perp CA, KF\perp AB, KD=x, KE=y, KF=z$. Prove that:***

$$\frac{m_a^2}{x\cdot h_a}+\frac{m_b^2}{y\cdot h_b}+\frac{m_c^2}{z\cdot h_c}=\frac{3}{16S^2}\cdot(a^2+b^2+c^2)^2$$

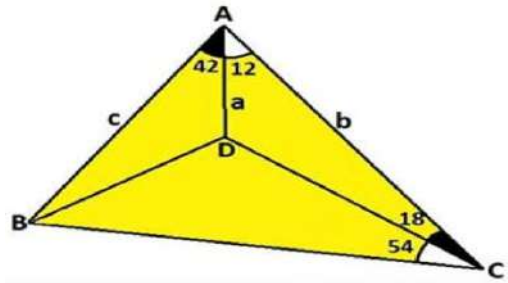
Mehmet Şahin

1.106 ***In ΔABC the following relationship holds:***

$$\frac{h_a+h_b}{r_a+r_b}+\frac{h_b+h_c}{r_b+r_c}+\frac{h_c+h_a}{r_c+r_a}=\frac{2(R+r)}{R}$$

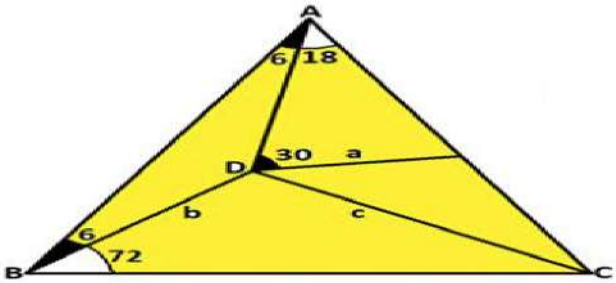
Bogdan Fuştei

1.107 ***Prove that:*** $a^2 + b^2 = c^2$



Mohamed Ozcelic

1.108



Prove that: $c^2 = a^2 + b^2$

Mohamed Ozcelik

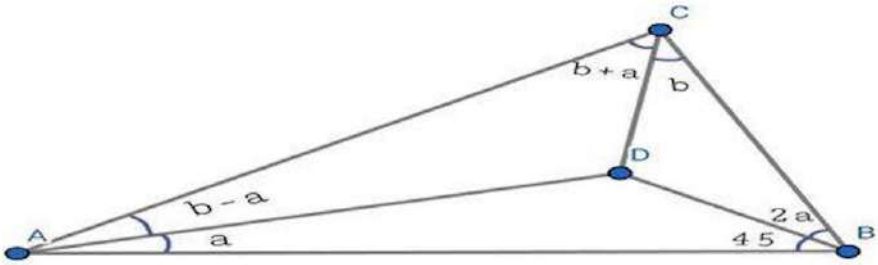
1.109 ***In*** $\Delta ABC, K - Lemoine's\ point, KD \perp BC, KE \perp CA, KF \perp AB,$

$KD = x, KE = y, KF = z$ ***. Prove that:***

$$\frac{xh_a}{r_b r_c} + \frac{yh_b}{r_c r_a} + \frac{zh_c}{r_a r_b} = \frac{x}{r_a} + \frac{y}{r_b} + \frac{z}{r_c}$$

Mehmet Şahin

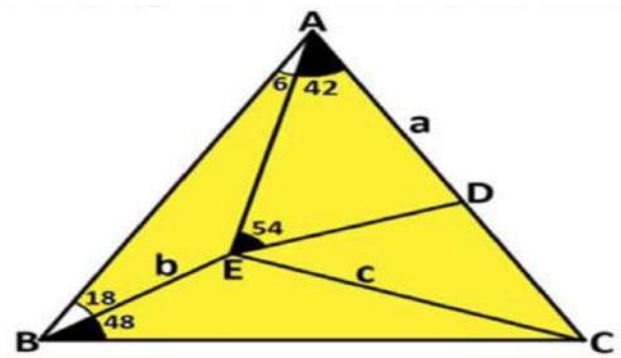
1.110



$a = ?$

Murat Oz

1.111



$$Prove\ that: c^2 = a^2 + b^2$$

Mohamed Ozcelic

1.112 ***In*** ΔABC , $K - Lemoine's\ point$, $KD \perp BC, KE \perp CA, KF \perp AB,$

$KD = x, KE = y, KF = z$. ***Prove that:***

$$\frac{xr_a + yr_b + zr_c}{x + y + z} = \frac{ar_a + br_b + cr_c}{a + b + c} = 2R - r$$

Mehmet Şahin

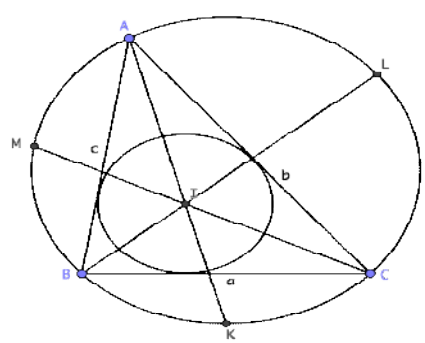
1.113 ***In*** $\Delta ABC, R_a, R_b, R_c$***-circumradii of*** $\Delta VI_bI_c,\ \Delta VI_cI_a,\ \Delta VI_aI_b$

I_a, I_b, I_c***-excenters,*** V ***−Bevan's point. Prove that:***

$$\sum_{cyc} \frac{1}{R_a^2} = \frac{2R - r}{2R^3}, \qquad \prod_{cyc} R_a = \frac{4R^4}{r}, \qquad \frac{a}{R_a^2} + \frac{b}{R_b^2} + \frac{c}{R_c^2} = \frac{[I_aI_bI_c] - 2[ABC]}{2R^3}$$

Mehmet Şahin

1.114 ***Prove:*** $(IK)(IL)(IM) = \frac{2R^2S}{s}$



Thanasis Gakopoulos

1.115 ***Prove that $x_1 = \cos A$, $x_2 = \cos B$, $x_3 = \cos C$ are the roots of equation:***

$$4R^2x^3 - 4R(R+r)x^2 + (s^2 + r^2 - 4R^2)x + (2R+r)^2 - s^2 = 0$$

Marian Ursărescu

1.116 ***In acute ΔABC, o_a, o_b, o_c – circumcevians. Prove that:*** $\frac{1}{o_a} + \frac{1}{o_b} + \frac{1}{o_c} = \frac{2}{R}$

Mustafa Tarek

1.117 ***If in ΔABC, I_a, I_b, I_c – excenters then:*** $AI_c^2 + BI_a^2 + CI_b^2 = 16R^2 - s^2 - r^2$

Mehmet Șahin

1.118 ***ΔDEF – pedal triangle of I – incenter in ΔABC, φ – inradii of ΔDEF, I_a, I_b, I_c – excenters, $\varphi_a, \varphi_b, \varphi_c$ – inradii of $\Delta I_aBC, \Delta I_bCA, \Delta I_cAB$. Prove that:***

$$\frac{\varphi^2}{\varphi_a^2} + \frac{\varphi^2}{\varphi_b^2} + \frac{\varphi^2}{\varphi_c^2} = \frac{s^2 + r^2 - 8Rr}{4R^2}$$

Mehmet Șahin

1.119 ***In ΔABC the following relationship holds:***

$$\frac{bc}{h_a^2} + \frac{ca}{h_b^2} + \frac{ab}{h_c^2} = \frac{r + r_a}{h_a} + \frac{r + r_b}{h_b} + \frac{r + r_c}{h_c}$$

Mehmet Șahin

1.120 ***In ΔABC, ΔDEF – is pedal triangle of I – incenter. If $IM \perp FE$, $IK \perp DF$, $IL \perp DE$, $M \in (EF)$, $K \in (DF)$, $L \in (DE)$ then:***

$$IM^2 + IK^2 + IL^2 = r^2\left(1 - \frac{r}{2R}\right)$$

Mehmet Șahin

1.121 ***In ΔABC, R_k is the circumradii of pedal triangle of Lemoine's point. Prove that:***

$$R_k = \frac{m_a m_b m_c}{m_a^2 + m_b^2 + m_c^2}$$

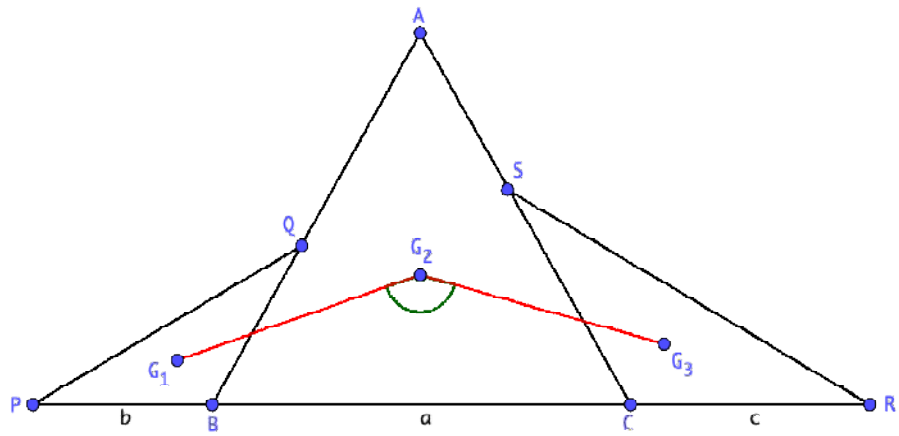
Mehmet Șahin

1.122 ***If ΔDEF is pedal triangle of I – incenter in ΔABC and R_a, R_b, R_c are the circumradii of $\Delta AFE, \Delta BDF, \Delta CDE$ then:***

$$R_a^2 + R_b^2 + R_c^2 = \frac{s^2 + r^2 - 8Rr}{4}$$

Mehmet Şahin

1.123 ***ΔABC equilateral, ΔBQP isosceles, ΔCRS isosceles $BC = a, BP = b, CR = c$, G_1, G_2, G_3 centroids $3a(a-b-c) = bc$. Calculate $m(\sphericalangle F)$***



Thanasis Gakopoulos

1.124 ***In $\Delta ABC, \Delta DEF$ – is pedal triangle of I – incenter, I_a, I_b, I_c – excenters***

If $IM \perp FE, IK \perp DF, IL \perp DE, M \in (EF), K \in (DF), L \in (DE)$ then:

$$\frac{1}{IM^2 \cdot h_a^2} + \frac{1}{IK^2 \cdot h_b^2} + \frac{1}{IL^2 \cdot h_c^2} = \frac{[I_aI_bI_c]}{[ABC]^3} \cdot \frac{r_a + r_b + r_c}{r}$$

Mehmet Şahin

1.125 ***In ΔABC the following relationship holds:***

$$r_ar_b(r_a + r_b) + r_br_c(r_b + r_c) + r_cr_a(r_c + r_a) = 2s^2(2R - r)$$

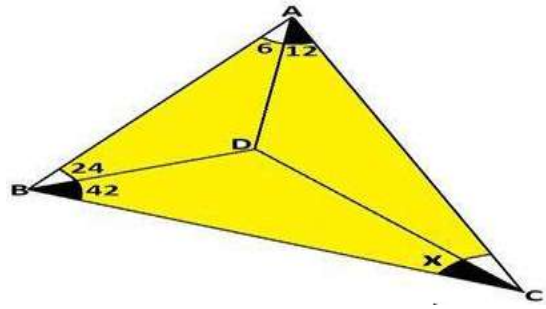
Bogdan Fuştei

1.126 ***In $\Delta ABC, I$ – incenter, AA', BB', CC' - internal bisectors, $\frac{IA}{IA'}, \frac{IB}{IB'}, \frac{IC}{IC'} \in \mathbb{N}^*$***

Find: $\Omega = \frac{w_aw_bw_c}{m_am_bm_c} + \frac{h_ah_bh_c}{w_aw_bw_c} + \frac{m_am_bm_c}{h_ah_bh_c}$

Daniel Sitaru

1.127



Find "x".

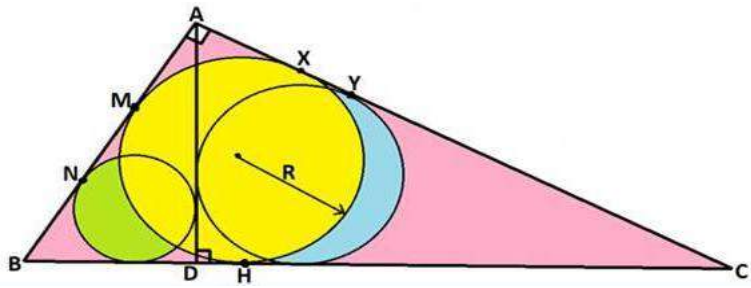
Mohamed Ozcelik

1.128 ***In ΔABC the following relationship holds:***

$$\sum \frac{1}{b\cos B + c\cos C - a\cos A} = \frac{R}{2S\cos A\cos B\cos C} + \frac{1}{a\cos A + b\cos B + c\cos C}$$

Daniel Sitaru

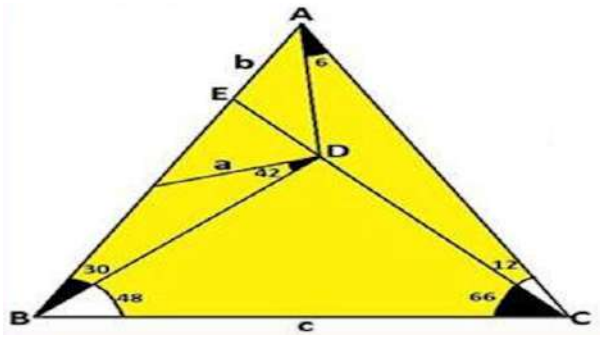
1.129



Prove that: $AD - MN - XY = R$

Muhammad Ozcelik

1.130



E, D, C ***collinears. Prove that:*** $a^2 + b^2 = c^2$

Mohamed Ozcelik

1.131 ΔDEF ***pedal triangle of*** I ***– incenter in*** $\Delta ABC, R_a, R_b, R_c$ ***– circumradii of*** $\Delta AEF, \Delta BFD, \Delta CDE, \varphi_a, \varphi_b, \varphi_c$ ***– circumradii in*** $\Delta BIC, \Delta CIA, \Delta AIB$***. Prove that:***

$$\frac{R_a \cdot R_b \cdot R_c}{\varphi_a \cdot \varphi_b \cdot \varphi_c} = \sin\frac{A}{2} \cdot \sin\frac{B}{2} \cdot \sin\frac{C}{2}$$

Mehmet Şahin

1.132 ***In*** ΔABC ***the following relationship holds:***

$$\frac{\sin\frac{A}{2}\sin\frac{B}{2} + \sin\frac{B}{2}\sin\frac{C}{2} + \sin\frac{C}{2}\sin\frac{A}{2}}{\frac{w_a + w_b}{h_c} + \frac{w_b + w_c}{h_a} + \frac{w_c + w_a}{h_b}} = \frac{r}{4R}$$

Mustafa Tarek

1.133 ***In*** $\Delta ABC, M$ ***– Mittenpunkt,*** I_a, I_b, I_c ***– excenters. Prove that:***

$$\frac{[I_b M I_c]}{\cos^2\frac{A}{2}} = \frac{[I_c M I_a]}{\cos^2\frac{B}{2}} = \frac{[I_a M I_b]}{\cos^2\frac{C}{2}}$$

Mustafa Tarek

QUADRILATERAL

2.1 ***In $ABCD$ convexe quadrilateral: $AB = a, BC = b, CD = c, DA = d$. Prove that:***

$$\sum \sqrt{a^2 + b^2 + c^2} > 2\sqrt{3 \cdot AC \cdot BD}$$

Daniel Sitaru

2.2 ***In $ABCD$ cyclic quadrilateral, $AB = a, BC = b, CD = c, DA = d, s -$ semiperimeter:***

$$\sin A \sin B \leq \left(1 - \frac{s}{a}\right)\left(1 - \frac{s}{b}\right)\left(1 - \frac{s}{c}\right)\left(1 - \frac{s}{d}\right)$$

Daniel Sitaru

2.3 ***In $ABCD$ cyclic quadrilateral, $AB = a, BC = b, CD = c, DA = d, S - area\,[ABCD]$***

$$\sin A + \sin B + \sin C + \sin D \leq \frac{4S}{\sqrt{abcd}}$$

Daniel Sitaru

2.4 ***In $ABCD$ convexe quadrilateral:***

$$\frac{\sin(\widehat{ABD})}{\sin(\widehat{DBC})} + \frac{\sin(\widehat{BDC})}{\sin(\widehat{ADB})} + \frac{\sin(\widehat{ACB})}{\sin(\widehat{ACD})} + \frac{\sin(\widehat{DAC})}{\sin(\widehat{CAB})} \geq 4$$

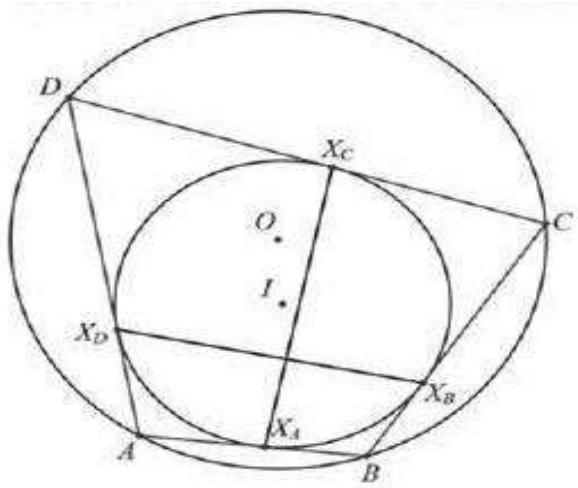
Daniel Sitaru

2.5 ***In $ABCD$ cyclic quadrilateral, a, b, c, d – sides, R - circumradius:***

$$\frac{\sqrt[4]{a^3b^3c^3d^3}}{(a + b + c + d)^2} \leq \frac{R}{8\sqrt{2}}$$

Adil Abdullayev

2.6 ***I – incentre, R – circumradii. Prove that: $IA \cdot IB \cdot IC \cdot IC \leq R^4$***



Mustafa Tarek

2.7 ***$ABCD$ – bicentric quadrilateral, $AC = e, BD = f, R$ – circumradii, r – inradii***

Prove that: $ef \neq 4Rr$

Mustafa Tarek

2.8 ***If a, b, c, d – sides in a bicentric quadrilateral, s – semiperimeter, e, f – diagonals, R – circumradii then:***

$$\frac{2(ab + ac + ad + bc + bd + cd)}{s^3(s^2 + ef)} \leq \frac{efR^2}{abcd(abc + abd + bcd + cda)}$$

Daniel Sitaru

TETRAHEDRON

3.1 ***Let $A_1A_2A_3A_4$ be a tetrahedron and let M be its interior point. Denote respectively by S_i and d_i the are and distance from M to face opposite to vertex A_i. Prove that:***

$$\sum_{1\le i<j\le 4} S_iS_jd_id_j \le \frac{27}{8}V^2 \text{ where } V \text{ is the volume of the tetrahedron.}$$

Nguyen Viet Hung

3.2 ***In $ABCD$ tetrahedron, h_A, h_B, h_C, h_D – altitudes, R – circumradius, r – inradius:***

$$R(h_A + h_B + h_C + h_D) \ge 48r^2$$

Daniel Sitaru

3.3 ***If in $ABCD$ – tetrahedron, h_A, h_B, h_C, h_D - altitudes, r – inradii then:***

$$\frac{h_A - r}{h_A + r} + \frac{h_B - r}{h_B + r} + \frac{h_C - r}{h_C + r} + \frac{h_D - r}{h_D + r} \ge \frac{12}{5}$$

D.M. Bătinețu – Giurgiu and Neculai Stanciu

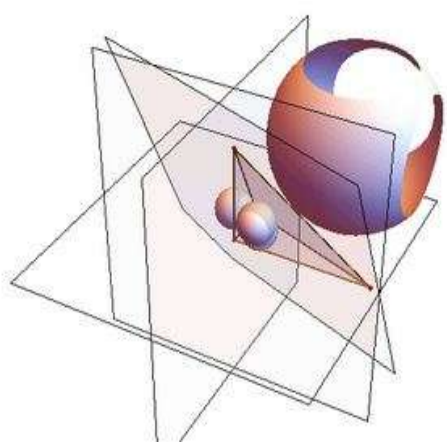
3.4 ***If in $ABCD$ - tetrahedron $AD = BC = a, BD = AC = b, CD = AB = c$,***

R - radii of circumsphere then: $8(4R^2 - a^2)(4R^2 - b^2)(4R^2 - c^2) \le a^2b^2c^2$

Daniel Sitaru

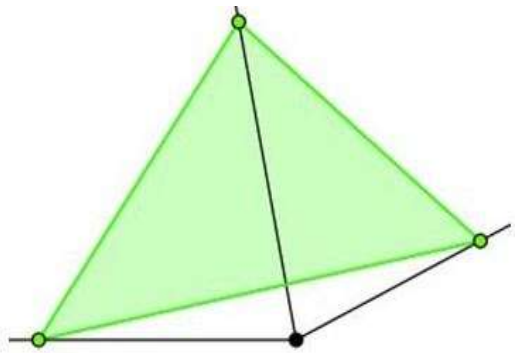
3.5 ***$ABCD$ – tetrahedron, r – inradii, r_A, r_B, r_C, r_D – exradii. Prove that:***

$$\frac{1}{\sqrt{r_A}} + \frac{1}{\sqrt{r_B}} + \frac{1}{\sqrt{r_C}} + \frac{1}{\sqrt{r_D}} \le \frac{4}{\sqrt{2r}}$$



Marian Ursărescu

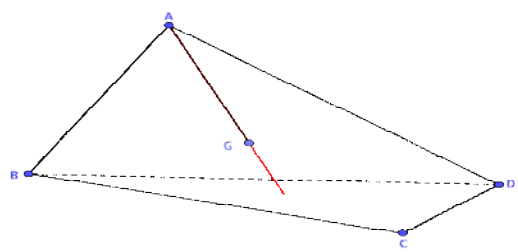
3.6 $A(a,0,0), B(0,b,0), C(0,0,c), O(0,0,0), a,b,c > 0, H$ ***– orthocenter, G – centroid of ΔABC. Prove that:*** $3OH^2 \leq \sqrt[3]{a^2b^2c^2} \leq 3OG^2$



Daniel Sitaru

3.7 $ABCD$ ***tetrahedron, G centroid of $ABCD$,*** $AB = 3, AC = 5, AD = 4$

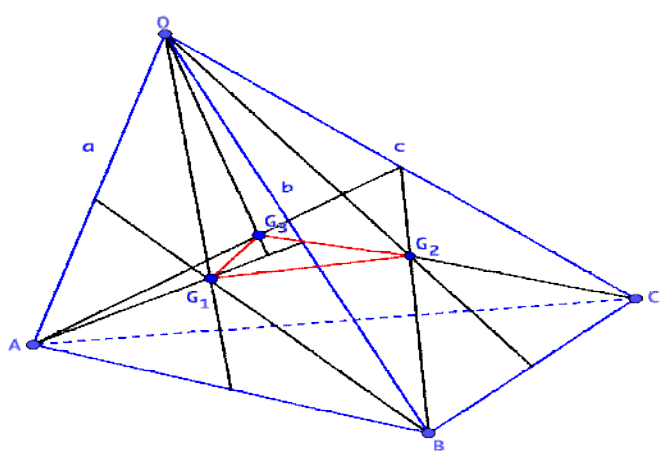
$\widehat{BAC} = 45°, \widehat{CAD} = 30°, \widehat{BAD} = 60°$. ***Find*** AG.



Thanasis Gakopoulos

3.8 $OABC$ ***tetrahedron,*** $OA = a, OB = b, OC = c$, G_1 ***centroid of*** ΔOAB, G_2 ***centroid of*** ΔOBC, G_3 ***centroid of*** ΔOAC, $\Delta AOB = \Delta BOC = \Delta AOC = 9 = 120°$

Prove that: $(G_1G_2)^2 + (G_2G_3)^2 + (G_1G_3)^2 = \frac{3(a^2+b^2+c^2)-(a+b+c)^2}{18}$

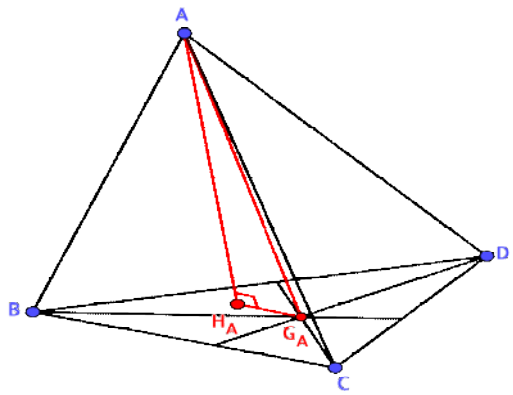


Thanasis Gakopoulos

3.9 $AB = 3, AC = 4, AD = 5, \widehat{BAC} = \vartheta_1 = cos^{-1}\left(\frac{1}{3}\right), \widehat{CAD} = \vartheta_2 = cos^{-1}\left(\frac{1}{4}\right)$

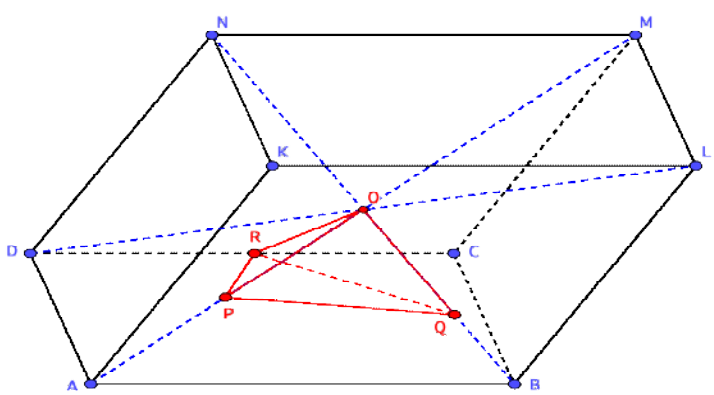
$\widehat{BAD} = \vartheta_3 = cos^{-1}\left(\frac{1}{5}\right)$, $AH_A \perp$ ***plane*** (BCD),G_A ***- centroid of*** ΔBCD

Find: $H_AG_A = ?$ ***(length)*** $S(AH_AG_a) = ?$ ***(area)***



Thanasis Gakopoulos

3.10 $ABCDKLMN$ ***parallelepiped,*** $AB = 36, AK = 24, AD = 12, \widehat{BAD} = \widehat{DAK} = \widehat{KAB} = 60°$, O ***– center of*** $ABCDKLMN$, R ***– midpoint of*** DC, P ***– midpoint of*** OA, $OQ = 2 \cdot QB$.***Find:*** V ***– the volume of tetrahedron*** $OPQR$, V_0 ***– the volume of parallelepiped*** $ABCDKLMN$, d ***– distance of the point*** O ***from the plane*** PQR



Thanasis Gakopoulos

3.11 ***Let*** $OABC$ ***be a tetrahedron with*** $\angle AOB = \angle BOC = \angle COA = 90^\circ$ ***and let*** P ***be any point inside the triangle*** ABC. ***Denote respectively by*** d_a, d_b, d_c ***the distances from*** P ***to faces*** $(OBC), (OCA), (OAB)$. ***Prove that:***

(a) $d_a^2 + d_b^2 + d_c^2 = OP^2$.

(b) $d_a d_b d_c \leq \frac{OA \cdot OB \cdot OC}{27}$

(c) $OA \cdot d_a^3 + OB \cdot d_b^3 + OC \cdot d_c^3 \geq OP^4$.

Nguyen Viet Hung

TRIGONOMETRY

4.1 ***Solve for real numbers:*** $[\tan x]\cdot(\cot x-[\cot x])=(\tan x-[\tan x])\cdot[\cot x]$

$[*]-$ ***great integer function***

Rovsen Pirguliyev

4.2 ***Prove that:***

$$\sin^2\frac{7\pi}{18}\cdot\sin\frac{5\pi}{18}-\sin^2\frac{\pi}{18}\cdot\sin\frac{7\pi}{18}+\sin^2\frac{5\pi}{18}\cdot\sin\frac{\pi}{18}=\frac{3}{4}$$

George Apostolopoulos

4.3 ***Prove that:***

$$\csc\left(\frac{\pi}{14}\right)-4\cos\left(\frac{2\pi}{7}\right)=2$$

Vasile Mircea Popa

4.4 ***Prove that:***

$$2+2\sum_{n=1}^{\infty}2^n\cdot\sin^2\left(\frac{90}{2^n}\right)\cdot\tan\left(\frac{45}{2^n}\right)=\pi$$

Radeshyam Takur

4.5 ***Prove that:***

$$\sin 3^\circ=\frac{1}{4}\left(\sqrt{8-\sqrt{10-2\sqrt{5}}-\sqrt{3}-\sqrt{15}}\right)=\frac{1}{4}\left(\sqrt{8-\sqrt{\sqrt{10}\left(\sqrt{10}-\sqrt{2}\right)}-\sqrt{3}-\sqrt{15}}\right)$$

Naren Bhandari

4.6 ***Prove that:***

$$\tan\left[\frac{\pi}{64}\right]^2\tan\left[\frac{\pi}{32}\right]^2\tan\left[\frac{\pi}{16}\right]^2\tan\left[\frac{\pi}{8}\right]^2=$$

$$= \frac{\left(2-\sqrt{2}\right)\left(2-\sqrt{2+\sqrt{2}}\right)\left(2-\sqrt{2+\sqrt{2+\sqrt{2}}}\right)\left(2-\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}\right)}{\left(2+\sqrt{2}\right)\left(2+\sqrt{2+\sqrt{2}}\right)\left(2+\sqrt{2+\sqrt{2+\sqrt{2}}}\right)\left(2+\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}\right)}$$

John Horton Conway

4.7 ***Prove that:***

$$\cos\left(\frac{2\pi}{13}\right)\cdot\cos\left(\frac{5\pi}{13}\right)\cdot\cos\left(\frac{6\pi}{13}\right) = \frac{\sqrt{13}-3}{16}$$

Vasile Mircea Popa

4.8 ***Prove that:***

$$\sqrt[3]{1+\cos\frac{4\pi}{19}+\cos\frac{6\pi}{19}-\cos\frac{9\pi}{19}}+\sqrt[3]{1+\cos\frac{2\pi}{19}-\cos\frac{3\pi}{19}-\cos\frac{5\pi}{19}}+$$

$$+\sqrt[3]{1-\cos\frac{\pi}{19}-\cos\frac{7\pi}{19}+\cos\frac{8\pi}{19}} = \frac{1}{2}\sqrt[3]{12\sqrt[3]{19}-4}$$

Vasile Mircea Popa

4.9 ***Prove that:***

$$\sin\frac{2\pi}{13}-\sin\frac{5\pi}{13}+\sin\frac{6\pi}{13} = \frac{1}{2}\sqrt{\frac{13-3\sqrt{13}}{2}}$$

Vasile Mircea Popa

4.10 ***If*** $x, y, z \in \left(0, \frac{\pi}{2}\right)$ ***then:***

$$\frac{\tan x}{\sin y+\sin z}+\frac{\tan y}{\sin z+\sin x}+\frac{\tan z}{\sin x+\sin y} > \frac{3}{2}$$

D.M. Bătinețu – Giurgiu; Neculai Stanciu

4.11 ***If*** $a, b, c > 0, a+b+c = \sqrt{3}$ ***then:***

$$\sum\left(2\arctan\left(\frac{a+b}{2}\right)+\arctan c\right)\le\frac{3\pi}{2}$$

Daniel Sitaru

4.12 ***In $\Delta\, ABC$:***

$$\frac{\sin^2 A}{\sin^{-1}\frac{4}{5}}+\frac{\sin^2 B}{\sin^{-1}\frac{5}{13}}+\frac{\sin^2 C}{\sin^{-1}\frac{16}{65}}\ge\frac{2s^2}{\pi R^2},\qquad \frac{\sin^2 A}{\tan^{-1}\frac{1}{2}}+\frac{\sin^2 B}{\tan^{-1}\frac{1}{5}}+\frac{\sin^2 C}{\tan^{-1}\frac{1}{8}}\ge\frac{4s^2}{\pi R^2}$$

Daniel Sitaru

4.13 ***If $x, y > 0; z \in \mathbb{R}$ then:***

$$\frac{(x+y)^2}{(x\sin^2 z+y\cos^2 z)(x\cos^2 z+y\sin^2 z)}+\frac{x}{y}+\frac{y}{x}\ge 6$$

Daniel Sitaru

4.14 ***If $a, b, c, x \in \mathbb{R}$ then:***

$$a^2+b^2+c^2+(\sin x+\cos x+\sin x\cos x)(ab+bc+ca)\ge 0$$

Daniel Sitaru

4.15 ***If $a, b \in \left(0, \frac{\pi}{2}\right)$ then:***

$$\frac{\cos a}{1+\cos^4 a}+\frac{\sin a\cos b}{1+\sin^4 a\cos^4 b}+\frac{\sin a\sin b}{1+\sin^4 a\sin^4 b}<\frac{9\sqrt{3}}{10}$$

Daniel Sitaru

4.16 ***If $a, b, c \ge 0, a^2+b^2+c^2=2$ then:***

$$\sqrt{ab}\csc\frac{\pi}{7}+\sqrt{bc}\csc\frac{2\pi}{7}+\sqrt{ca}\csc\frac{3\pi}{7}\le 4$$

Daniel Sitaru

4.17 ***Prove that:***

$$\sqrt[3]{2\left(\cos\frac{\pi}{13}+\cos\frac{5\pi}{13}\right)}-\sqrt[3]{2\left(\cos\frac{2\pi}{13}-\cos\frac{3\pi}{13}\right)}-\sqrt[3]{2\left(\cos\frac{4\pi}{13}+\cos\frac{6\pi}{13}\right)}=\sqrt[3]{7-3\sqrt[3]{13}}$$

Vasile Mircea Popa

4.18 ***Prove that:***

$$\frac{\tan 36^\circ + \tan 72^\circ}{4\cos 54^\circ} = \varphi$$

Alpaslan Ceran

4.19 ***If*** $x, y \in \left(0, \frac{\pi}{2}\right)$ ***then:***

$$\sin(x+y) < \sin x\left(\frac{\sin y}{y}\right)^3 + \sin y\left(\frac{\sin x}{x}\right)^3$$

Daniel Sitaru

4.20 ***If*** $m \in \mathbb{N}, m \geq 2$ ***then:***

$$m + \tan^2\frac{\pi}{36} + \tan^2\frac{11\pi}{36} + \tan^2\frac{13\pi}{36} + \tan^2\frac{21\pi}{36} > 2 + \left(\frac{3}{16}\right)^{\frac{m-2}{m}}$$

Daniel Sitaru

4.21 ***If*** $A = \tan\beta\tan\gamma + 5, B = \tan\gamma\tan\alpha + 5, C = \tan\alpha\tan\beta + 5,$

$\alpha, \beta, \gamma = \frac{\pi}{2}$, ***then:*** $\sqrt{A} + \sqrt{B} + \sqrt{C} \leq 4\sqrt{3}$

Boris Colakovic

4.22 ***If in*** $\Delta\, ABC, a \geq b \geq c$ ***then:***

$$m_a\cos^2\frac{\pi}{7} + m_b\cos^2\frac{2\pi}{7} + m_c\cos^2\frac{3\pi}{7} < \frac{5s}{6}$$

Daniel Sitaru

4.23 ***If*** $a, b, c \in (4, \infty), abc = 2^{11}$ ***then:***

$$\prod\left(a^2\sin\frac{2\pi}{a} + (a+1)^2\sin\frac{2\pi}{a+1}\right) > 2^{16}$$

Daniel Sitaru

4.24 ***If*** $0 < A < B + C, 0 < B < C + A, 0 < C < A + B, A + B + C < \pi$ ***then:***

$$(\sin A + \sin B - \sin C)(\sin B + \sin C - \sin A)(\sin C + \sin A - \sin B) > 0$$

Daniel Sitaru

4.25 ***Prove that:***

$$2^{\cos x} + 2^{\sin x} \geq 2^{\frac{\sqrt{2}-1}{\sqrt{2}}}, \forall x \in \mathbb{R}.$$

Ibrahim Abdulazeez

4.26 ***Prove that:***

$$\sin 2° + \sin 3° + \sin 4° + \cdots + \sin 10° < 54 \cdot \sin 1°$$

Ilkin Guliyev

4.27 ***If*** $x, y, z \in \left[0, \frac{\pi}{2}\right)$ ***then:***

$$xyz(\sin x + \sin y + \sin z) \leq y^2 z \sin x + z^2 x \sin y + x^2 y \sin z$$

Daniel Sitaru

4.28 ***If*** $a, b, c, d > 0$ ***then:***

$$\sum a \tan^{-1} a \geq \sum \sqrt[3]{bcd} \tan^{-1} a \geq 4\sqrt[4]{abcd \prod \tan^{-1} a}$$

Daniel Sitaru

SOLUTIONS

TRIANGLE

1.1 *Solution 1(Khanh Hung Vu)*

We put $\frac{BA_1}{CA_1} = \frac{CB_1}{AB_1} = \frac{AC_1}{BC_1} = \frac{CA_1}{BA_1} = \frac{AB_1}{CB_1} = \frac{BC_1}{AC_1} = k$ ***and*** $BC = a, CA = b, AB = c$

$$\Rightarrow BA_1 = CA_2 = \frac{ak}{k+1}, CB_1 = AB_2 = \frac{bk}{k+1} \text{ and } AC_1 = BC_2 = \frac{ck}{k+1}$$

By cosine's law of triangle ABA_1 ***and*** ACA_2***, we have:***

$$AA_1^2 - AA_2^2 = AB^2 + BA_1^2 - 2AB \cdot BA_1 \cdot \cos B - \left(AC^2 + CA_2^2 - 2AC \cdot CA_2 \cdot \cos C\right) =$$

$$= AB^2 - AC^2 - 2AB \cdot BA_1 \cdot \cos B + 2AC \cdot A_2C \cdot \cos C$$

$$\Rightarrow AA_1^2 - AA_2^2 = c^2 - b^2 - 2c \cdot \frac{ak}{k+1} \cdot \frac{a^2 + c^2 - b^2}{2ab} + 2b \cdot \frac{ak}{k+1} \cdot \frac{a^2 + b^2 - c^2}{2ab} =$$

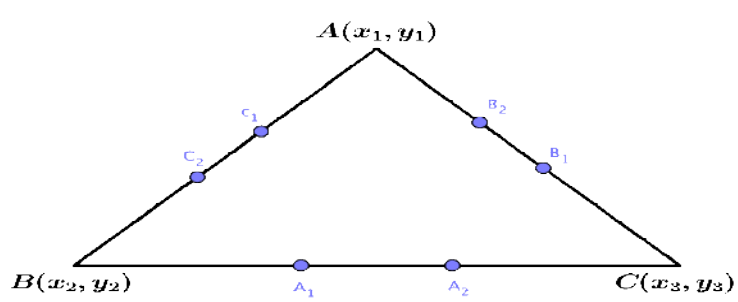
$= c^2 - b^2 + \frac{2(b^2-c^2)\cdot k}{k+1}$. ***Similarly, we have*** $BB_1^2 - BB_2^2 = a^2 - c^2 + \frac{2(c^2-a^2)\cdot k}{k+1}$ ***and***

$CC_1^2 - CC_2^2 = b^2 - a^2 + \frac{2(a^2-b^2)\cdot k}{k+1}$. ***Similarly, we have*** $BB_1^2 - BB_2^2 = a^2 - c^2 + \frac{2(c^2-a^2)\cdot k}{k+1}$

and $CC_1^2 - CC_2^2 = b^2 - a^2 + \frac{2(a^2-b^2)\cdot k}{k+1}$

$$\Rightarrow AA_1^2 - AA_2^2 + BB_1^2 - BB_2^2 + CC_1^2 - CC_2^2 = 0 \Rightarrow AA_1^2 + BB_1^2 + CC_1^2 = AA_2^2 + BB_2^2 + CC_2^2$$

Solution 2(Ravi Prakash)



Let $k = \frac{BA_1}{CA_1} = \frac{CB_1}{AB_1} = \frac{AC_1}{BC_1} = \frac{CA_2}{BA_2} = \frac{AB_2}{CB_2} = \frac{BC_2}{AC_2}$***. Coordinates of*** $A_1\left(\frac{x_2+kx_3}{k+1}, \frac{y_2+ky_3}{k+1}\right)$

$$AA_1^2 = \left(x_1 - \frac{x_2 - kx_3}{k+1}\right)^2 + \left(y_1 - \frac{y_2 + ky_3}{k+1}\right)^2$$

$$= \frac{1}{(k+1)^2}[(x_1 - x_2) + k(x_1 - x_3)]^2 + \frac{1}{(k+1)^2}[(y_1 - y_2) + k(y_1 - y_3)]^2$$

$$= \frac{1}{(k+1)^2}\begin{bmatrix}(x_1 - x_2)^2 + (y_1 - y_2)^2 + k^2\{(x_1 - x_3)^2 + (y_1 - y_3)^2\} + \\ +2k(x_1 - x_2)(x_1 - x_3) + 2k(y_1 - y_2)(y_1 - y_3)\end{bmatrix}$$

$$= \frac{1}{(k+1)^2}[AB^2 + k^2AC^2 + 2k(x_1 - x_2)(x_1 - x_3) + 2k(y_1 - y_2)(y_1 - y_3)]$$

$$\therefore AA_1^2 + BB_1^2 + CC_1^2 =$$

$$= \frac{1}{(k+1)^2}\begin{bmatrix}(k^2+1)(AB^2 + BC^2 + AC^2) + \\ +2k\left(x_1^2 + x_2^2 + x_3^2 + y_1^2 + y_2^2 + y_3^2 — x_1x_2 - x_2x_3 - x_3x_1 - y_1y_2 - y_2y_3 - y_3y_1\right)\end{bmatrix}$$

$= E$ ***(say).Similarly*** $AA_2^2 + BB_2^2 + CC_2^2 = E$

$$\therefore AA_1^2 + BB_1^2 + CC_1^2 = AA_2^2 + BB_2^2 + CC_2^2$$

1.2 *Solution (Daniel Sitaru)*

$$m(\sphericalangle A) = 135^\circ \to R = \frac{a}{2\sin A} = \frac{a\sqrt{2}}{2}; S = \frac{1}{2}bc\sin 135^\circ = \frac{\sqrt{2}bc}{4},$$

$$a^2 = b^2 + c^2 + \sqrt{2}bc$$

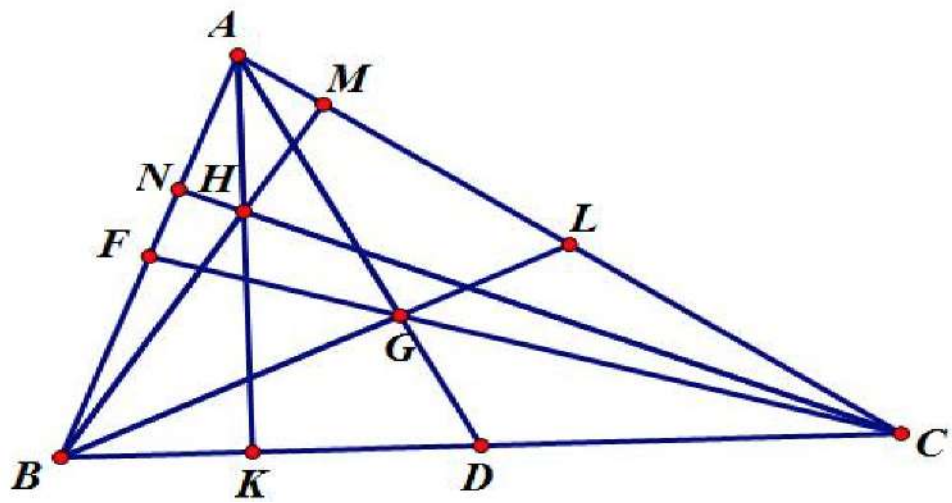
$$\frac{s+r}{R+r} = \sqrt{2} \leftrightarrow s = \sqrt{2}R + r(\sqrt{2} - 1) \leftrightarrow s = a + r(\sqrt{2} - 1)$$

$$a + b + c = 2a + 2r(\sqrt{2} - 1) \leftrightarrow b + c - a = (\sqrt{2} - 1)\cdot\frac{\sqrt{2}bc}{2s}$$

$$(a + b + c)(b + c - a) = (2 - \sqrt{2})bc \leftrightarrow (b + c)^2 - a^2 = (2 - \sqrt{2})bc$$

$$(b + c)^2 - b^2 - c^2 - \sqrt{2}bc = (2 - \sqrt{2})bc \leftrightarrow (2 - \sqrt{2})bc = (2 - \sqrt{2})bc$$

1.3 *Solution (Khanh Hung Vu)*



We have $h_b = \frac{2S}{b} \Rightarrow h_b^2 = \frac{4S^2}{b^2} = \frac{4\cdot\frac{2(a^2b^2+b^2c^2+c^2a^2)-(a^4+b^4+c^4)}{16}}{b^2} =$

$$= \frac{2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4)}{4b^2}$$

We have $\tan\beta = \frac{ML}{BM} = \frac{\sqrt{m_a^2 - h_b^2}}{h)b} = \frac{\sqrt{\frac{2a^2+2c^2-b^2}{4} - \frac{2(a^2b^2+b^2c^2+c^2a^2)-(a^4+b^4+c^4)}{4b^2}}}{\frac{2S}{b}} =$

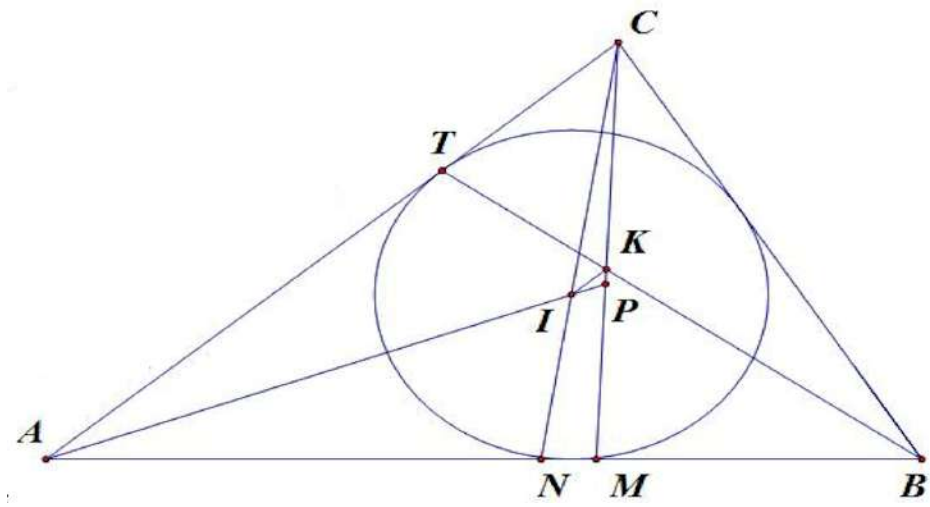
$$= \frac{\sqrt{\frac{(2a^2 + 2c^2 - b^2)\cdot b^2 - 2(a^2b^2 + b^2c^2 + c^2a^2) + (a^4 + b^4 + c^4)}{4b^2}}}{\frac{2S}{b}}$$

$\Rightarrow \tan\beta = \frac{\sqrt{\frac{a^4+c^4-2a^2c^2}{4b^2}}}{\frac{2S}{b}} = \frac{\frac{a^2-c^2}{2b}}{\frac{2S}{b}} = \frac{a^2-c^2}{4S}$ ***(1) (Since*** $a > c$***)***

Similarly, we have $\tan\alpha = \frac{b^2-c^2}{4S}$ ***and*** $\tan\gamma = \frac{a^2-b^2}{4S} \Rightarrow \tan\alpha + \tan\gamma = \frac{a^2-c^2}{4S}$ ***(2)***

(1) and (2) $\Rightarrow \tan\beta = \tan\alpha + \tan\gamma$

1.4 *Solution (Khanh Hung Vu)*



1) Put $b = \frac{a^2+c^2}{a+c}$ ***and*** $P = AI \cap CK, M = CP \cap AB, N = CI \cap AB$

We have $\frac{AM}{MB} = \frac{b^2}{a^2}$ ***and*** $AM + MB = AB = c \Rightarrow AM = \frac{cb^2}{a^2+b^2}$

ΔMAC ***has*** AP ***is a bisector of*** $\angle A \Rightarrow \frac{MP}{PC} = \frac{AM}{AC} = \frac{\frac{cb^2}{a^2+b^2}}{b} = \frac{bc}{a^2+b^2}$

On the other hand, we have $MP + PC = MC = \frac{ab\sqrt{2a^2+2b^2-c^2}}{a^2+b^2}$

$$\Rightarrow CP = \frac{ab\sqrt{2a^2+2b^2-c^2}\cdot(a^2+b^2)}{(a^2+b^2)(a^2+b^2+bc)} = \frac{ab\sqrt{2a^2+2b^2-c^2}}{a^2+b^2+bc} \quad \textbf{(1)}$$

We have $\frac{CK}{KM} = \frac{a^2+b^2}{c^2}$ ***and*** $CK + KM = CM = \frac{ab\sqrt{2a^2+2b^2-c^2}}{a^2+b^2}$

$$\Rightarrow CK = \frac{ab\sqrt{2a^2+2b^2-c^2}\cdot(a^2+b^2)}{(a^2+b^2)(a^2+b^2+c^2)} = \frac{ab\sqrt{2a^2+2b^2-c^2}}{a^2+b^2+c^2} \quad \textbf{(2)}$$

(1) and (2) $\Rightarrow \frac{CK}{CP} = \frac{ab^2+b^2+c^2}{a^2+b^2+bc}$ ***(3)***

On the other hand, ΔANC ***has*** AI ***is a bisector of*** $\angle A \Rightarrow AI = \frac{2AN\cdot AC\cdot\cos\frac{A}{2}}{AN+AC}$

Similarly, we have $AP = \frac{2AM \cdot AC \cdot \cos\frac{A}{2}}{AM+AC} \Rightarrow \frac{AI}{AP} = \frac{AN(AM+AC)}{AM(AN+AC)} = \frac{\frac{bc}{b+c}\left(\frac{cb^2}{a^2+b^2}+b\right)}{\frac{cb^2}{a^2+b^2}\left(\frac{bc}{b+c}+b\right)} = \frac{a^2+b^2+bc}{b(a+b+c)}$ **(4)**

Since $b = \frac{a^2+c^2}{a+c} \Rightarrow a^2 + b^2 + c^2 = b^2 + b(a+c) = b(a+b+c)$

(3) and (4) $\Rightarrow \frac{CK}{CP} = \frac{AI}{AP} \Rightarrow IK \parallel AC$ **(Thales theorem)**

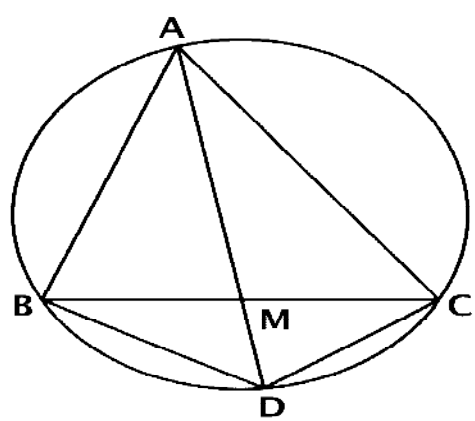
2) We need to prove that $\frac{AT}{TC} = \frac{c^2}{a^2}$ **(5)**

$$\Rightarrow \frac{p-a}{p-c} = \frac{c^2}{a^2} \Rightarrow \frac{b+c-a}{a+b-c} = \frac{c^2}{a^2} \Rightarrow ba^2 + ca^2 - a^3 = ac^2 + bc^2 - c^3$$

$$\Rightarrow b = \frac{ac^2-a^2c+a^3-c^3}{a^2-c^2} \Rightarrow b = \frac{(a-c)(a^2+c^2)}{(a-c)(a+c)} \Rightarrow b = \frac{a^2+c^2}{a+c}$$ **(True)**

(5) $\Rightarrow T$ **in symmedian line from** $B \Rightarrow T \in BK \Rightarrow B, K, T$ **are collinear**

1.5 *Solution (Marian Ursărescu)*



From Ptolemy's theorem $\Rightarrow AD \cdot BC = AD \cdot CD + AC \cdot BD \Rightarrow AD \cdot a = CD \cdot c + BD \cdot b \Rightarrow$

$$\Rightarrow AD = \frac{C \cdot CD + b \cdot BD}{a}$$ **(1)**

We must show: $AD = BD + DC$**, from (1)** $\Rightarrow$

$$\Rightarrow \frac{b \cdot BD + c \cdot CD}{a} = BD + DC \Leftrightarrow b \cdot BD + c \cdot CD = a \cdot BD + a \cdot DC \Leftrightarrow$$

$$\Leftrightarrow BD(b-a)+CD(c-a)=0 \Leftrightarrow BD\left(b-\frac{b^2+c^2}{b+c}\right)+CD\left(c-\frac{b^2+c^2}{b+c}\right)=0 \Leftrightarrow$$

$$\Leftrightarrow BD(b^2+bc-b^2-c^2)+CD(bc+c^2-b^2-c^2)=0 \Leftrightarrow$$

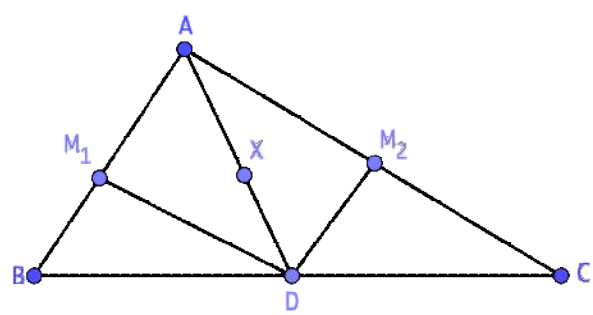
$$\Leftrightarrow BD\cdot c(b-c)+CD\cdot b(c-b)=0 \text{ but } \Delta ABC \text{ non-isosceles} \Rightarrow$$

$$BD\cdot c-b\cdot CD \Rightarrow \text{we must show } BD\cdot c=b\cdot CD \text{ (2)}$$

$$\text{But } AM \text{ median} \Rightarrow S_{ABM}=S_{ACM} \text{ and } S_{DBM}=S_{CDM} \Rightarrow$$

$$\left.\begin{array}{r} S_{ABD}=S_{ACD} \Rightarrow BD\cdot c\cdot \sin ABD=b\cdot DC\cdot \sin(ACD) \\ but\ \sin(ABD)=\sin(\pi-ABD)=\sin\widehat{ACD} \end{array}\right\} \Rightarrow BD\cdot c=b\cdot DC \Rightarrow 2 \text{ its true.}$$

1.6 *Solution (Marian Ursărescu)*



$$1.\ x=k \Rightarrow \frac{BD}{DC}=\frac{c^2}{b^2} \Rightarrow \frac{BD}{BC}=\frac{c^2}{b^2+c^2} \Rightarrow BD=\frac{ac^2}{b^2+c^2} \text{ and } DC=\frac{ab^2}{b^2+c^2} \quad (1)$$

From median theorem we have:

$$DM_1^2=\frac{2(BD^2+AD^2)-c^2}{4} \text{ and } DM_2^2=\frac{2(DC^2+AD^2)-b^2}{4} \text{ and } DM_1=DM_2$$

$$\Rightarrow 2BD^2-c^2=2DC^2-b^2 \Leftrightarrow$$

$$2\left(\frac{a^2c^4}{(b^2+c^2)^2}-\frac{a^2b^4}{(b^2+c^2)^2}\right)-(c^2-b^2)=0 \Leftrightarrow$$

$$2\frac{a^2}{(b^2+c^2)^2}(c^4-b^4)-(c^2-b^2)=0 \Leftrightarrow \frac{2a^2(c^2+b^2)}{(b^2+c^2)^2}-1=0 \Leftrightarrow$$

$$\frac{2a^2}{b^2+c^2} = 1 \Leftrightarrow 2a^2 = b^2 + c^2 \ \textit{true (1)}$$

$$\textit{2. } X = I \Rightarrow \frac{BD}{DC} = \frac{c}{b} \Rightarrow \frac{BD}{BC} = \frac{c}{b+c} \Rightarrow BD = \frac{ac}{b+c} \textit{ and } CD = \frac{ab}{b+c}$$

Again, median theorem $\Rightarrow$

$$2BD^2 - c^2 = 2CD^2 - b^2 \Leftrightarrow 2\left(\frac{a^2c^2}{(b+c)^2} - \frac{a^2b^2}{(b+c)^2}\right) - (c^2 - b^2) = 0$$

$$\Leftrightarrow \frac{2a^2}{(b+c)^2}(c^2 - b^2) - (c^2 - b^2) = 0 \Leftrightarrow \frac{2a^2}{(b+c)^2} - 1 = 0 \Leftrightarrow 2a^2 = (b+c)^2 \ \textit{(true)}$$

$$\textit{3. If } x = \Gamma \Rightarrow BD = s - b \textit{ and } s - c = CD$$

Again, from median theorem $\Rightarrow 2BD^2 - c^2 = 2CD^2 - b^2 \Leftrightarrow 2(s-b)^2 - c^2 = 2(s-c)^2 - b^2 \Leftrightarrow$

$$\Leftrightarrow 2((s-b)^2 - (s-c)^2) - (c^2 - b^2) = 0 \Leftrightarrow\Leftrightarrow 2(s-b-s+c)(s-b+s-c) - (c^2 - b^2) = 0 \Leftrightarrow 2a(c-b) - (c^2 - b^2) = 0 \Leftrightarrow 2a = b + c \ \textit{true.}$$

1.7 *Solution (Marian Ursărescu)*

We can choose ΔABC ***with circumcenter*** O ***in origin of axis.***

Let $z_A, z_B, z_C \in \mathbb{C}$***, so that*** $A(z_A), B(z_B), C(z_C)$

$$S_{ABC} = \left|\frac{i}{4}\begin{vmatrix} z_A & \overline{z_A} & 1 \\ z_B & \overline{z_B} & 1 \\ z_C & \overline{z_C} & 1 \end{vmatrix}\right| \ \textit{(1)}$$

But $z_{H_1} = z_p + z_A + z_B, z_{H_2} = z_p + z_B + z_C, z_{H_3} = z_p + z_A + z_C \Rightarrow$

$$S_{H_1H_2H_3} = \left|\frac{i}{4}\begin{vmatrix} z_p + z_A + z_B & \overline{z_p} + \overline{z_A} + \overline{z_B} & 1 \\ z_p + z_B + z_C & \overline{z_p} + \overline{z_B} + \overline{z_C} & 1 \\ z_p + z_A + z_C & \overline{z_p} + \overline{z_A} + \overline{z_C} & 1 \end{vmatrix}\right| \ \textit{(2)}$$

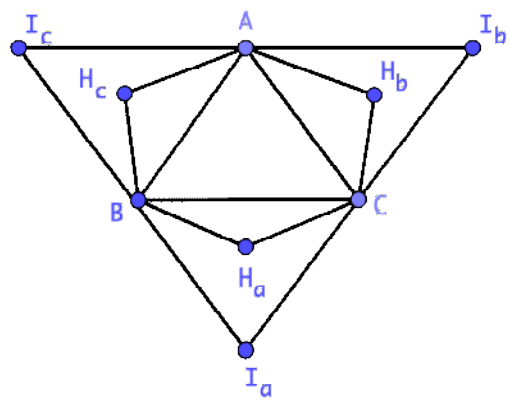
Now, using two properties:

$$\begin{vmatrix} a'_{11}+a''_{11} & a_{12} & a_{13} \\ a'_{21}+a''_{21} & a_{22} & a_{23} \\ a'_{31}+a''_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a'_{11} & a_{12} & a_{13} \\ a'_{21} & a_{22} & a_{23} \\ a'_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a''_{11} & a_{12} & a_{13} \\ a''_{21} & a_{22} & a_{23} \\ a''_{31} & a_{32} & a_{33} \end{vmatrix} \quad \textbf{(3)}$$

$$\textbf{and} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -\begin{vmatrix} a_{12} & a_{11} & a_{13} \\ a_{22} & a_{21} & a_{23} \\ a_{32} & a_{31} & a_{33} \end{vmatrix} \quad \textbf{(4)}$$

$$\textbf{From (1)+(2)+(3)+(4)} \Rightarrow S_{ABC} = S_{H_1H_2H_3}$$

1.8 *Solution (Marian Ursărescu)*



$$\Delta AI_bC \sim \Delta CI_aB \sim \Delta BI_cA \sim \Delta I_aI_bI_c \textbf{ with angles: } \frac{\pi}{2}-\frac{A}{2}, \frac{\pi}{2}-\frac{B}{2}, \frac{\pi}{2}-\frac{C}{2} \Rightarrow \frac{S_{I_aBC}}{S_{I_aI_bI_c}} = \left(\frac{I_aC}{I_aI_c}\right)^2 \quad \textbf{(1)}$$

$$\textbf{But } I_cC \perp I_aI_b \Rightarrow \frac{I_aC}{I_aI_c} = \cos(BI_aC) = \cos\left(\frac{\pi}{2}-\frac{\pi}{2}\right) = \sin\frac{A}{2} \quad \textbf{(2)}$$

$$\left.\begin{array}{l} \textit{From } (1)+(2) \Rightarrow S_{I_aBC} = S_{I_aI_bI_c}\sin^2\dfrac{A}{2} \\ \textit{Similarly } S_{I_bAC} = S_{I_AI_BI_C} \textit{ and } S_{I_cAB} = S_{I_aI_bI_c}\sin^2\dfrac{C}{2} \end{array}\right\} \overset{(3)}{\Rightarrow}$$

$$\textbf{From (3)} \Rightarrow S_{I_aI_bI_c} = S_{ABC} + S_{I_aI_bI_c}\left(\sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2}\right) \Rightarrow$$

$$S = S_{I_AI_BI_C}\left(1-\sin^2\frac{A}{2}-\sin^2\frac{B}{2}-\sin^2\frac{C}{2}\right) \quad \textbf{(4)}$$

$$\textbf{But } \sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2} = 1-\frac{r}{2R} \quad \textbf{(5)}$$

From (4)+(5) $\Rightarrow S = S_{I_AI_BI_c} \cdot \frac{r}{2R} \Rightarrow s_{I_aI_bI_c} = \frac{S \cdot 2R}{r} = 2Rs$ ***(5)***

Now, we have $m(H_aBC) = \frac{C}{2}, m(H_aCB) = \frac{B}{2}$ ***and*** $m(BH_aC) = \frac{\pi}{2} + \frac{A}{2}$***, but*** $m\Delta BCI$ ***we have***

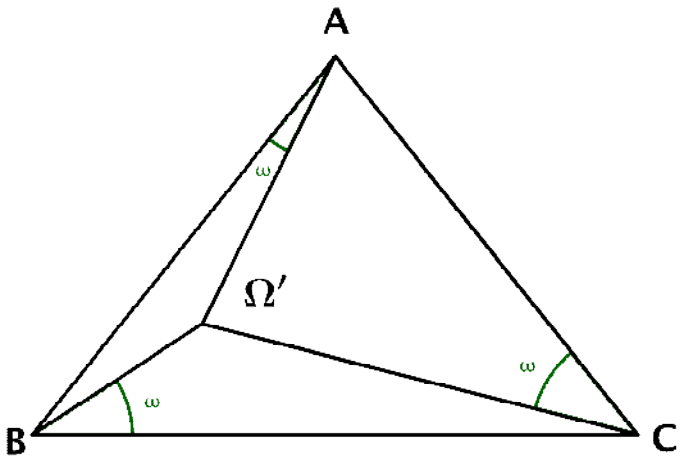
$$m(IBC) = \frac{B}{2}, m(ICB) = \frac{C}{2}, m(BIC) = \frac{\pi}{2} + \frac{A}{2} \Rightarrow$$

$\Rightarrow S_{BH_AC} = \frac{ar}{2}$***, and similarly,*** $S_{AH_BC} = \frac{br}{2}$ ***and*** $S_{AH_CB} = \frac{cr}{2} \Rightarrow$

$S_{BH_AC} + S_{AH_BC} + S_{AH_CB} = \frac{(a+b+c)r}{2} = sr = S$ ***(7). From (7)*** $\Rightarrow S_{AH_CBH_aH_bA} = 2sr$ ***(8)***

From (6)+(8) $= \frac{S_{AH_cBH_aCH_bA}}{S_{I_aI_bI_c}} = \frac{2sr}{2Rs} = \frac{r}{R} \quad \left(s = \frac{a+b+c}{2}\right)$

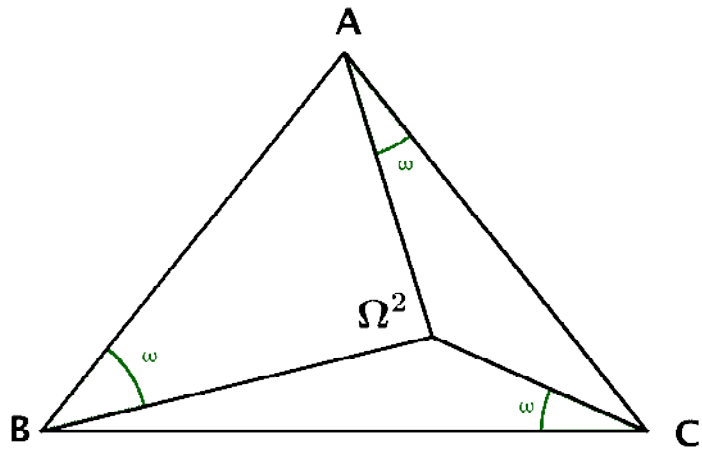
1.9 *Solution (Mehmet Şahin)*



$$\frac{B\Omega'}{\sin\omega} = 2R_c \Rightarrow B\Omega' = 2R_c \sin\omega, \frac{C\Omega'}{\sin\omega} = 2R_a \Rightarrow C\Omega' = 2R_a \cdot \sin\omega$$

$$\frac{A\Omega'}{\sin\omega} = 2R_b \Rightarrow A\Omega' = 2R_b \cdot \sin\omega$$

$$A\Omega' \cdot B\Omega' \cdot C\Omega' = 8R_aR_bR_c \cdot \sin^3\omega = 8R^3 \cdot \sin^3\omega = (2R\sin\omega)^3 \quad (1)$$



$$\frac{B\Omega^{(2)}}{\sin\omega} = 2R_a' \Rightarrow B\Omega^{(2)} = 2R_a' \cdot \sin\omega, \frac{C\Omega^{(2)}}{\sin\omega} = 2R_a' \Rightarrow C\Omega^{(2)} = 2R_a' \cdot \sin\omega$$

$$\frac{A\Omega^{(2)}}{\sin\omega} = 2R_c' \Rightarrow A\Omega^{(2)} = 2R_c' \cdot \sin\omega$$

$$A\Omega^{(2)} \cdot B\Omega^{(2)} \cdot C\Omega^{(2)} = 8R_a' \cdot R_b' \cdot R_c' \cdot \sin^3\omega = 8 \cdot R^3 \cdot \sin^3\omega = (2R\sin\omega)^3 \quad \textbf{\textit{(2)}}$$

From (1) and (2) $A\Omega' \cdot B\Omega' \cdot C\Omega' = A\Omega^{(2)} \cdot B\Omega^{(2)} \cdot C\Omega^{(2)}$ ***as desired***

1.10 *Solution(Daniel Sitaru)*

$$b = 2R\sin B = 2R\sin\frac{B}{3}\left(3 - 4\sin^2\frac{B}{3}\right) = 2R\sin\frac{B}{3}\left(3\cos^2\frac{B}{3} - \sin^2\frac{B}{3}\right) =$$

$$= 8R\sin\frac{B}{3}\left(\frac{3}{4}\cos^2\frac{B}{3} - \frac{1}{4}\sin^2\frac{B}{3}\right) = 8R\sin\frac{B}{3}\left(\sin^2\frac{\pi}{3}\cos^2\frac{B}{3} - \sin^2\frac{B}{3}\cos^2\frac{\pi}{3}\right) =$$

$$= 8R\sin\frac{B}{3}\left(\frac{\pi}{3}\cos\frac{B}{3} - \sin\frac{B}{3}\cos\frac{\pi}{3}\right)\left(\sin\frac{\pi}{3}\cos\frac{B}{3} + \sin\frac{B}{3}\cos\frac{\pi}{3}\right)$$

$$b = 8R\sin\frac{B}{3}\sin\left(\frac{\pi}{3} - \frac{B}{3}\right)\sin\left(\frac{\pi}{3} + \frac{B}{3}\right)$$

Analogous:

$$c = 8R\sin\frac{C}{3}\sin\left(\frac{\pi}{3} - \frac{C}{3}\right)\sin\left(\frac{\pi}{3} + \frac{C}{3}\right)$$

$$\textit{In } \Delta\ ARC: \frac{RC}{\sin\frac{A}{3}} = \frac{AC}{\sin\left(\pi - \frac{A}{3} - \frac{C}{3}\right)}$$

$$RC = \frac{b\sin\frac{C}{3}}{\sin\frac{A+C}{3}} = \frac{2R\sin\frac{B}{3}\sin\left(\frac{\pi}{3} - \frac{B}{3}\right)\sin\left(\frac{\pi}{3} + \frac{B}{3}\right)\sin\frac{C}{3}}{\sin\left(\frac{\pi}{3} - \frac{B}{3}\right)}$$

$$RC = 2R\sin\frac{B}{3}\sin\left(\frac{\pi}{3} + \frac{B}{3}\right)\sin\frac{C}{3}$$

$$OB \cdot RC \cdot MA = 8R^3\left(\sin\frac{A}{3}\sin\frac{B}{3}\sin\frac{C}{3}\right)^2 \prod \sin\frac{\pi + A}{3} = OC \cdot RA \cdot MB$$

1.11 *Solution(Daniel Sitaru)*

$$S = S[ABC], S[BCO_1] = \frac{a^2\sqrt{3}}{12}, S[ACO_2] = \frac{b^2\sqrt{3}}{12}, S[ABO_3] = \frac{c^2\sqrt{3}}{12}$$

$$S[AO_3BO_1CO_2] = S[ABC] + S[BCO_1] + S[ACO_2] + S[ABO_3]$$

$$S[AO_3BO_1CO_2] = S + \frac{\sqrt{3}}{12}(a^2 + b^2 + c^2)$$

$$S[BO_1O_3] = \frac{1}{2} \cdot \frac{a\sqrt{3}}{3} \cdot \frac{c\sqrt{3}}{3}\sin\left(B + \frac{\pi}{3}\right) = \frac{ac}{6}\left(\frac{1}{2}\sin B + \frac{\sqrt{3}}{2}\cos B\right) =$$

$$= \frac{S}{6} + \frac{\sqrt{3}}{24}(a^2 + c^2 - b^2)$$

$$S[O_1O_2O_3] = S[AO_3BO_1CO_2] - \sum S[BO_1O_3]$$

$$2S[O_1O_2O_3] = 2S + \frac{\sqrt{3}}{6}(a^2 + b^2 + c^2) - 2 \cdot 3 \cdot \frac{S}{6} - \frac{\sqrt{3}}{12}\sum(a^2 + c^2 - b^2)$$

$$2S[O_1O_2O_3] = S + \frac{\sqrt{3}}{12}(a^2 + b^2 + c^2) = S[AO_3BO_1CO_2]$$

1.12 *Solution (Daniel Sitaru)*

$$\cos C = \frac{1}{2} = \frac{a^2 + b^2 - c^2}{2ab} \to c^2 = a^2 - ab + b^2$$

$$\cos A \cos B = \frac{b^2 + c^2 - a^2}{2bc} \cdot \frac{a^2 + c^2 - b^2}{2ac} =$$

$$= \frac{b^2 + a^2 - ab + b^2 - a^2}{2bc} \cdot \frac{a^2 + a^2 - ab + b^2 - b^2}{2ac} =$$

$$= \frac{2b^2 - ab}{2bc} \cdot \frac{2a^2 - ab}{2ac} = \frac{(2b-a)(2a-b)}{4c^2} \quad \textbf{(1)}$$

$$R = \frac{c}{2\sin 60°} = \frac{c}{\sqrt{3}}, r = \frac{S}{s} = \frac{\frac{1}{2}ab \cdot \frac{\sqrt{3}}{2}}{s} = \frac{ab\sqrt{3}}{2(a+b+c)}$$

$$\frac{r}{R} = \frac{\frac{ab\sqrt{3}}{2(a+b+c)}}{\frac{c}{\sqrt{3}}} = \frac{3ab}{2c(a+b+c)} = \frac{3ab}{4sc}, \left(\frac{r}{R}\right)^2 + \frac{r}{R} - \frac{1}{2} = \left(\frac{3ab}{4sc}\right)^2 + \frac{3ab}{4sc} - \frac{1}{2} \quad \textbf{(2)}$$

$$\textbf{(1)} = \textbf{(2)} \leftrightarrow \frac{(2b-a)(2a-b)}{4c^2} = \frac{9a^2b^2 + 12abcs - 8s^2c^2}{16s^2c^2} \leftrightarrow$$

$$\leftrightarrow 4s^2(4ab - 2b^2 - 2a^2 + ab) = 9a^2b^2 + 12abcs - 8s^2c^2$$

$$\leftrightarrow 20abs^2 - 8b^2s^2 - 8a^2s^2 = 9a^2b^2 + 12abcs - 8s^2a^2 - 8s^2b^2 + 8abs^2$$

$$\leftrightarrow 12abs^2 = 9a^2b^2 + 12abcs \leftrightarrow 4s^2 = 3ab + 4cs \leftrightarrow$$

$$\leftrightarrow (a + b + c)^2 = 3ab + 2c(a + b + c) \leftrightarrow$$

$$\leftrightarrow a^2 + b^2 + c^2 - ab = 2c^2 \leftrightarrow c^2 = a^2 - ab + b^2$$

1.13 *Solution 1(Daniel Sitaru)*

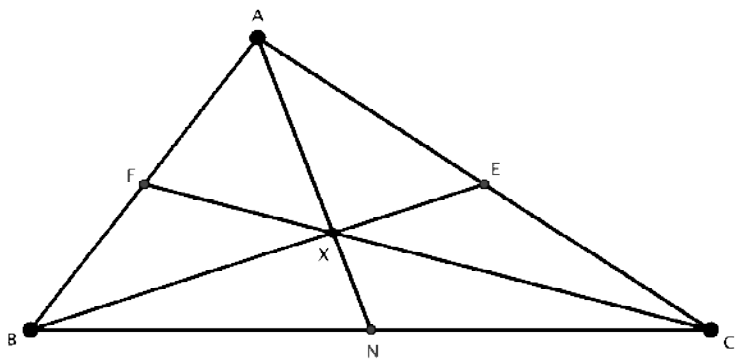
$$\frac{AF}{FB} = x, \frac{BD}{DC} = y, \frac{CE}{AE} = z, \qquad xyz = 1$$

$$\sum \frac{1}{\frac{1}{\frac{AF}{FB}+\frac{AE}{EC}}+1} = \sum \frac{1}{\frac{1}{x+\frac{1}{z}}+1} = \sum \frac{x+\frac{1}{z}}{1+x+\frac{1}{z}} =$$

$$= \sum \frac{x+xy}{1+x+xy} = \sum \frac{1+x+xy-1}{1+x+xy} = 3 - \sum \frac{1}{1+x+xy} = 2 \leftrightarrow$$

$$\leftrightarrow \sum \frac{1}{1+x+xy} = 1 \ (true\ if\ xyz = 1)$$

Solution 2(Daniel Sitaru)



$$\frac{AF}{FB} = x, \frac{BD}{DC} = y, \frac{CE}{AE} = z$$

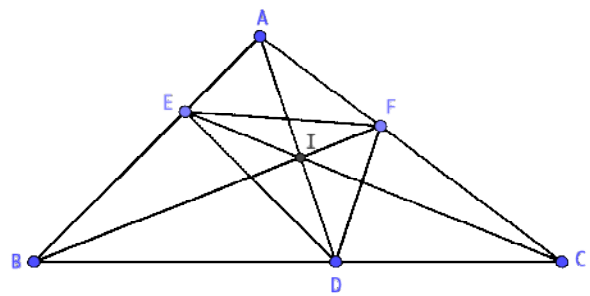
By Gergonne's theorem:

$$\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = 1, \qquad (1)$$

$$\sum \frac{1}{\frac{1}{\frac{AF}{FB}+\frac{AE}{EC}}+1} \overset{VAN\ AUBEL}{=} \sum \frac{1}{\frac{1}{\frac{AX}{XN}}+1} = \sum \frac{x}{1+x} =$$

$$= \sum \frac{1+x-1}{1+x} = 3 - \sum \frac{1}{1+x} \overset{(1)}{=} 3 - 1 = 2$$

1.14 *Solution (Marian Ursărescu)*



$$\frac{BD}{DC} = \frac{c}{b} \Rightarrow \frac{BD}{a} = \frac{c}{b+c} \Rightarrow BD = \frac{ac}{b+c} \text{ and } DC = \frac{ab}{b+c} \quad (1)$$

$$\frac{AE}{BE} = \frac{b}{a} \Rightarrow \frac{AE}{c} = \frac{b}{a+b} \Rightarrow AE = \frac{bc}{a+b} \text{ and } BE = \frac{ac}{a+b} \quad (2)$$

From (1)+(2)$\Rightarrow DE^2 = BD^2 + EB^2 - 2BE \cdot ED =$

$$= \frac{a^2c^2}{(b+c)^2} + \frac{a^2c^2}{(a+b)^2} - \frac{2a^2c^2}{(b+c)(a+b)} \cdot \cos B =$$

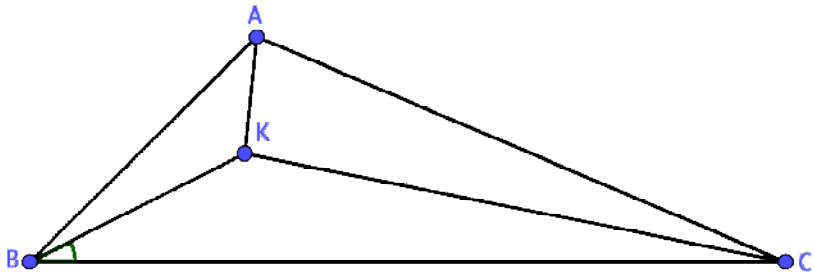
$$= \frac{a^2c^2}{(b+c)^2(a+b)^2}\left((a+b)^2 + (b+c)^2 - (a+b)(b+c)\left(\frac{a^2+c^2-b^2}{2ac}\right)\right) \Rightarrow$$

$$DE = \frac{abc\sqrt{3+2(\cos A - \cos B + \cos C)}}{(b+c)(b+a)} \quad (3) \text{ and } DF = \frac{abc\sqrt{3+2(\cos A + \cos B - \cos C)}}{(c+a)(c+b)} \quad (4)$$

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R} \quad (5)$$

From (3)+(4)+(5) and $DE = DF \Rightarrow \frac{a}{b+c} = \frac{b}{c+a} + \frac{c}{a+b}$

1.15 *Solution (Marian Ursărescu)*



$$AK = \frac{2bcm_a}{a^2+b^2+c^2} \quad (1)$$

$$a^2 = b^2 + c^2 - 2bcmA = 4c^2 + c^2 + 2c^2 = 7c^2 \Rightarrow a^2 = 7c^2 \quad (2)$$

$$m_a^2 = \frac{2(b^2+c^2)-a^2}{4} = \frac{10c^2-7c^2}{4} = \frac{3c^2}{4} \quad (3)$$

$$\text{From (1)+(2)+(3)} \Rightarrow AK = \frac{4c^2 \cdot \frac{\sqrt{c}c}{2}}{7c^2+4c^2+c^2} = \frac{\sqrt{3}c}{6}$$

$$CK = \frac{2abm_c}{a^2+b^2+c^2} \quad (4)$$

$$m_c^2 = \frac{2(a^2+b^2)-c^2}{4} = \frac{2(7c^2+4c^2)-c^2}{4} = \frac{21c^2}{4} \quad (5)$$

$$\text{From (4)+(5)} \Rightarrow CK = \frac{2\sqrt{7}c \cdot 2c \cdot \frac{\sqrt{21}c}{2}}{12c^2} = \frac{7\sqrt{3}c}{6} \quad (6)$$

$$\text{From (5)+(6)} \Rightarrow KC^2 = AK^2 + AC^2 \Rightarrow A = \frac{\pi}{2}$$

$$BK = \frac{2acm_b}{a^2+b^2+c^2} = \frac{2a\sqrt{7}c\sqrt{3}c}{12c^2} = \frac{\sqrt{21}c}{6} \quad (7)$$

$$\Delta BKC \text{ we have } KC^2 = BK^2 + BC^2 - 2BK \cdot BC \cdot \cos\varphi$$

$$\Rightarrow \frac{147c^2}{36} = 7c^2 + \frac{21}{36}c^2 - 2 \cdot \sqrt{7} \cdot \frac{c\sqrt{21}}{6} c \cos\varphi \Rightarrow \cos\varphi = \frac{\sqrt{3}}{2} \Rightarrow \varphi = 30^\circ$$

1.16 *Solution (Mehmet Sahin)*

In $\Delta\ ABC$, the following equalities are valid:

$$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}, w_a = \frac{2\sqrt{bc}}{b+c}\sqrt{s(s-a)}$$

$$8R \cdot \sum \frac{\sin\frac{A}{2}}{w_a} = 2 + \sum \frac{a}{s-a} \Leftrightarrow 8R \cdot \sum \frac{\sin\frac{A}{2}}{w_a} - \sum \frac{a}{s-a} = 2$$

$$T = 8R \cdot \left[\frac{\sqrt{(s-b)(s-c)}}{\sqrt{bc}} \cdot \frac{b+c}{2\sqrt{bc} \cdot \sqrt{s(s-a)}} + \frac{\sqrt{(s-c)(s-a)}}{\sqrt{ca}} \cdot \frac{c+a}{2\sqrt{ca} \cdot \sqrt{s(s-c)}} + \frac{\sqrt{(s-a)(s-b)}}{\sqrt{ab}} \cdot \frac{a+b}{2\sqrt{ab}\sqrt{s(s-a)}}\right]$$

$$T = 8R \cdot \left[\frac{\Delta}{2bc} \cdot \frac{b+c}{s(s-a)} + \frac{\Delta}{2ca} \cdot \frac{c+a}{s(s-c)} + \frac{\Delta}{2ab} \cdot \frac{a+b}{s(s-c)}\right]$$

$$= 4RD \cdot \left[\frac{b+c}{\frac{4R\Delta}{a}} \cdot \frac{1}{s(s-a)} + \frac{c+a}{\frac{4R\Delta}{b}} \cdot \frac{1}{s(s-b)} + \frac{a+b}{\frac{4RD}{c}} \cdot \frac{1}{s(s-c)}\right]$$

$$T = \frac{a(b+c)}{s(s-a)} + \frac{b(c+a)}{s(s-b)} + \frac{c(a+b)}{s(s-c)},\ P = \sum \frac{a}{s-a} = \frac{a}{s-a} + \frac{b}{s-b} + \frac{c}{s-c}$$

We have to show that $T - P = 2$.

$$T - P = \left(\frac{a(b+c)}{s(s-a)} - \frac{a}{s-a}\right) + \left(\frac{b(c+a)}{s(s-b)} - \frac{b}{s-b}\right) + \left(\frac{c(a+b)}{s(s-c)} - \frac{c}{s-c}\right)$$

$$= \frac{ab+ac-as}{s(s-a)} + \frac{bc+ba-bs}{s(s-b)} + \frac{ca+cb-cs}{s(s-c)}$$

$$= \frac{ab+ac-a\left(\frac{a+b+c}{2}\right)}{s(s-a)} + \frac{bc+ba-b\left(\frac{a+b+c}{2}\right)}{s(s-b)} + \frac{ca+cb-c\left(\frac{a+b+c}{2}\right)}{s(s-c)}$$

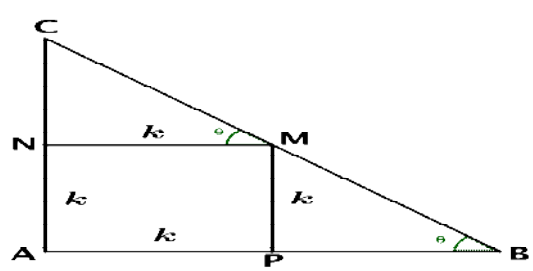
$$= \frac{2ab+2ac-a^2-ab-ac}{2s(s-a)} + \frac{2bc+2ba-ab-b^2-bc}{2s(s-b)} + \frac{2ca+2cb-ca-cb-c^2}{2s(s-c)}$$

$$T - P = \frac{ab+ac-a^2}{2s(s-a)} + \frac{bc+ba-b^2}{2s(s-b)} + \frac{ca+cb-c^2}{2s(s-c)}$$

$$T - P = \frac{a(b+c-a)}{2s(s-a)} + \frac{b(a+c-b)}{2s(s-b)} + \frac{c(a+b-c)}{2s(s-c)}$$

$$= \frac{a \cdot 2(s-a)}{2s(s-a)} + \frac{b \cdot 2(s-b)}{2s(s-b)} + \frac{c \cdot 2(s-c)}{2s(s-c)} = \frac{a}{s} + \frac{b}{s} + \frac{c}{s} = \frac{a+b+c}{s} = \frac{2s}{s} = 2$$

1.17 *Solution (Ravi Prakash)*



$$\frac{CN}{NM} = \tan\theta, CN = k\tan\theta, \frac{MP}{PB} = \tan\theta \Rightarrow PB = k\cot\theta$$

$$b = AN + NC = k + k\tan\theta = k(1 + \tan\theta), c = AP + PB = k + k\cot\theta = k(1 + \cot\theta)$$

$$a = BM + MC = k(\sec\theta + \csc\theta), \frac{b}{c} = \frac{1+\tan\theta}{1+\cot\theta} = \tan\theta$$

$$\therefore \left(\frac{b}{c} + 1\right)^2 + \left(\frac{c}{b} + 1\right)^2 = (\tan\theta + 1)^2 + (\cot\theta + 1)^2 =$$

$$= (\tan\theta + 1)^2 + \left(1 + \cot^2\theta\right) = (\tan\theta + 1)^2\csc^2\theta = (\sec\theta + \csc\theta)^2 = \left(\frac{a}{k}\right)^2$$

1.18 *Solution (Daniel Sitaru)*

$$HI^2 = 4R^2 + 4Rr + 3r^2 - s^2$$

$$H - lies\ on\ the\ incircle \rightarrow HI = r \rightarrow 4R^2 + 4Rr + 3r^2 - s^2 = r^2 \rightarrow$$

$$\rightarrow s^2 = 4R^2 + 4Rr + 2r^2$$

$$\prod cosA = \frac{s^2 - (2R + r)^2}{4R^2} = \frac{4R^2 + 4Rr + 2r^2 - 4R^2 - 4Rr - r^2}{4R^2} = \frac{r^2}{4R^2}$$

1.19 *Solution (Daniel Sitaru)*

$$\Omega O^2 \overset{CRELLE}{=} R^2 \cdot \frac{\sum a^4}{\sum a^2b^2} - R^2$$

$$\frac{1}{R^2 - \Omega O^2} = \frac{1}{2R^2 - R^2 \cdot \frac{\sum a^4}{\sum a^2b^2}} = \sum \frac{1}{a^2} \leftrightarrow$$

$$\leftrightarrow R^2 \sum \frac{1}{a^2}\left(2 - \frac{\sum a^4}{\sum a^2b^2}\right) = 1 \leftrightarrow \frac{(abc)^2}{16S^2} \cdot \frac{\sum a^2b^2}{(abc)^2}\left(2 - \frac{\sum a^4}{\sum a^2b^2}\right) = 1 \leftrightarrow$$

$$2\sum a^2b^2 - \sum a^4 = 16S^2$$

$$2\sum a^2b^2 - \sum a^4 = 2(s^4 - 2s^2r(4R - r) + r^2(4R + r)^2) - 2(s^4 - 2s^2r(4R + 3r) + r^2(4R + r)^2)$$

$$= -4s^2r(4R - r - 4R - 3r) = 16r^2s^2 = 16S^2$$

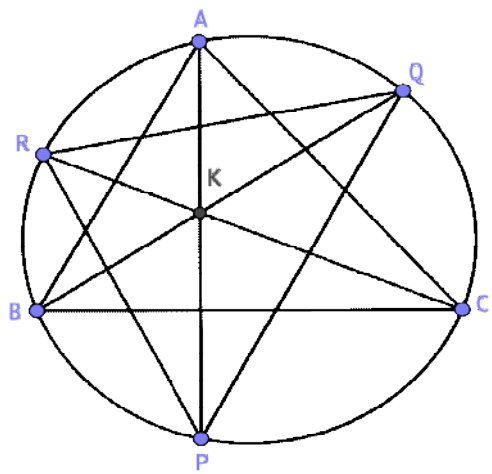
1.20 *Solution (Daniel Sitaru)*

$$IH^2 = 4R^2 + 4Rr + 3r^2 - s^2 = r^2 \rightarrow s^2 = 4R^2 + 4Rr + 2r^2$$

$$\sum tanA = \frac{2rs}{s^2 - (2R + r)^2} =$$

$$= \frac{2rs}{4R^2 + 4Rr + 2r^2 - 4R^2 - 4Rr - r^2} = \frac{2rs}{r^2} = \frac{2s}{r}$$

1.21 *Solution (Marian Ursărescu)*



k – symmedian points of ΔABC, P, Q, R points of intersection of AK, BK and CK with circumcircle of ΔABC. Prove: $a^2 + b^2 = 5c^2 \Rightarrow PR^2 + QR^2 = PQ^2$

$$PR^2 + QR^2 = PQ^2 \Leftrightarrow \widehat{PRQ} = \frac{\pi}{2} \Leftrightarrow PQ \text{ its diameter of } \Delta ABC \ (1)$$

Let AA_1, BB_1, CC_1 medians of ΔABC.

$$\left.\begin{aligned} AA_1^2 + BB_1^2 &= \frac{2(b^2 + c^2) - a^2}{4} + \frac{2(a^2 + c^2) - b^2}{4} = \frac{4c^2 + a^2 + b^2}{4} \\ CC_1^2 &= \frac{2(a^2 + b^2) - c^2}{4} \end{aligned}\right\} \Rightarrow$$

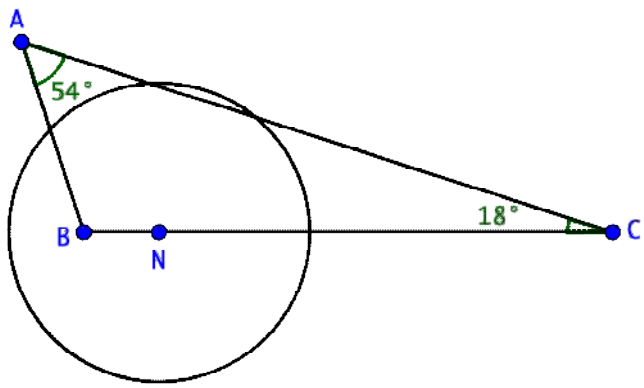
$$AA_1^2 + BB_1^2 = CC_1^2 \Leftrightarrow 4c^2 + a^2 + b^2 = 2a^2 + 2b^2 - c^2 \Leftrightarrow 5c^2 = a^2 + b^2 \ \textbf{(true)}$$

$\Rightarrow AA_1 \perp BB_1 \Rightarrow$ ***We have $AA_1 \perp BB_1$ (medians) (2)***

AA_1 and AP its isogonal (3), BB_1 and BQ its isogonal (4)

From (2)+(3)+(4)$\Rightarrow$ PQ its diameter $\Rightarrow$ (1) its true.

1.22 *Solution (Marian Ursărescu)*



Lemma: $AN = \frac{R}{2}\sqrt{1 + 8\cos A \sin B \sin C}$ ***(1)***

Demonstration: ΔANO, AN ***its median*** $\Rightarrow$

$$\Rightarrow AN^2 = \frac{2(AO^2 + AN^2) - ON^2}{4} = \frac{2(R^2 + 4R^2\cos^2 A) - R^2(1 - 8\cos A\cos B\cos C)}{4}$$

$$= \frac{R^2}{4}[1 + 8\cos A\,(\cos B\cos C - \cos(B + C))] = \frac{R^2}{4}(1 + 8\cos A\sin B\sin C) \Rightarrow$$

$$\Rightarrow AN^2 = \frac{R}{2}\sqrt{1 + 8\cos A\sin B\sin C}$$

In our case: $A = 54° = \frac{3\pi}{10}, B = 18° = \frac{\pi}{10}°, C = 108° = \frac{3\pi}{5}$

$$BC = 2R\sin A = 2R\sin\frac{3\pi}{10} = 2R\sin\frac{(\sqrt{5}+1)}{4} = \frac{R(\sqrt{5}+1)}{2} \quad \textbf{(2)}$$

$$BN \overset{(1)}{=} \frac{R}{2}\sqrt{1 + 8\cos B\sin A\sin C} = \frac{R}{2}\sqrt{1 + 8\cos\frac{\pi}{10}\cdot\sin\frac{3\pi}{10}\cdot\sin\frac{3\pi}{5}} =$$

$$= \frac{R}{2}\sqrt{1 + 8\frac{\sqrt{10 + 2\sqrt{5}}}{4}\cdot\frac{(\sqrt{5} + 1)}{4}\cdot\frac{\sqrt{10 + 2\sqrt{5}}}{4}} = \frac{R}{2}\sqrt{1 + \frac{1}{8}(10 + 2\sqrt{5})(\sqrt{5} + 1)}$$

$$= \frac{R}{2}\sqrt{1 + \frac{1}{4}(5 + \sqrt{5})(\sqrt{5} + 1)} = \frac{R}{2}\sqrt{\frac{7+3\sqrt{5}}{2}} \quad \textbf{(3)}$$

$$CN \overset{(1)}{=} \frac{R}{2}\sqrt{1+8\cos C\sin A\sin B} = \frac{R}{2}\sqrt{1-8\cos\frac{2\pi}{5}\sin\frac{3\pi}{10}\cdot\sin\frac{\pi}{10}}$$

$$= \frac{R}{2}\sqrt{1-\frac{1}{8}\left(\sqrt{5}-1\right)\left(\sqrt{5}+1\right)\left(\sqrt{5}-1\right)} = \frac{R}{2}\sqrt{\frac{3-\sqrt{5}}{2}} \quad \textbf{\textit{(4)}}$$

From (3)+(4) $\Rightarrow BN + CN = \frac{R}{2}\left(\sqrt{\frac{7+3\sqrt{5}}{2}} + \sqrt{\frac{3-\sqrt{5}}{2}}\right)$ ***(5)***

From (2)+(5) we must show: $\sqrt{7+3\sqrt{5}} + \sqrt{3-\sqrt{5}} = \sqrt{10} + \sqrt{2} \Leftrightarrow$

$$7 + 3\sqrt{5} + 3 - \sqrt{5} + 2\sqrt{\left(7+3\sqrt{5}\right)\left(3-\sqrt{5}\right)} = 12 + 4\sqrt{5} \Leftrightarrow$$

$$\Leftrightarrow \sqrt{6+2\sqrt{5}} = \sqrt{5}+1 \Leftrightarrow \sqrt{5}+1 = \sqrt{5}+1 \textbf{\textit{ true.}}$$

1.23 *Solution (Soumava Chakraborty)*

$$\frac{s-a}{AI} = \frac{4R\cos\frac{A}{2}\left(\sin\frac{B}{2}\sin\frac{C}{2}\sin\frac{A}{2}\right)}{r} = \frac{4R\cos\frac{A}{2}\left(\frac{r}{4R}\right)}{r} \overset{(1)}{=} \cos\frac{A}{2}$$

Now, $\tan\frac{A}{4} = \frac{1-\cos\frac{A}{2}}{\sin\frac{A}{2}} \overset{by\ (1)}{=} \frac{AI}{r}\left(1-\frac{s-a}{AI}\right) \overset{(2)}{=} \frac{AI-s+a}{r}$ ***and***

$$\cot\frac{A}{4} = \frac{1+\cos\frac{A}{2}}{\sin\frac{A}{2}} \overset{by\ (1)}{=} \frac{AI}{r}\left(1+\frac{s-a}{AI}\right) \overset{(3)}{=} \frac{AI+s-a}{r}$$

Similarly, $\tan\frac{B}{4} \overset{(2i)}{=} \frac{BI-s+b}{r}$, $\cot\frac{B}{4} \overset{(3i)}{=} \frac{BI+s-b}{r}$ ***and*** $\tan\frac{C}{4} \overset{(2ii)}{=} \frac{CI-s+c}{r}$, $\cot\frac{C}{4} \overset{(3ii)}{=} \frac{CI+s-c}{r}$

(2)+(2i)+(2ii) $\Rightarrow \sum\tan\frac{A}{4} = \frac{\sum AI-3s+2s}{r} \Rightarrow \sum\tan\frac{A}{4} \overset{(a)}{=} \frac{\sum AI-s}{r}$ ***and***

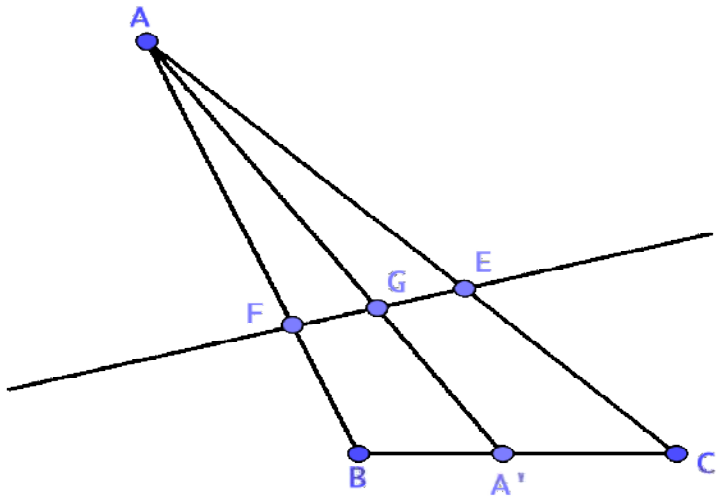
(3)+(3i)+(3ii) $\Rightarrow \sum\cot\frac{A}{4} \overset{(b)}{=} \frac{\sum AI+s}{r}$

(a),(b) $= \left(\sum\tan\frac{A}{4}\right)\left(\sum\cot\frac{A}{4}\right) = \frac{(\sum AI)^2-s^2}{r^2} \overset{(c)}{=} \left(\frac{\sum AI}{r}\right)^2 - \left(\frac{s}{r}\right)^2$

Now, RHS$= \left[\sum w_a\left(\frac{1}{h_b}+\frac{1}{h_c}\right)\right]^2 - \left(\sum\frac{r_b+r_c}{a}\right)^2 = \left[\sum\frac{2bc}{b+c}\cos\frac{A}{2}\left(\frac{b+c}{2rs}\right)\right]^2 - \left(\sum\frac{4R\cos^2\frac{A}{2}}{4R\sin\frac{A}{2}\cos\frac{A}{2}}\right)^2$

$$= \left(\sum\frac{4Rrs\cos\frac{A}{2}}{rs\cdot 4R\sin\frac{A}{2}\cos\frac{A}{2}}\right)^2 - \left(\sum\frac{s}{r_a}\right)^2 = \left(\frac{1}{r}\sum AI\right)^2 - \left(\frac{s}{r}\right)^2 \overset{by\ (c)}{=} LHS \ \textbf{(proved)}$$

1.24 *Solution 1 (Marian Ursărescu)*



From transversal theorem we have:

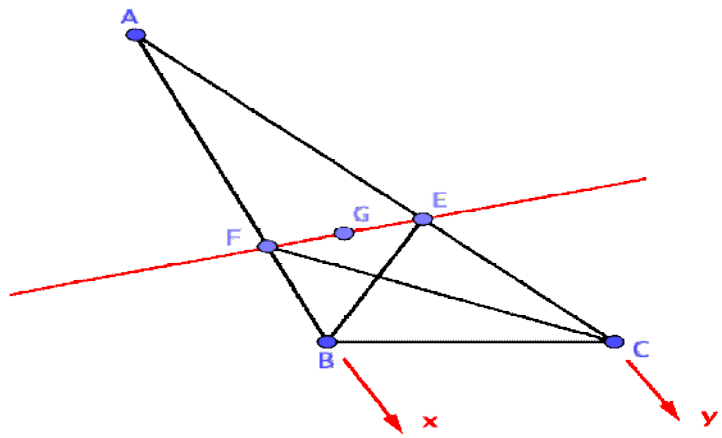
$$A'C\cdot\frac{FB}{FA}+BA'\cdot\frac{EC}{AE}=BC\cdot\frac{GA'}{GA}\ \textbf{(1)},\ A'C=BA'=\frac{9}{2}\ \textbf{(2)}$$

From bisector theorem: $\frac{FB}{FA}=\frac{a}{b}\wedge\frac{EC}{AE}=\frac{a}{c}$ ***(3),*** $\frac{GA'}{GA}=\frac{\frac{1}{3}AA'}{\frac{2}{3}GA}=\frac{1}{2}$ ***(4)***

From (1)+(2)+(3)+(4)⇒

$$\frac{a}{2}\cdot\frac{a}{b}+\frac{a}{2}\cdot\frac{a}{c}=a\cdot\frac{1}{2}\Leftrightarrow\frac{a}{b}+\frac{a}{c}=1\Leftrightarrow\frac{1}{b}+\frac{1}{c}=\frac{1}{a}$$

Solution 2 (Thanasis Gakopoulos)



$$G \text{ centroid, } BE, CF \text{ bisectors, } F\left(\frac{bc}{a+b}, 0\right), E\left(0, \frac{bc}{a+c}\right), G\left(\frac{c}{3}, \frac{a}{3}\right)$$

$$F, E, G \text{ collinear} \Leftrightarrow \begin{vmatrix} 1 & 1 & 1 \\ \frac{bc}{a+b} & 0 & \frac{c}{3} \\ 0 & \frac{bc}{a+c} & \frac{a}{3} \end{vmatrix} = 0 \Leftrightarrow \frac{1}{a} = \frac{1}{b} + \frac{1}{c}$$

1.25 *Solution (Khanh Hung Vu)*

Lemma: If $(AMN), (BEF)$ and (CUV) are 3 externally tangent circles and internally tangent circles with (O) and put R_1, R_2, R_3 are the radii of $(AMN), (BEF)$ and (CUV) and

$$AB = c, BC = a, CA = b. \text{ We have } R_1 = \frac{2RS}{a^2+2S}, R_2 = \frac{2RS}{b^2+2S}, R_3 = \frac{2RS}{c^2+2S}$$

Put R_1, R_2, R_3 is the radii of circle have diameter A_1M, MB and AB

Put $A_1B = a_1, A_2B = a_2, \dots, A_nB = a_n$ and $A_{n+1}B = a_{n+1}$ and put $A_1A_2 = b_1$,

$$A_2A_3 = b_2, \dots, A_{n-1}A_n = b_{n-1}, A_nA_{n+1} = b_n$$

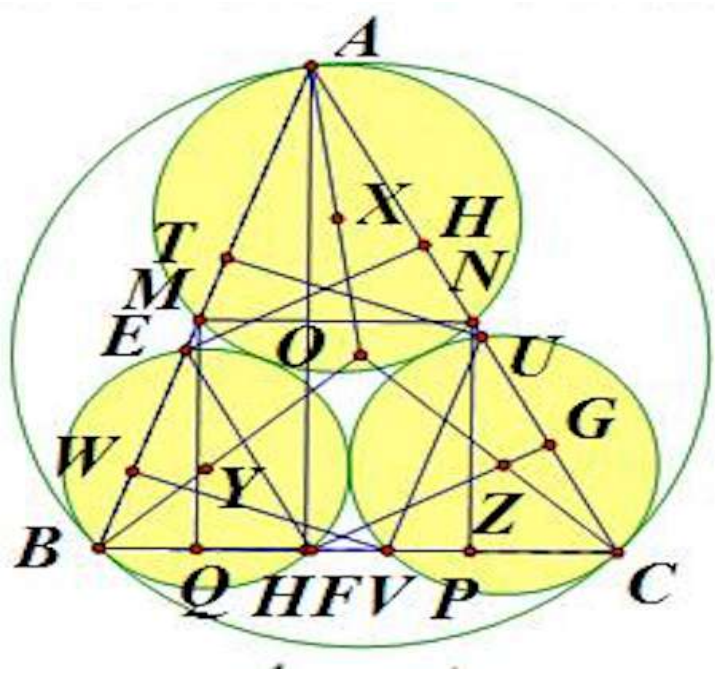
$$\text{We have } A_1M = \frac{2R}{7}, MB = \frac{2(6-m)R}{7} \text{ and } AB = \frac{2(7-m)R}{7}$$

Find $r_{n,m,l}$. Applying the lemma, we have

$$R_1 = \frac{2\cdot R_3\cdot S_{A_2A_1B}}{a_2^2+2S_{A_2A_2B}}, R_2 = \frac{2R_3\cdot S_{A_2A_1B}}{b_1^2+2S_{A_2A_1B}} \text{ and } r_{n,m,l} = \frac{2R_3\cdot S_{A_2A_1B}}{A_1B_2^2+2S_{A_2A_1B}}$$

On the other hand, we have $a_2^2 + b_1^2 = a_1^2 \Rightarrow \frac{1}{R_1} + \frac{1}{R_2} - \frac{1}{r_{n,m,l}} = \frac{1}{R_3} \Rightarrow$

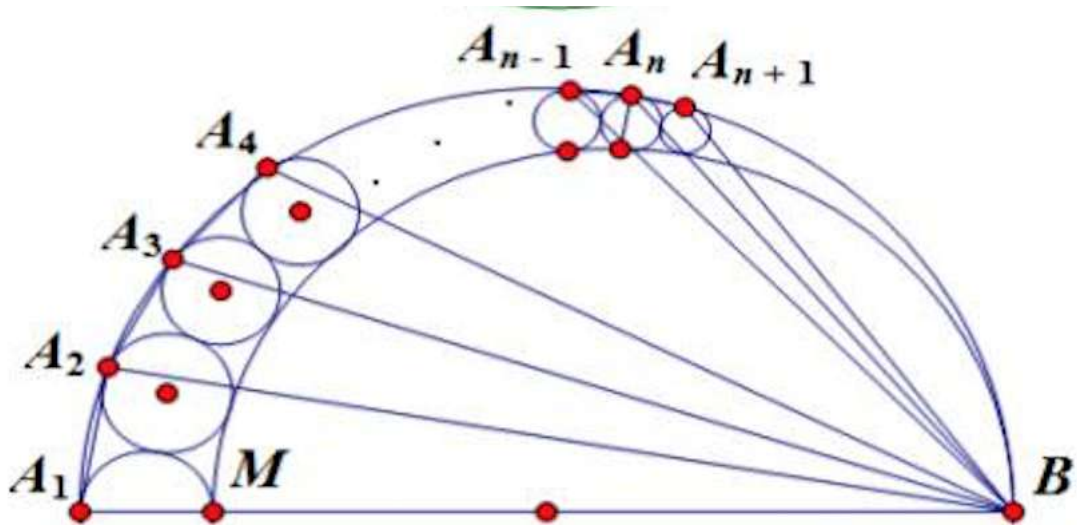
$$\Rightarrow r_{n,m,l} = \frac{1}{\frac{7}{R} + \frac{7}{(6-m)R} - \frac{7}{(7-m)R}} = \frac{R(6-m)(7-m)}{7(m^2-13m+43)}$$



Find $r_{n,m,2}$

We have $r_{n,m,l} = \frac{2R_3S_{A_1A_2B}}{a_1^2+2S_{A_2A_1B}} = \frac{2R_3S_{A_2A_3B}}{a_3^2+2S_{A_2A_3B}} \Rightarrow S_{A_2A_3B} = \frac{a_2b_1a_3^2}{2a_1^2}$ ***(1)***

We have $R_2 = \frac{2R_3S_{A_1A_2B}}{b_1^2+2S_{A_2A_1B}} = \frac{2R_3S_{A_2A_3B}}{b_2^2+2S_{A_2A_3B}} \Rightarrow S_{A_2A_3B} = \frac{a_2b_1b_2^2}{2b_1^2}$ ***(2)***



(1) and (2) $\Rightarrow \frac{a_3}{a_1} = \frac{b_2}{b_1}$

On the other hand, we have $R_1 = \frac{2R_3 S_{A_1A_2B}}{a_2^2+2S_{A_2A_1B}} \Rightarrow \frac{R}{7} = \frac{a_2b_1R_3}{a_2^2+a_2b_1} \Rightarrow \frac{b_1}{a_2} = \frac{1}{6-m}$

And $b_1^2 + a_2^2 = \frac{(14-2m)^2R^2}{49}$***. So,*** $b_1 = \frac{2(7-m)R}{7\sqrt{m^2-12m+37}}$ ***and*** $a_2 = \frac{2(7-m)(6-m)R}{7\sqrt{m^2-12m+37}}$***. By Cos's law for triangle*** BA_2A_3***, we have*** $b_2^2 + a_3^2 + 2b_2a_3 \cdot \frac{b_1}{c_1} = a_2^2$***. On the other hand, we have*** $\frac{a_3}{a_1} = \frac{b_2}{b_1}$

$$\Rightarrow \frac{a_3}{b_2} = \frac{a_1}{b_1} = \frac{\frac{2(7-m)R}{7}}{\frac{2(7-m)R}{7\sqrt{m^2-12m+37}}} = \sqrt{m^2-12m+37}$$

So $b_2 = \frac{2R(7-m)(6-m)}{7\sqrt{(m^2-12m+37)(m^2-12m+40)}}$ ***and*** $a_3 = \frac{2(7-m)(6-m)R}{7\sqrt{m^2-12m+40}}$

We have $r_{n,m,2} = \frac{2R_3A_{A_2A_3B}}{a_2^2+2S_{A_2A_3B}} = \frac{R_3 \cdot \frac{a_2b_1b_2^2}{b_1^2}}{a_2^2+\frac{a_2b_1b_2^2}{b_1^2}}$

$$= \frac{\frac{4(7-m)^2(6-m)}{49(m^2-12m+37)}R^2 \cdot \frac{(6-m)^2}{m^2-12m+40} \cdot \frac{(7-m)R}{7}}{\frac{4(7-m)^2(6-m)^2}{49(m^2-12m+37)}R^2 + \frac{4(7-m)^2(6-m)}{49(m^2-12m+37)}R^2 \cdot \frac{(6-m)^2}{m^2-12m+40}}$$

$$= \frac{R(6-m)(7-m)}{7(m^2-13m+46)}$$

Find $r_{n,m,3}$

We have $r_{n,m,2} = \frac{2R_3 S_{A_2A_3B}}{a_2^2+2S_{A_2A_3B}} = \frac{2R_3 S_{A_3A_4B}}{a_4^2+2S_{A_3A_4B}} \Rightarrow S_{A_3A_4B} = \frac{a_2 b_1 a_3^2 a_4^2}{2a_1^2 a_2^2}$ ***(3)***

We have $R_2 = \frac{2R_3 S_{A_2A_3B}}{b_2^2+2S_{A_2A_3B}} = \frac{2R_3 S_{A_3A_4B}}{b_3^2+2S_{A_3A_4B}} \Rightarrow S_{A_3A_4B} = \frac{a_2 b_1 b_2^2 b_3^2}{2b_1^2 b_2^2}$ ***(4)***

(3) and (4) $\Rightarrow \frac{a_4}{a_2} = \frac{b_3}{b_2}$

$$\Rightarrow \frac{a_4}{b_3} = \frac{a_2}{b_2} = \frac{\dfrac{2(7-m)(6-m)R}{7\sqrt{m^2-12m+37}}}{\dfrac{2R(7-m)(6-m)}{7\sqrt{(m^2-12m+37)(m^2-12m+40)}}} = \sqrt{m^2-12m+40}$$

On the other hand, by Cos's law for triangle BA_3A_4***, we have***

$$b_3^2 + a_4^2 + 2b_3a_4 \cdot \frac{b_2^2 + a_2^2 - a_3^2}{2b_2a_2} = a_3^2$$

So, $b_3 = \frac{2R(7-m)(6-m)}{7\sqrt{(m^2-12m+40)(m^2-12m+45)}}$ ***and*** $a_4 = \frac{2(7-m)(6-m)R}{7\sqrt{m^2-12m+45}}$

We have $r_{n,m,3} = \frac{2R_3 S_{A_3A_4B}}{a_3^2+2S_{A_3A_4B}} = \frac{\frac{a_2 b_1 a_3^2 a_4^2}{a_1^2 a_2^2} \cdot R_3}{a_3^2+\frac{a_2 b_1 a_3^2 a_4^2}{a_1^2 a_2^2}} = \frac{R_3 b_1 a_4^2}{a_2 b_1^2 + b_1 a_4^2}$

$$= \frac{\dfrac{(7-m)R}{7} \cdot \dfrac{2(7-m)R}{7\sqrt{m^2-12m+37}} \cdot \dfrac{4(7-m)^2(6-m)^2R^2}{49(m^2-12m+45)}}{\dfrac{2(7-m)(6-m)R}{7\sqrt{m^2-12m+37}} \cdot \dfrac{4(7-m)^2R^2}{49} + \dfrac{2(7-m)R}{7\sqrt{m^2-12m+37}} \cdot \dfrac{4(7-m)^2(6-m)^2R^2}{49(m^2-12m+45)}}$$

Prove that $r_{n,m,j} = \frac{(n-m)^2-(n-m)}{(n-m)^2-(n-m)+j^2} \cdot \frac{R}{n} = \frac{R(7-m)(6-m)}{7(m^2-13m+42+j^2)}$

We need to prove that $b_j = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+(j-1)^2} \cdot \sqrt{(6-m)^2+j^2}}$

And $a_j = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+(j-1)^2}}$ ***by introduction*** $j = 2$***, we have***

$$b_2 = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+1^2}\cdot\sqrt{(6-m)^2+2^2}} \text{ and } a_2 = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+1^2}} \text{ (True)}$$

Suppose that it is true until $j = k$

We need to prove with $j = k + 1$

$$\text{We have } R_2 = \frac{2R_3 S_{A_{k-1}A_kB}}{b_{k-1}^2+2S_{A_{k-1}A_kB}} = \frac{2R_3 S_{A_kA_{k+1}B}}{b_k^2+2S_{A_kA_{k+1}B}} \Rightarrow \frac{S_{A_kA_{k+1}B}}{S_{A_{k-1}A_kB}} = \frac{b_k^2}{b_{k-1}^2}$$

$$\Rightarrow \frac{b_k \cdot a_{k+1}}{b_{k-1} \cdot a_{k-1}} = \frac{b_k^2}{b_{k-1}^2} \Rightarrow \frac{a_{k+1}}{a_{k-1}} = \frac{b_k}{b_{k-1}} \Rightarrow a_{k+1} = \frac{a_{k-1}b_k}{b_{k-1}} = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2 + k^2}}$$

$$\text{So } a_j = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+(j-1)^2}} \text{ when } j \geq 2$$

$$\text{Similarly, we have } \frac{a_{k+2}}{a_k} = \frac{b_{k+1}}{b_k} \Rightarrow b_{k+1} = \frac{a_{k+2}b_k}{a_k} = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+k^2}\cdot\sqrt{(6-m)^2+(k+1)^2}}$$

$$\text{So } a_j = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+(j-1)^2}}, \text{ when } j \geq 2$$

$$\text{Similarly, we have } \frac{a_{k+2}}{a_k} = \frac{b_{k+1}}{b_k} \Rightarrow b_{k+1} = \frac{a_{k+2}b_k}{a_k} = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+k^2}\cdot\sqrt{(6-m)^2+(k+1)^2}}$$

$$\text{So, } b_j = \frac{2R(7-m)(6-m)}{\sqrt{(6-m)^2+(j-1)^2}\cdot\sqrt{(6-m)^2+j^2}} \text{ when } j \geq 2$$

$$\text{We need to prove that } r_{n,m,j} = \frac{(n-m)^2-(n-m)}{(n-m)^2-(n-m)+j^2} \cdot \frac{R}{n} = \frac{R(7-m)(6-m)}{7(m^2-13m+42+j^2)} \text{ by introduction}$$

$$j = 2, \text{ we have } r_{n,m,l} = \frac{R(7-m)(6-m)}{7(m^2-13m+46)} \text{ (True)}$$

Suppose that it is true until $j = k$

We need to prove with $j = k + 1$

$$\text{We have } r_{n,m,k} = \frac{R(7-m)(6-m)}{7(m^2-13m+42+k^2)} \Rightarrow \frac{2R_3 S_{A_{k-1}A_kB}}{a_k^2+2S_{A_{k-1}A_kB}} = \frac{R(7-m)(6-m)}{7(m^2-13m+42+k^2)}$$

$$\Rightarrow S_{A_{k-1}A_kB} = \frac{2R^2(7-m)^2(6-m)^3}{49[(6-m)^2 + (j-1)^2][(6-m)^2 + j^2]}$$

$$\Rightarrow S_{A_kA_{k+1}B} = \frac{2R^2(7-m)^2(6-m)^3}{49[(6-m)^2+(j-1)^2][(6-m)^2+j^2]} \cdot \frac{(6-m)^2+(j-1)^2}{(6-m)^2+(j+1)^2}$$

$$\Rightarrow S_{A_kA_{k+1}B} = \frac{2R^2(7-m)^2(6-m)^3}{49[(6-m)^2+(j+1)^2][(6-m)^2+j^2]}$$

We have $r_{n,m,k+1} = \frac{2R_3S_{A_kA_{k+1}B}}{a_{k+1}^2+2S_{A_kA_{k+1}B}} = \frac{2\cdot\frac{(7-m)R}{7}\cdot\frac{2R^2(7-m)^2(6-m)^3}{49[(6-m)^2+(j+1)^2][(6-m)^2+j^2]}}{\frac{4R^2(7-m)^2(6-m)^2}{49[(6-m)^2+j^2]}+\frac{4R^2(7-m)^2(6-)^3}{49[(6-m)^2+(j+1)^2][(6-m)^2+j^2]}} =$

$$= \frac{R(7-m)(6-m)}{7[m^2-13m+42+(j+1)^2]}$$

So, by introduction, we have $r_{n,m,j} = \frac{(n-m)^2-(n-m)}{(n-m)^2-(n-m)+j^2} \cdot \frac{R}{n}$ ***(QED)***

1.26 *Solution (Soumava Chakraborty)*

$$r_a - r_b = s\left(\tan\frac{A}{2} - \tan\frac{B}{2}\right)$$

$$= s\left(\frac{\sin\frac{A}{2}}{\cos\frac{A}{2}} - \frac{\sin\frac{B}{2}}{\cos\frac{B}{2}}\right) = \frac{s\left(\sin\frac{A}{2}\cos\frac{B}{2} - \cos\frac{A}{2}\sin\frac{B}{2}\right)}{\cos\frac{A}{2}\cos\frac{B}{2}}$$

$$= \frac{s}{\cos\frac{A}{2}\cos\frac{B}{2}} \cdot \sin\left(\frac{A-B}{2}\right) = \frac{4s\cos\frac{C}{2}}{4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}} \cdot \sin\left(\frac{A-B}{2}\right) = \frac{4s\cos\frac{C}{2}}{\sum\sin A} \cdot \sin\left(\frac{A-B}{2}\right)$$

$$= \frac{4s\cos\frac{C}{2}}{\frac{a+b+c}{2R}} \cdot \sin\left(\frac{A-B}{2}\right) = \frac{4s\cos\frac{C}{2}\cdot 2R}{2s} \cdot \sin\left(\frac{A-B}{2}\right) = 4R\sin\left(\frac{A-B}{2}\right)\cos\frac{C}{2}$$

1.27 *Solution (Daniel Sitaru)*

$$\sum\frac{\sqrt{a}}{cosA'} = \sum\frac{bc\sqrt{a}}{b+c-a} = 2\sum\frac{bc\sqrt{a}}{2(s-a)} =$$

$$= abc\sum\frac{1}{\sqrt{a}(s-a)} = \sqrt{abc}\cdot\frac{\sum(s-a)(s-b)}{(s-a)(s-b)(s-c)} =$$

$$= \frac{\sqrt{4RS}}{S^2} \cdot r(4R + r) = \frac{\sqrt{4RS}}{S}(4R + r) = 2\sqrt{\frac{R}{S}}(r_a + r_b + r_c)$$

1.28 *Solution (Daniel Sitaru)*

Let $\Delta T_A T_B T_C$ be the Gergonne's triangle:

$$m_1^2 + m_2^2 + m_3^2 = \frac{3}{4}\sum T_A T_B{}^2 = \frac{3}{4}\sum(2(s-c)^2 - 2(s-c)^2 cosC) =$$

$$= \frac{3}{2}\sum(s-c)^2(1 - cosC) = 3\sum(s-c)^2 sin^2\frac{C}{2} = 3\sum\frac{s(s-a)(s-b)(s-c)^2}{sab} =$$

$$= \frac{3S^2}{sabc}\sum c(s-c) = \frac{3S}{4Rs}(2s^2 - 2s^2 + 2r^2 + 8Rr) =$$

$$= \frac{3S}{4Rs} \cdot 2r(4R + r) = \frac{3rs}{2Rs} \cdot r(4R + r) = \frac{3r^2}{2R}(r_a + r_b + r_c)$$

1.29 *Solution (Daniel Sitaru)*

$$S_1 = \frac{1}{2}AR \cdot AQ \cdot sinA = \frac{(s-a)^2 sinA}{2} = \frac{a(s-a)^2}{4R}, \qquad A = \frac{abcS}{2r^2 s}$$

$$4A^2 S_1 S_2 S_3 = 4\left(\frac{abcS}{2r^2 s}\right)^2 \cdot \frac{abc(s-a)^2(s-b)^2(s-c)^2}{64R^3} =$$

$$= \frac{4(abc)^3 \cdot S^2 \cdot S^4}{4r^4 s^4 \cdot 64R^3} = \left(\frac{abc}{4R}\right)^3 \cdot S^2 = S^5, \quad S_0 A = \frac{2r^2 sS}{abc} \cdot \frac{abcS}{2r^2 s} = S^2$$

1.30 *Solution (Daniel Sitaru)*

$$\left(\frac{h_1}{h_a}\right)^2 = \left(\frac{\frac{2S[C_A C_B C_C]}{C_B C_C}}{\frac{2S}{a}}\right)^2 = \left(\frac{S[C_A C_B C_C]}{C_B C_C} \cdot \frac{a}{S}\right)^2 = \frac{a^2}{C_B C_C{}^2} \cdot \left(\frac{\frac{S^2}{2Rs}}{S}\right)^2 =$$

$$= \frac{a^2}{\frac{4S^2(s-a)}{sbc}} \cdot \frac{S^2}{4R^2 s^2} = \frac{a \cdot 4RS}{4(s-a) \cdot 4R^2 s} = \frac{ar}{4R(s-a)}$$

$$\sum\left(\frac{h_1}{h_a}\right)^2 = \sum\frac{ar}{4R(s-a)} = \frac{r}{4R}\sum\frac{a}{s-a} = \frac{r}{4R} \cdot \frac{2(2R-r)}{r} = \frac{2R-r}{2R} = 1 - \frac{r}{2R}$$

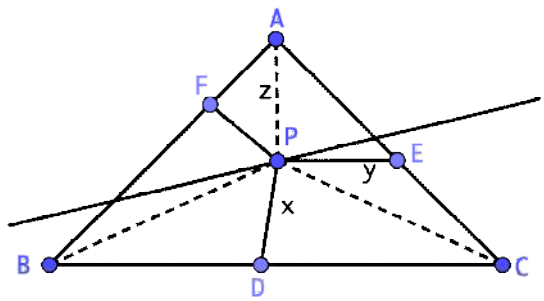
1.31 *Solution (Daniel Sitaru)*

Let $\Delta A_1B_1C_1$ ***be the intouch triangle*** $S_0 = S[A_1B_1C_1] = \frac{S^2}{2Rs}$.

$$\frac{\frac{1}{h_1^2}+\frac{1}{h_2^2}+\frac{1}{h_3^2}}{\frac{1}{h_a^2}+\frac{1}{h_b^2}+\frac{1}{h_c^2}} = \frac{\sum\frac{{A_1B_1}^2}{4S_0^2}}{\sum\frac{a^2}{4S^2}} = \frac{S^2}{S_0^2}\cdot\frac{\sum {A_1B_1}^2}{\sum a^2} =$$

$$= \frac{S^2}{\left(\frac{S^2}{2Rs}\right)^2}\cdot\frac{\frac{3}{4}\sum {A_1B_1}^2}{\frac{3}{4}\sum a^2} = \frac{4R^2s^2}{S^2}\cdot\frac{m_1^2+m_2^2+m_3^2}{m_a^2+m_b^2+m_c^2} = \frac{4R^2}{r^2}\cdot\frac{m_1^2+m_2^2+m_3^2}{m_a^2+m_b^2+m_c^2}$$

1.32 *Solution (Marian Ursărescu)*



We have: $x^2 = \frac{2(PB^2+PC^2)-a^2}{4}$ ***and similarly*** $\Rightarrow my^2 + nz^2 = (m+n)x^2 \Leftrightarrow$

$m\cdot\frac{2(PA^2+PC^2)-b^2}{4} + n\cdot\frac{2(PA^2+PB^2)-c^2}{4} = (m+n)\frac{2(PB^2+PC^2)-a^2}{4}$ ***and we have***

$mb^2 + nc^2 = (m+n)a^2 \Rightarrow$ ***we must show:***

$$m(PA^2+PC^2) + n(PA^2+PB^2) = (m+n)(PB^2+PC^2)$$

$$\Leftrightarrow (m+n)PA^2 = mPB^2 + nPC^2 \quad \textbf{(1)}$$

Let $P(z_p)\ A(z_A), B(z_B), C(z_C); z_P, z_A, z_C \in \mathbb{C}$

$P \in OH \Leftrightarrow \exists t \in [0,1]$ ***such that*** $z_p = t_{z+1} + (1-t)z_0$, ***but*** $z_0 = 0$,

$z_H = z_A + z_B + z_C \Rightarrow z_p = t(z_A + z_B + z_C) \Rightarrow$ ***from (1) we must show:***

$$(m+n)|z_P - z_A|^2 = m|z_P - z_B|^2 + n|z_P - z_C|^2 \Leftrightarrow$$

$$(m+n)|(t-1)z_A + z_B + z_C|^2 = m|(t-1)z_B + z_A + z_C|^2 + n|(t-1)z_B + z_A + z_C|^2 \quad \textbf{(2)}$$

Relation (2) it is true because:

$$mb^2 + nc^2 = (m+n)a^2 \Leftrightarrow m|z_A - z_C|^2 + n|z_A - z_B|^2 = (m+n)|z_B - z_C|^2$$

1.33 *Solution 1 (Daniel Sitaru)*

$$\begin{cases} BC^2 = CF^2 + FB^2 - 2CF \cdot FB \cdot \cos(\sphericalangle CFB) \\ CA^2 = FA^2 + CF^2 - 2FA \cdot FC \cdot \cos(\sphericalangle AFC) \\ AB^2 = FA^2 + FB^2 - 2FA \cdot FB \cdot \cos(\sphericalangle BFA) \end{cases}$$

$$BC = a, AC = b, AB = c$$

$$\begin{cases} a^2 = y^2 + FB^2 - 2y \cdot FB \cdot \cos 120^\circ = y^2 + FB^2 + y \cdot FB \\ b^2 = x^2 + y^2 - 2xy \cdot \cos 120^\circ = x^2 + y^2 + xy \\ c^2 = x^2 + FB^2 - 2x \cdot FB \cdot \cos 120^\circ = x^2 + FB^2 + x \cdot FB \end{cases}$$

$$c^2 = a^2 + b^2$$

$$x^2 + FB^2 + x \cdot FB = y^2 + FB^2 + y \cdot FB + x^2 + y^2 + xy$$

$$x^2 + x \cdot FB = 2y^2 + y \cdot FB + x^2 + xy, \quad x \cdot FB = 2y^2 + y \cdot FB + xy$$

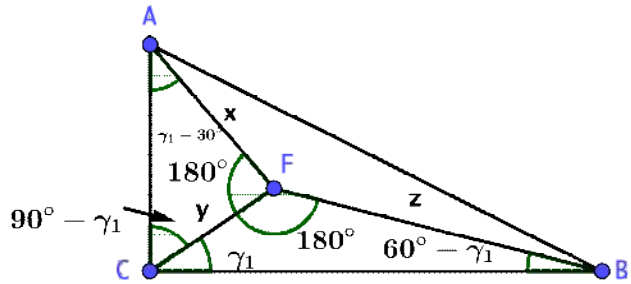
$$x \cdot FB - y \cdot FB = 2y^2 + xy, \ FB(x-y) = y(2y+x), x > y \to x - y > 0, FB = \frac{y(2y+x)}{x-y}$$

Solution 2 (Jovica Mikic)

Let $\sphericalangle BCF = \gamma_1$

Then $\sphericalangle CBF = 60^\circ - \gamma_1$

$\sphericalangle ACF = 90^\circ - \gamma_1$ ***and*** $\sphericalangle CAF = \gamma_1 - 30^\circ$



$$x > y$$

i) By sine law in ΔBFC

$$\frac{z}{\sin\gamma_1} = \frac{y}{\sin(60° - \gamma_1)} \Rightarrow z = y\frac{\sin\gamma_1 \,|: \sin\gamma_1}{\sin(60° - \gamma_1)\,|: \sin\gamma_1}$$

$$z = y\frac{\sin\gamma_1|:\sin\gamma_1}{\sin 60°\cos\gamma_1 - \cos 60°\sin\gamma_1|:\sin\gamma_1} \Rightarrow z = \frac{y}{\frac{\sqrt{3}}{2}\cot\gamma_1 - \frac{1}{2}} \quad (*)$$

$$\cot\gamma_1 = ?$$

ii) By sine law in ΔACF

$$\frac{x}{\sin(90° - \gamma_1)} = \frac{y}{\sin(\gamma_1 - 30°)} \Leftrightarrow \frac{x}{y} = \frac{\cos\gamma_1}{\sin(\gamma_1 - 30°)} = \frac{\cos\gamma_1}{\sin\gamma_1\frac{\sqrt{3}}{2} - \cos\gamma_1\frac{1}{2}}$$

$$\frac{x}{y} = \frac{\cos\gamma_1|:\sin\gamma_1}{\sin\gamma_1\cdot\frac{\sqrt{3}}{2} - \frac{1}{2}\cos\gamma_1|:\sin\gamma_1} \Rightarrow \frac{x}{y} = \frac{\cot\gamma_1}{\frac{\sqrt{3}}{2} - \frac{1}{2}\cot\gamma_1} \quad (**)$$

By (**): $\frac{x}{y}\cdot\frac{\sqrt{3}}{2} - \frac{1}{2}\cdot\frac{x}{y}\cot\gamma_1 = \cot\gamma_1$

$$\Leftrightarrow \frac{x}{y}\cdot\frac{\sqrt{3}}{2} = \cot\gamma_1\left(1 + \frac{1}{2}\cdot\frac{x}{y}\right) \Leftrightarrow \frac{x}{y}\cdot\frac{\sqrt{3}}{2} = \cot\gamma_1\cdot\frac{2y + x}{2y} \Leftrightarrow$$

$$\cot\gamma_1 = \frac{x\sqrt{3}}{x+2y} \quad (***)$$

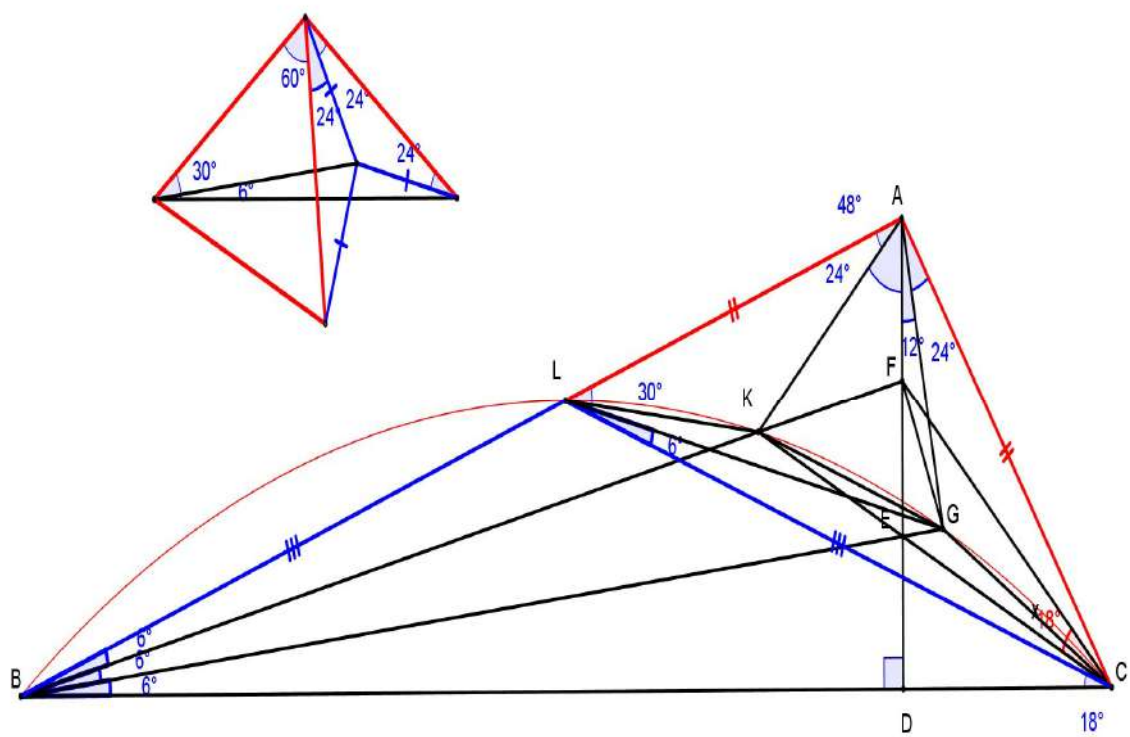
By (*) and (***), it follows:

$$z = \frac{y}{\frac{\sqrt{3}}{2}\cdot \cot\gamma_1 - \frac{1}{2}} = \frac{y}{\frac{\sqrt{3}}{2}\cdot\frac{x\sqrt{3}}{x+2y} - \frac{1}{2}} = \frac{y}{\frac{3x}{2(x+2y)} - \frac{1}{2}} = \frac{2y}{\frac{3x}{x+2y} - 1}$$

$$= \frac{\left(\frac{2y}{1}\right)}{\left(\frac{3x-(x+2y)}{x+2y}\right)} = \frac{\left(\frac{2y}{1}\right)}{\frac{2x-2y}{x+2y}} \Leftrightarrow z = \frac{2y(x+2y)}{2(x-y)}$$

$z = \frac{y(x+2y)}{x-y}$; $x > y \Rightarrow x - y > 0$ ***as expected. Q.E.D.***

1.34 *Solution (Apostolis Manoloudis)*



Let $GA = GC$ ***with*** $\measuredangle GAC = 24° = \measuredangle GCA$ ***and*** $AC = AL$ $(\Rightarrow BL = LC) \Rightarrow$

$\Rightarrow \measuredangle GLA = 30° = GCB \Rightarrow BLGC =$ ***cyclic*** $\Rightarrow GLC = 6° = \measuredangle GBC \Rightarrow B, E, G =$ ***collinear. But***

$\measuredangle BAG = 84° = \measuredangle BGA = \measuredangle AEG \Rightarrow AB = BG, AE = AG$ ***and*** $AF = FG$,

$\measuredangle FAG = 12° = \measuredangle FGA = \measuredangle EFG = 24° = \measuredangle ACG \Rightarrow AFGC =$ ***cyclic*** $\Rightarrow$

$$\Rightarrow \measuredangle ACF = \measuredangle AGF = 12^\circ = \measuredangle FCG. \textbf{ Let } \measuredangle BAK = 24^\circ \Rightarrow \Delta ABK = \Delta GBK \Rightarrow$$

$$\Rightarrow AK = KG = AG = GC. \textbf{ Is } AKG = 60^\circ = 2\measuredangle ALG \Rightarrow$$

$$\Rightarrow LK = KG = GC, \measuredangle KLG = 6^\circ = KGL \textbf{ and } BLKGC = \textbf{cyclic. But } A = \textbf{circumcenter of}$$

$$\Delta KEG \Rightarrow \measuredangle EKG = \measuredangle \frac{EAG}{2} = \frac{12^\circ}{2} = 6^\circ = \measuredangle CKG \Rightarrow K, E, C = \textbf{collinear. So,}$$

$$\measuredangle GKC = 6^\circ = \measuredangle GCK \Rightarrow x = \measuredangle FCE = \measuredangle FCG + \measuredangle GCK = 12^\circ + 6^\circ = 18^\circ$$

1.35 *Solution (Daniel Sitaru)*

$$\rho(D) = h_1(2R - h_1) = BD \cdot DC = \frac{a}{2} \cdot \frac{a}{2} = \frac{a^2}{4}, \quad h_1^2 - 2Rh_1 + \frac{a^2}{4} = 0$$

$$h_1 = \frac{2R - \sqrt{4R^2 - a^2}}{2} = \frac{2R - \sqrt{4R^2 - 4R^2 sin^2 A}}{2} = \frac{2\left(R - R\sqrt{1 - sin^2 A}\right)}{2}$$

$$h_1 = R - RcosA$$

$$2\left(h_1^2 + h_2^2 + h_3^2\right) = 2\sum_{cyc}(R - RcosA)^2 = 2R^2\left(3 - 2\sum_{cyc} cosA + \sum_{cyc} cos^2 A\right)$$

$$= 2R^2\left(3 - 2\left(1 + \frac{r}{R}\right) + \frac{6R^2 + 4Rr + r^2 - s^2}{2R^2}\right) =$$

$$= 6R^2 - 4R^2 - 4Rr + 6R^2 + 4Rr + r^2 - s^2 = 8R^2 + r^2 - s^2$$

1.36 *Solution (Daniel Sitaru)*

$$\Delta AFC: \ b^2 = y^2 + z^2 - 2yzcos120^\circ = y^2 + z^2 + yz$$

$$\Delta BFC: \ a^2 = x^2 + z^2 - 2xzcos120^\circ = x^2 + z^2 + xz$$

$$\Delta AFB: c^2 = x^2 + y^2 - 2xycos120^\circ = x^2 + y^2 + xy$$

$$\Delta ABC: b^2 = a^2 + c^2 - 2accos60^\circ = a^2 + c^2 - ac$$

$$y^2 + z^2 + yz = x^2 + z^2 + xz + x^2 + y^2 + xy - \sqrt{ac}$$

$$\sqrt{ac} = 2x^2 + xz + xy - yz$$

$$\sqrt{(x^2 + z^2 + xz)(x^2 + y^2 + xy)} = 2x^2 + xz + xy - yz$$

$$(x^2+z^2+xz)(x^2+y^2+xy)=(2x^2+xz+xy-yz)^2$$

$$x^4+x^3y-xyz^2-xy^2z+x^3z-x^2yz=0$$

$$x^3(x+y+z)-xyz(x+y+z)=0, x(x+y+z)(x^2-yz)=0,\ x^2=yz$$

1.37 *Solution (Daniel Sitaru)*

$$tan\omega=\frac{sinAsinBsinC}{1+cosAcosBcosC}\ ,tan\left(\frac{\pi}{2}-A\right)=\frac{sinAsinBsinC}{1+cosAcosBcosC}$$

$$\frac{cosA}{sinA}=\frac{sinAsinBsinC}{1+cosAcosBcosC}$$

$$cosA+cos^2AcosBcosC=sin^2AsinBsinC$$

$$-\cos(B+C)+cos^2AcosBcosC=sin^2AsinBsinC$$

$$sinBsinC-cosBcosC+cos^2AcosBcosC=sin^2AsinBsinC$$

$$sinBsinC(1-sin^2A)=cosBcosC(1-cos^2A)$$

$$sinB\cdot sinC\cdot cos^2A=cosB\cdot cosC\cdot sin^2A$$

$$\frac{sin^2A}{cos^2A}=\frac{sinB\cdot sinC}{cosB\cdot cosC}\ ,\qquad tan^2A=tanBtanC$$

1.38 *Solution (Daniel Sitaru)*

$$\Delta BCM_a:\ M_aB=M_aC, \sphericalangle(BM_aC)=\pi-A, \sphericalangle(BCM_a)=\sphericalangle(CBM_a)=\frac{A}{2}$$

$$\frac{BC}{\sin(\pi-A)}=\frac{M_aB}{sin\frac{A}{2}}\rightarrow M_aB=\frac{BC\cdot sin\frac{A}{2}}{sin(\pi-A)}=\frac{a\cdot sin\frac{A}{2}}{sinA}=2Rsin\frac{A}{2}$$

$$S_1=\frac{1}{2}\cdot M_aB\cdot M_aC\cdot sinA=\frac{1}{2}\cdot\frac{a\cdot sin\frac{A}{2}}{sinA}\cdot 2Rsin\frac{A}{2}\cdot sinA=R\cdot a\cdot sin^2\frac{A}{2}$$

$$S_1+S_2+S_3=R\left(a\cdot sin^2\frac{A}{2}+b\cdot sin^2\frac{B}{2}+c\cdot sin^2\frac{C}{2}\right)$$

$$S_1+S_2+S_3=R\cdot\frac{s(R-r)}{R}=s(R-r)=\frac{S}{r}(R-r)$$

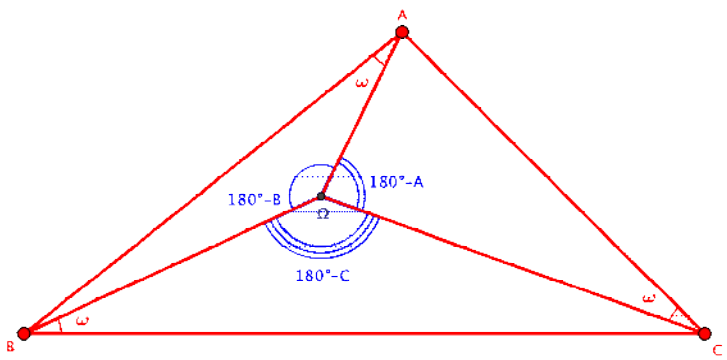
$$\frac{S_1+S_2+S_3}{S}=\frac{R-r}{r}=\frac{R}{r}-1$$

1.39 *Solution (Daniel Sitaru)*

$$\sum \frac{1}{S[IEF]} = \sum \frac{2}{r^2 sinA} = \frac{2}{r^2}\sum \frac{1}{sinA} = \frac{2}{r^2}\cdot\frac{s^2+r^2+4Rr}{2sr} \geq$$

$$\overset{GERRETSEN}{\geq} \frac{1}{r^2}\cdot\frac{16Rr-5r^2+r^2+4Rr}{S} = \frac{20Rr-4r^2}{r^2S} = \frac{4(5R-r)}{rS}$$

1.40 *Solution (Soumava Chakraborty)*



$$\measuredangle A\Omega B = 180° - (B - \omega + \omega) = 180° - B$$

$$In\ \Delta\ A\Omega B, \frac{B\Omega}{\sin\omega} = \frac{C}{\sin(180°-B)} = \frac{C}{\sin B} \Rightarrow B\Omega \overset{(1)}{=} \frac{C}{\sin B}\sin\omega$$

$$\because \frac{1}{2}ca\sin B = \Delta, \therefore \frac{1}{\sin B} \overset{(2)}{=} \frac{ca}{2\Delta}$$

$$\therefore B\Omega = \frac{c^2a}{2\Delta}\cdot\frac{2\Delta}{\sqrt{\sum a^2b^2}} = \frac{c^2a}{\sqrt{\sum a^2b^2}}\ \textbf{(using (1), (2))} \Rightarrow B\Omega^2 = \frac{c^4a^2}{\sum a^2b^2} \Rightarrow \frac{1}{B\Omega^2} = \frac{\sum a^2b^2}{c^4a^2}\ \textbf{(a)}$$

$$\textbf{Similarly, } \frac{1}{C\Omega^2} = \frac{\sum a^2b^2}{a^4b^2}\ \textbf{(b) and, } \frac{1}{A\Omega^2} = \frac{\sum a^2b^2}{b^4c^2}\ \textbf{(c)}$$

$$\textbf{(a)+(b)+(c)} \Rightarrow LHS = \frac{\sum a^2b^2}{a^2b^2c^2}\left(\frac{a^2b^2c^2}{c^4a^2}+\frac{a^2b^2c^2}{a^4b^2}+\frac{a^2b^2c^2}{b^4c^2}\right) = \left(\sum\frac{1}{a^2}\right)\left(\frac{a^2}{b^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}\right)$$

1.41 *Solution (Daniel Sitaru)*

$$\sum\left(\frac{b}{c}+\frac{c}{b}\right)cosA = \frac{1}{2}\sum\frac{(b^2+c^2)(b^2+c^2-a^2)}{b^2c^2} =$$

$$= \frac{1}{2}\sum\frac{b^4+2b^2c^2+c^4-b^2a^2-a^2c^2}{b^2c^2} =$$

$$= \frac{1}{2}\sum\frac{b^2}{c^2} + \frac{2}{2}\cdot 3 + \frac{1}{2}\sum\frac{c^2}{b^2} - \frac{1}{2}\sum\frac{a^2}{c^2} - \frac{1}{2}\sum\frac{a^2}{b^2} = 3$$

1.42 *Solution (Marian Ursărescu)*

We know areal coordinates for k: $\forall M \in P \Rightarrow$

$\overrightarrow{MK} = \frac{a^2\overrightarrow{MA}+b^2\overrightarrow{MB}+c^2\overrightarrow{MC}}{a^2+b^2+c^2}$***. Let*** $M = 0 \Rightarrow \overrightarrow{OK} = \frac{a^2\overrightarrow{OA}+b^2\overrightarrow{OB}+c^2\overrightarrow{OC^2}}{a^2+b^2+c^2}|^2$ ***and using inner product***

$$OK^2 = \frac{a^4OA^2 + b^4OB^2 + c^4OC^2 + 2a^2b^2\overrightarrow{OA}\overrightarrow{OB} + 2a^2c^2\overrightarrow{OA}\cdot\overrightarrow{OC} + 2b^2c^2\overrightarrow{OK}\cdot\overrightarrow{OC}}{(a^2 + b^2 + c^2)^2} \Rightarrow$$

$$OK^2 = \frac{R^2(a^4+b^4+c^4)+2a^2b^2\overrightarrow{OA}\cdot\overrightarrow{OB}+2a^2c^2\overrightarrow{OA}\cdot\overrightarrow{OC}+2b^2c^2\overrightarrow{OB}\cdot\overrightarrow{OC}}{(a^2+b^2+c^2)^2} \quad \textbf{(1)}$$

But $OA = OB = OC = R$ ***and the use:*** $\overrightarrow{AB}\cdot\overrightarrow{AC} = \frac{AB^2+AC^2-BC^2}{4}$***, and we have:***

$$\overrightarrow{OA}\cdot\overrightarrow{OB} = \frac{OA^2+OB^2-AB^2}{2} = \frac{2R^2-AB^2}{2} = R^2 - \frac{C^2}{2} \quad \textbf{(2)}$$

From (1)+(2)$\Rightarrow OK^2 = \frac{R^2(a^4+b^4+c^4)+a^2b^2(2R^2-c^2)+a^2c^2(2R^2-b^2)+b^2c^2(2R^2-a^2)}{(a^2+b^2+c^2)^2} \Rightarrow$

$$OK^2 = \frac{R^4(a^4 + b^4 + c^4) + 2a^2b^2 + 2a^2c^2 + 2b^2c^2) - 3a^2b^2c^2}{(a^2 + b^2 + c^2)^2} \Rightarrow$$

$$OK^2 = \frac{R^2(a^2 + b^2 + c^2)^2 - 3a^2b^2c^2}{(a^2 + b^2 + c^2)^2} \Rightarrow OK^2 = R^2 - \frac{3a^2b^2c^2}{(a^2 + b^2 + c^2)^2}$$

1.43 *Solution(Daniel Sitaru)*

$$\sum_{cyc}\sqrt{\frac{h_bh_c}{rr_a}} = \sum_{cyc}\sqrt{\frac{\frac{2S}{b}\cdot\frac{2S}{c}}{\frac{S}{s}\cdot\frac{S}{s-a}}} = 2\sum_{cyc}\sqrt{\frac{s(s-a)}{bc}} =$$

$$= 2\sum_{cyc}cos\frac{A}{2} = 2\sum_{cyc}\left(\frac{2bc}{b+c}cos\frac{A}{2}\cdot\frac{b+c}{2bc}\right) =$$

$$= \sum_{cyc}\left(w_a \cdot \frac{b+c}{bc}\right) = \sum_{cyc}\left(w_a\left(\frac{1}{b}+\frac{1}{c}\right)\right) = \sum_{cyc}\left(\frac{w_a}{b}+\frac{w_a}{c}\right) = \sum_{cyc}\frac{w_b+w_c}{a}$$

1.44 *Solution (Daniel Sitaru)*

$$cosC = \frac{DC}{b} \to DC = bcosC \to DL = bcosC - \frac{a}{2} \to$$

$$\to tan\alpha = \frac{2bcosC - a}{2h_a} = \frac{2abcosC - a^2}{4S} = \frac{a^2+b^2-c^2-a^2}{4S} = \frac{b^2-c^2}{4S}$$

$$tan\beta = \frac{a^2-c^2}{4S}, tan\theta = \frac{a^2-b^2}{4S}$$

$$tan\beta = \frac{a^2-c^2}{4S} = \frac{b^2-c^2}{4S} + \frac{a^2-b^2}{4S} = tan\alpha + tan\theta$$

1.45 *Solution (Daniel Sitaru)*

$$a = \frac{a+b+c}{4} \to 3a = b+c \to a = \frac{b+c}{3} \to s = \frac{2(b+c)}{3}$$

$$S^2 = \frac{2(b+c)}{3}\cdot\left(\frac{2(b+c)}{3}-\frac{b+c}{3}\right)\left(\frac{2(b+c)}{3}-b\right)\left(\frac{2(b+c)}{3}-c\right) = \frac{2(b+c)^2}{81}(2b-c)(2c-b)$$

$$r_{SPIEKER\ CIRCLE} = \frac{r}{2} = S_pI = \frac{3}{2}GI = \frac{1}{2}\sqrt{s^2+5r^2-16Rr} \to r^2 = 2s^2+5r^2-16Rr \leftrightarrow$$

$$\leftrightarrow 16.\frac{abc}{4S}\cdot r = s^2+4r^2 \leftrightarrow \frac{4abc}{s} = s^2+4\frac{S^2}{s^2} \leftrightarrow 4abcs = s^4+4S^2 \leftrightarrow$$

$$\leftrightarrow 4\cdot\frac{b+c}{3}\cdot bc\cdot\frac{2(b+c)}{3} = \frac{16}{81}\cdot(b+c)^4 + 4\cdot\frac{(b+c)^2}{81}\cdot 2(2b-c)(2c-a) \leftrightarrow$$

$$\leftrightarrow 8bc = \frac{16}{9}(b+c)^2 + \frac{8}{9}(2b-c)(2c-b) \leftrightarrow bc = \frac{2}{9}\left(b^2+2bc+c^2\right) + \frac{1}{9}(2b-c)(2c-a) \leftrightarrow$$

$$\leftrightarrow 9bc = 2b^2+4bc+2c^2+4bc-2b^2-2c^2+bc \leftrightarrow 0 = 0$$

1.46 *Solution 1 (Marian Ursărescu)*

Let M be a point of $\Delta\ ABC$ plane. We have:

$$\overrightarrow{MG} = \frac{\overrightarrow{MA}+\overrightarrow{MB}+\overrightarrow{MC}}{3} \text{ and } \overrightarrow{MC} = \frac{a^2\overrightarrow{MA}+b^2\overrightarrow{MB}+c^2\overrightarrow{MC}}{a^2+b^2+c^2}$$

$$\textbf{But } c^2 = \frac{a^2+b^2}{2} \Rightarrow \overrightarrow{MK} = \frac{2a^2\overrightarrow{MA}+2b^2\overrightarrow{MB}+(a^2+b^2)\overrightarrow{MC}}{3(a^2+b^2)}$$

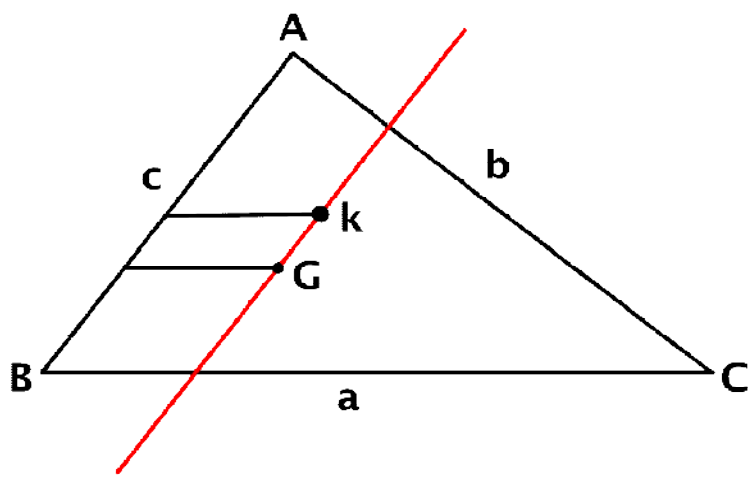
$$\textbf{Then } \overrightarrow{GK} = \overrightarrow{OK} - \overrightarrow{OG} = \frac{2a^2\overrightarrow{MA}+2b^2\overrightarrow{MB}+(a^2+b^2)\overrightarrow{MC}}{3(a^2+b^2)} - \frac{\overrightarrow{MA}+\overrightarrow{MB}+\overrightarrow{MC}}{3}$$

$$= \frac{2a^2\overrightarrow{MA} + 2b^2\overrightarrow{MB} + (a^2 + b^2)\overrightarrow{MC} - (a^2 + b^2)\overrightarrow{MA} - (a^2 + b^2)\overrightarrow{MB} - (a^2 + b^2)\overrightarrow{MC}}{3(a^2 + b^2)}$$

$$= \frac{(a^2 - b^2)\overrightarrow{MA} + (b^2 - a^2)\overrightarrow{MB}}{3(a^2 + b^2)} = \frac{(b^2 - a^2)(\overrightarrow{MB} - \overrightarrow{MA})}{3(a^2 + b^2)} \Rightarrow$$

$$\overrightarrow{GK} = \frac{(b^2-a^2)}{3(a^2+b^2)} \cdot \overrightarrow{AB} \Rightarrow \overrightarrow{GK} \textbf{ and } \overrightarrow{AB} \textbf{ colliniar} \Rightarrow GK \parallel AB$$

Solution 2 (Thanasis Gakopoulos)



$$BC \equiv Bx, BA \equiv By, G\left(\frac{a}{3}, \frac{c}{3}\right), k(k_1, k_2)$$

$$k_1 = \frac{ac^2}{a^2 + b^2 + c^2}, k_2 = \frac{a^2c}{a^2 + b^2 + c^2}$$

$$GK \parallel AB \Leftrightarrow k_1 = \frac{a}{3} \Leftrightarrow \frac{ac^2}{a^2 + b^2 + c^2} = \frac{a}{3} \Leftrightarrow 3c^2 = a^2 + b^2 + c^2 \Leftrightarrow a^2 + b^2 = 2c^2$$

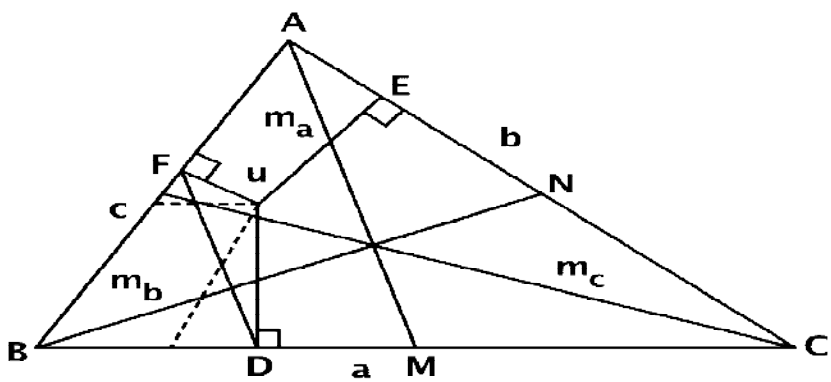
1.47 *Solution 1 (Daniel Sitaru)*

$$\begin{cases} EF = \dfrac{2S}{a^2+b^2+c^2}\sqrt{2c^2+2b^2-a^2} \\ DF = \dfrac{2S}{a^2+b^2+c^2}\sqrt{2a^2+2c^2-b^2} \end{cases}$$

$$\frac{m_a}{EF} = \frac{m_a}{\dfrac{2S}{a^2+b^2+c^2}\sqrt{2c^2+2b^2-a^2}} = \frac{m_a}{\dfrac{2s}{a^2+b^2+c^2}\cdot 2m_a} = \frac{a^2+b^2+c^2}{4}$$

$$\frac{m_b}{DF} = \frac{m_b}{\dfrac{2S}{a^2+b^2+c^2}\sqrt{2a^2+2c^2-b^2}} = \frac{m_b}{\dfrac{2S}{a^2+b^2+c^2}\cdot 2m_b} = \frac{a^2+b^2+c^2}{4}$$

Solution 2 (Thanasis Gakopoulos)



$$BC \equiv Bx, BA = By,\ B(0,0), N\left(\frac{a}{2},\frac{c}{2}\right), K(K_1,K_2)$$

$$K_1 = \frac{ac^2}{a^2+b^2+c^2}, K_2 = \frac{a^2c}{a^2+b^2+c^2} \quad \textbf{(1)},\ \cos B = \frac{a^2+c^2-b^2}{2ac} \quad \textbf{(2)}$$

$$m_b^2 = \left(\frac{a}{2}\right)^2 + \left(\frac{c}{2}\right)^2 + 2\cdot\frac{a}{2}\cdot\frac{c}{2}\cos B \overset{(2)}{\rightarrow} m_b^2 = \frac{2a^2-b^2+2c^2}{4} \quad \textbf{(3)}$$

$$D(K_1 + K_2\cdot\cos B, 0), F(0, K_2 + K_1\cos B)$$

$$(DF)^2 = (K_1 + K_2\cdot\cos B)^2 + (K_2 + K_1\cos B)^2 - 2(K_1 + K_2\cos B)(K_2 + K_1\cos B)\cdot\cos B$$

$$= \left(K_1^2 + K_2^2 + 2K_1K_2\cos B\right)(1-\cos^2 B) \underset{(2)}{\overset{(1)}{=}} \frac{16S^2}{\left(a^2+b^2+c^2\right)^2}\cdot\frac{\left(2a^2-b^2+2c^2\right)}{4} \quad \textbf{(4)}$$

(3)/(4) $\rightarrow \frac{m_b^2}{(DF)^2} = \frac{(a^2+b^2+c^2)^2}{16S^2} \rightarrow \frac{m_b}{DF} = \frac{a^2+b^2+c^2}{4S}$, ***cyclically similar*** $\frac{m_a}{EF} = \frac{m_b}{DF} = \frac{m_c}{DE} = \frac{a^2+b^2+c^2}{4S}$

1.48 *Solution (Rajsekhar Azaad)*

$$R_A = \frac{s-a}{2\cos\frac{A}{2}} \text{ and } R_a = \frac{a}{2\cos\frac{A}{2}}$$

$$\therefore R_a \cdot R_A \cdot R_b \cdot R_B \cdot R_c \cdot R_C = \frac{abc\prod(s-a)}{64\prod\cos^2\frac{A}{2}} = \frac{abc\cdot\prod(s-a)}{64S^3\prod(s-a)} \cdot a^2b^2c^2 =$$

$$= \frac{a^3b^3c^3}{64S^3} = \frac{64R^3r^3S^3}{64S^3} = R^3r^3$$

1.49 *Solution (Daniel Sitaru)*

$$1.\ 2cotB = cotA + cotC \leftrightarrow 2\cdot\frac{a^2+c^2-b^2}{4S} = \frac{b^2+c^2-a^2}{4S} + \frac{a^2+b^2-c^2}{4S} \leftrightarrow$$

$$2b^2 = a^2 + c^2$$

$$2.\ 2cos2B = cos2A + cos2C \leftrightarrow 2(1-2sin^2B) = 1-2sin^2A + 1-2sin^2C \leftrightarrow$$

$$\leftrightarrow -4\cdot\frac{b^2}{4R^2} = -2\cdot\frac{a^2}{4R^2} - 2\cdot\frac{c^2}{4R^2} \leftrightarrow 2b^2 = a^2 + c^2$$

$$3.\ 2m_b^2 = m_a^2 + m_c^2 \leftrightarrow 2\cdot\frac{2(a^2+c^2)-b^2}{4} = \frac{2(b^2+c^2)-a^2}{4} + \frac{2(a^2+b^2)-c^2}{4} \leftrightarrow$$

$$\leftrightarrow 6b^2 = 3(a^2+c^2) \leftrightarrow 2b^2 = a^2 + c^2$$

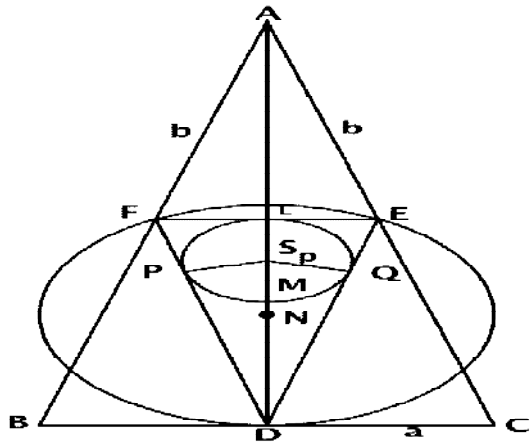
$$4.\ 2BH^2 = AH^2 + CH^2 \leftrightarrow 8R^2cos^2B = 4R^2cos^2A + 4R^2cos^2C \leftrightarrow$$

$$2cos^2B = cos^2A + cos^2C \leftrightarrow$$

$$\leftrightarrow 2\cdot\frac{1-cos2B}{2} = \frac{1-cos2A}{2} + \frac{1-cos2C}{2} \leftrightarrow 2cos2B = cos2A + cos2C \overset{(2)}{\leftrightarrow}$$

$$2b^2 = a^2 + c^2$$

1.50 *Solution (Thanasis Gakopoulos)*



$$AC > BC, (AB) = (AC) = b, (BC) = a,\ N - \textit{Nine point center},\ S_p - \textit{Spieker center}$$

$$(MN) = (MS_p) = (LS_p), AB \equiv Ax, AC \equiv Ay$$

$$N(e,e)\quad e = \frac{b(\cos A - \cos 2A)}{4\sin^2 A}\ \textbf{(1)},\ S_p(S_p, S_p)\ \ S_p = \frac{b(a+b)}{2(a+2b)}\ \textbf{(2)}$$

$$a^2 = 2b^2 - 2b^2\cos A \to \cos A = 1 - \frac{a^2}{2b^2}\ \textbf{(3)},\ ED: y = \frac{b}{2} \to Ox + 2y - b = 0$$

$$d(SpQ) = \sqrt{S_p} = \frac{|2S_p - b|\sin A}{\sqrt{2^2}} \to \sqrt{S_p} = \frac{b - 2S_p}{2}\sin A\ \textbf{(4)}$$

$$(NL) = 2(e - l)\cos\frac{A}{2}\ \textbf{(5)}, L(l,l), l = \frac{b}{4}\ \textbf{(6)}$$

$$3\sqrt{S_p} = (NL) \underset{(5)}{\overset{(4)}{\to}} 3\cdot\frac{b - 2S_p}{2}\cdot 2\sin\frac{A}{2}\cdot\cos\frac{A}{2} = 2(e - l)\cos\frac{A}{2} \to$$

$$\to 3(b - 2S_p)\sin\frac{A}{2} = (e - l)\cdot 2 \overset{(2)}{\to}$$

$$\underset{(6)}{\overset{(1)}{\to}} 3\cdot\frac{b^2}{a + 2b}\cdot\frac{a}{2b} = 2\left(\frac{\cos A - \cos 2A}{\sin^2 A} - 1\right)\frac{b}{4} \to \frac{3a}{a + 2b} = \frac{\cos A}{1 + \cos A} \overset{(3)}{\to}$$

$$\to \frac{3a}{a + 2b} = \frac{1 - \frac{a^2}{2b^2}}{1 + 1 - \frac{a^2}{2b^2}} \to \frac{2(b^2 - 3ab + a^2)}{4b^2 - a^2} = 0 \to \frac{b^2}{a^2} - 3\frac{b}{a} + 1 = 0 \to$$

$$\rightarrow \frac{b}{a} = \frac{3+\sqrt{5}}{2} \rightarrow \frac{b}{a} = \left(\frac{1+\sqrt{5}}{2}\right)^2 \rightarrow \sqrt{\frac{b}{a}} = \frac{1+\sqrt{5}}{2} \rightarrow \sqrt{\frac{AC}{BC}} = \phi \text{ (Golden Ratio)}$$

1.51 *Solution 1 (Mohammad Alhafi)*

Using the well known symmedian property we have: $\frac{\sin\angle CPK}{\sin\angle DPK} = \frac{CP}{PD}$

Using the ratio lemma and since PG ***is a median in*** ΔAPB ***we have:*** $\frac{\sin\angle APG}{\sin\angle BPG} = \frac{BP}{AP}$

but by a power of a point we have

$$PC \cdot AP = PD \cdot BP \Rightarrow \frac{CP}{PD} = \frac{BP}{AP} \Rightarrow \frac{\sin\angle CPK}{\sin\angle DPK} = \frac{\sin\angle APG}{\sin\angle BPG}$$

since $\angle CPK + \angle DPK = \angle APG + \angle BPG$ ***we have*** $\angle APG = \angle CPK$

so: G, P, K ***are collinear.***

Solution 2 (Omran Kouba)

Let $x = PA, y = PB, z = PC, t = PD$***, and let*** $a = CD$***. Recall that*** $xz = yt$***.***

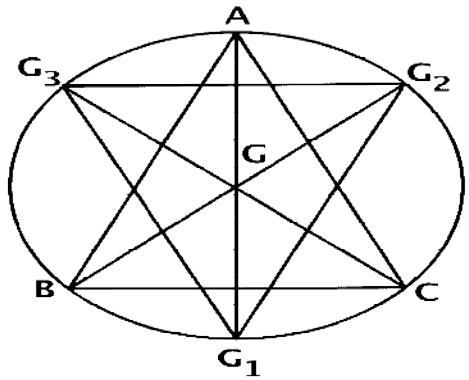
Since K ***is the barycenter of*** $(P; a^2), (C; t^2), (D; z^2)$ ***we conclude that***

$$(t^2 + z^2 + a^2)\overrightarrow{PK} = t^2\overrightarrow{PC} + z^2\overrightarrow{PD} = -\frac{zt^2}{x}\overrightarrow{PA} - \frac{tz^2}{y}\overrightarrow{PB} = -zt\left(\frac{t}{x}\overrightarrow{PA} + \frac{z}{y}\overrightarrow{PB}\right)$$

$$= -zt\left(\frac{t}{x}\overrightarrow{PA} + \frac{t}{x}\overrightarrow{PB}\right) = -\frac{zt^2}{x}\left(\overrightarrow{PA} + \overrightarrow{PB}\right) = -\frac{3zt^2}{x}\overrightarrow{PG}$$

1.52 *Solution (Marian Ursărescu)*

$$\frac{S_{G_1G_2G_3}}{S_{ABC}} = \frac{G_1G_2 \cdot G_1G_3 \cdot G_2G_3}{4R} \cdot \frac{4R}{AB \cdot BC \cdot AC}$$



$$\Rightarrow \frac{S_{G_1G_2G_3}}{S_{ABC}} = \frac{G_1G_2 \cdot G_1G_3 \cdot G_2G_3}{AB \cdot BC \cdot AC} \quad \textbf{(1)}$$

$$\Delta\, GAB \sim GG_2G_1 \Rightarrow \frac{GA}{GG_2} = \frac{AB}{G_1G_2} \Rightarrow G_1G_2 = \frac{c \cdot GG_2}{GA} = c \cdot \frac{GB \cdot GG_2}{GAGB} = \frac{c \cdot \rho(G)}{GA \cdot GB}$$

Similarly, $G_2G_3 = \frac{a\rho(G)}{GB \cdot GC}, G_1G_3 = \frac{b\rho(G)}{GA \cdot GC}$ ***(2)***

From (1)+(2)$\Rightarrow \frac{S_{G_1G_2G_3}}{S_{ABC}} = \frac{(\rho(G))^3}{(GA \cdot GB \cdot GC)^2}$ ***(3), But*** $\rho(G) = R^2 - OG^2 = \frac{1}{9}(a^2 + b^2 + c^2)$ ***(4)***

$GA^2 = \frac{4}{9}ma^2$ ***(5). From (3)+(4)+(5)***$\Rightarrow \frac{S_{G_1G_2G_3}}{S_{ABC}} = \frac{(a^2+b^2+c^2)^3}{4^3 \cdot m_a^2 \cdot m_b^2 \cdot m_c^2}$ ***(6)***

Now, from the same reason $\Rightarrow \frac{S_{K_1K_2K_3}}{S_{ABC}} = \frac{\rho(k)^3}{(KAKBKC)^3}$ ***(7)***

But $\rho(k) = R^2 - OK^2 = \frac{3a^2b^2c^2}{(a^2+b^2+c^2)^2}$ ***(8)***

$$KA^2 = \left(\frac{b^2+c^2}{a^2+b^2+c^2} S_a\right)^2 = \left(\frac{b^2+c^2}{(a^2+b^2+c^2)} \cdot \frac{2bc}{b^2+c^2} \cdot m_a\right)^2 = \frac{4b^2c^2}{(a^2+b^2+c^2)^2} \cdot m_a^2 \quad \textbf{(9)}$$

From (7)+(8)+(9)$\Rightarrow \frac{S_{K_1K_2K_3}}{S_{ABC}} = \frac{\frac{2+a^6b^6c^6}{(a^2+b^2+c^2)^6}}{\frac{64 \cdot a^4b^4c^4}{(a^2+b^2+c^2)^6} \cdot m_a^2 \cdot m_b^2 \cdot m_c^2} \Rightarrow$

$\frac{S_{K_1K_2K_3}}{S_{ABC}} = \frac{27}{64} \cdot \frac{a^2b^2c^2}{m_a^2m_b^2m_c^2}$ ***(10). From (6)+(10)***$\Rightarrow$ ***(to split)***$\Rightarrow$

$$\frac{S_{G_1G_2G_3}}{S_{K_1K_2K_3}} = \frac{(a^2+b^2+c^2)^3}{27a^2b^2c^2}$$

1.53 *Solution 1 (Daniel Sitaru)*

$$\frac{BD}{DC} = \frac{c^2}{b^2} \to \frac{BD}{BD+DC} = \frac{c^2}{b^2+c^2} \to BD = \frac{ac^2}{b^2+c^2}, DC = \frac{bc^2}{b^2+c^2}$$

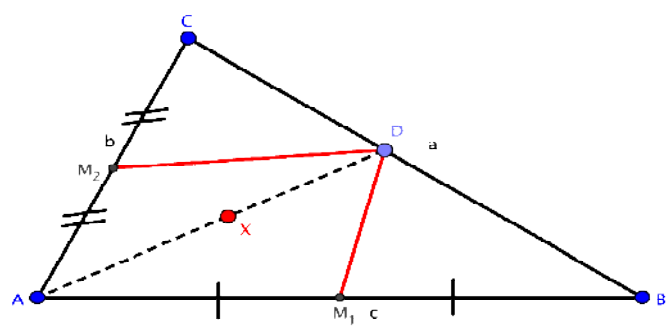
$$DM_1 = DM_2 \to DM_1^2 = DM_2^2 \to \frac{1}{2}(AD^2+BD^2) - \frac{c^2}{4} = \frac{1}{2}(AD^2+DC^2) - \frac{b^2}{4} \to$$

$$\to \frac{1}{2}\left(\frac{ac^2}{b^2+c^2}\right)^2 - \frac{c^2}{4} = \frac{1}{2}\left(\frac{ab^2}{b^2+c^2}\right)^2 - \frac{b^2}{4} \to$$

$$2a^2c^4 - c^2(b^2+c^2)^2 = 2a^2b^4 - b^2(b^2+c^2)^2 \to (b^2+c^2)^2(c^2-b^2) =$$

$$= 2a^2(b^2+c^2)(c^2-b^2)\ ;\ \ c^2-b^2 \neq 0 \to b^2+c^2 = 2a^2$$

Solution 2 (Thanasis Gakopoulos)



$$DM_1 = DM_2, M_1\left(\frac{c}{2},0\right), M_2\left(0,\frac{b}{2}\right), X(x_1,x_2), D(d_1,d_2)$$

$$\left.\begin{array}{l} AX: y = \frac{x_2}{x_1}\cdot x \\ BC: \frac{x}{c} + \frac{y}{b} = 1 \end{array}\right\} \to d_1 = \frac{b\cdot c\cdot x_1}{bx_1+cx_2}\ \textbf{(1)}, d_2 = \frac{b\cdot c\cdot x_2}{bx_1+c\cdot x_2}\ \textbf{(2)}$$

(II) If X incenter $x_1 = x_2 = \frac{bc}{a+b+c}$, ***(1),(2)*** $\to d_1 = d_2 = \frac{bc}{b+c}$ ***(3),r. cosine:*** $\cos\theta = \frac{b^2+c^2-a^2}{2bc}$ ***(4)***

$$(DM_1)^2 = (DM_2)^2 \to \left(\frac{c}{2} - d_1\right)^2 + (0 - d_2)^2 - 2\left(\frac{c}{2} - d_1\right) d_2 \cdot \cos\theta =$$

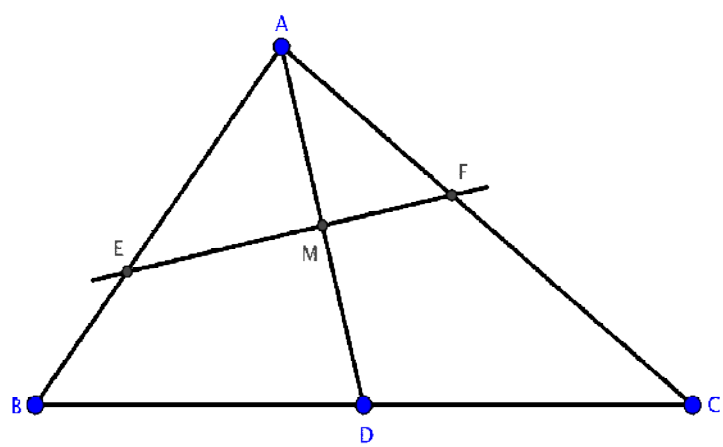
$$= (0 - d_1)^2 + \left(\frac{b}{2} - d_2\right)^2 - 2\left(\frac{b}{2} - d_2\right) d_1 \cdot \cos\theta \underset{(4)}{\overset{(3)}{\to}} (b + c)^2 = 2a^2$$

(I) ***If X symmedian point,*** $x_1 = \frac{b^2c}{a^2+b^2+c^2}, x_2 = \frac{bc^2}{a^2+b^2+c^2}$ ***and*** $d_1 = f_1(a, b, c), d_2 = f(a, b, c)$***, rule cosine,***

$$(DM_1)^2 = (DM_2)^2 \to \cdots \to b^2 + c^2 = 2a^2$$

1.54 *Solution (Marian Ursărescu)*

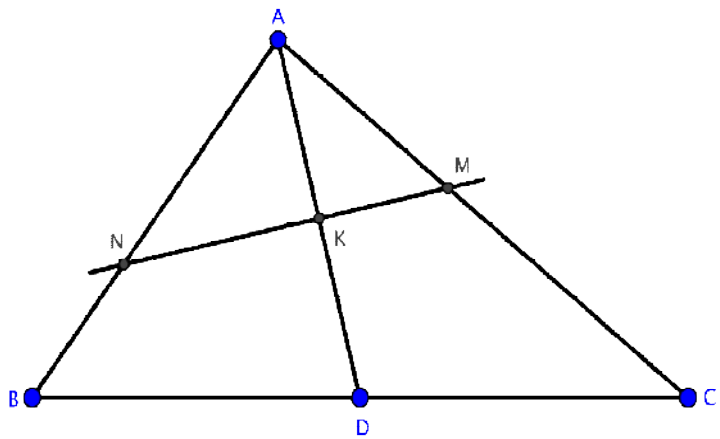
We use transversal theorem: $E \in (AB), F \in (AC)$



$$M \in (EF) \Leftrightarrow CD \cdot \frac{EB}{EA} + BD \cdot \frac{FC}{FA} = BC \cdot \frac{MD}{MA}$$

In our case $M = K \Rightarrow K \in (EF) \Leftrightarrow CD \cdot \frac{EB}{EA} + BD \cdot \frac{FC}{FA} = BC \cdot \frac{KD}{KA}$ ***(1)***

But $\frac{KA}{KM} = \frac{b^2+c^2}{a^2} \Rightarrow \frac{KM}{KA} = \frac{a^2}{b^2+c^2}$ ***(2)***



Now, $\frac{BD}{DC} = \frac{c^2}{b^2} \Rightarrow \frac{BD}{BC} = \frac{c^2}{b^2+c^2} \Rightarrow BD = \frac{ac^2}{b^2+c^2}$ ***and*** $CD = \frac{ab^2}{b^2+c^2}$ ***(3)***

From (1)+(2)+(3) $\Rightarrow K \in (EF) \Leftrightarrow \frac{ab^2}{b^2+c^2} \cdot \frac{FB}{EA} + \frac{ac^2}{b^2+c^2} \cdot \frac{FC}{FA} = a \cdot \frac{a^2}{b^2+c^2} \Rightarrow$

K, E, F ***collinear*** $\Leftrightarrow b^2 \cdot \frac{EB}{EA} + c^2 \cdot \frac{FC}{FA} = a^2$ ***(4)***

In our case $\frac{EB}{EA} = \frac{BN}{NA} = \frac{a}{b}$ ***(5) and*** $\frac{FC}{FA} = \frac{MC}{MA} = 1$ ***(6)***

From (4)+(5)+(6) we show this:

N, K, M ***colliniar*** $\Leftrightarrow b^2 \cdot \frac{a}{b} + c^2 = a^2 \Leftrightarrow N, K, M$ ***colliniar*** $\Leftrightarrow ab + c^2 = a^2$ ***(7)***

But $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$

(7) $\Leftrightarrow \sin A \sin B + \sin^2 C = \sin^2 A \Leftrightarrow \sin A \sin B = \sin^2 A - \sin^2 C \Leftrightarrow$

$$\Leftrightarrow \sin A \sin B = \frac{1 - \cos 2A}{2} - \frac{1 - \cos^2 C}{2} \Leftrightarrow \sin A \sin B = \frac{1}{2}(\cos^2 C - \cos^2 A) \Leftrightarrow$$

$$\Leftrightarrow \sin A \sin B = \sin(A + C) \cdot \sin(A - C) \Leftrightarrow \sin A \sin B = \sin B \cdot \sin(A - C) \Leftrightarrow$$

$\Leftrightarrow \sin A = \sin(A - C)$ ***(8). But*** $C = \frac{\pi}{7}, B = \frac{2\pi}{7}, A = \frac{4\pi}{7}$

$8 \Leftrightarrow \sin\frac{4\pi}{7} = \sin\frac{3\pi}{7}$, ***true because*** $\sin\left(\frac{4\pi}{7}\right) = \sin\left(\pi - \frac{3\pi}{7}\right) = \sin\frac{3\pi}{7}$